

# ECON 605: Macroeconomic Theory I: Lecture Notes

October 6, 2026

## 1 Introduction

### Detrending, Stationarity, and Filters

#### Why we detrend

The business cycle is about *fluctuations around a trend*. A convenient working decomposition is

$$y_t = \tau_t + x_t,$$

where  $y_t$  is the observed series (e.g., real GDP or employment),  $\tau_t$  is a smooth “trend” (growth, demographics, technology, seasonality if unadjusted), and  $x_t$  is the cyclical/stationary component we want to study. The goal of detrending is to construct  $x_t$  so that its distribution does not drift over time, making correlation, impulse responses, and model fitting meaningful.

**Intuition.** Think of  $\tau_t$  as the escalator and  $x_t$  as your steps on it. If you want to study how you walk (accelerations, pauses), you first subtract the escalator’s steady motion.

### 1.1 Trends: deterministic vs. stochastic

#### Deterministic trends

A deterministic trend is a known (or well-approximated) function of time, e.g.,

$$\tau_t = a_0 + a_1 t \quad (\text{linear}), \quad \tau_t = a_0 + a_1 t + a_2 t^2 \quad (\text{quadratic}).$$

Regressing  $y_t$  on time polynomials and using the residuals yields a detrended  $x_t$ .

#### Stochastic trends (unit roots)

A unit root process has a persistent, random drift:

$$y_t = y_{t-1} + \varepsilon_t, \quad \varepsilon_t \text{ i.i.d. (or weakly dependent).}$$

*First-differencing* removes the unit root:  $\Delta y_t = \varepsilon_t$  becomes stationary (up to short memory). This motivates using differences (or log-differences for growth rates).

**Slide tie-in.** Your slides explicitly write  $y_t = \tau_t + x_t$  and list polynomial forms for  $\tau_t$  on separate bullets; they also emphasize differencing when  $\tau_t$  is stochastic and note unit-root testing (ADF) for stationarity

diagnostics.<sup>1</sup>

## 1.2 Difference vs. log-difference

What each does

- **Difference**  $\Delta y_t = y_t - y_{t-1}$  removes a unit root in levels.
- **Log-difference**  $\Delta \ln y_t \approx \%$  growth. If  $y_t$  has a stochastic trend in logs,  $\Delta \ln y_t$  is stationary under standard conditions.

**Intuition.** If the variable grows roughly exponentially (GDP, employment), taking logs makes the trend roughly linear; differencing logs then measures growth rates. For unit-root removal per se, both differences and log-differences purge the random walk part; choose logs when you want elasticities/growth interpretation.

## 1.3 Testing for stationarity in practice

Augmented Dickey–Fuller (ADF) tests and their variants assess whether a series has a unit root. In applied work:

1. Plot  $y_t$  and  $\Delta y_t$  (or  $\Delta \ln y_t$ ).
2. Start with an ADF on  $y_t$  with a trend if plots suggest a trend.
3. If you fail to reject a unit root in levels, difference and re-test; usually you will then reject (i.e., differences are stationary).

**Caution.** Power is limited in small samples; combine tests with economic reasoning and visuals.

## 1.4 The Hodrick–Prescott (HP) filter

The HP trend  $\{\tau_t\}_{t=1}^T$  solves

$$\min_{\{\tau_t\}_{t=1}^T} \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} (\Delta^2 \tau_t)^2, \quad \Delta^2 \tau_t \equiv (\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}),$$

i.e., it trades off fit against the curvature of the trend (penalizing changes in the *slope*). The cycle is  $x_t = y_t - \tau_t$ ; the smoothing parameter  $\lambda$  controls how smooth the trend is (larger  $\lambda \Rightarrow$  smoother trend).

**Intuition.** HP says: “Fit the data closely, but prefer a trend with nearly constant growth.” Think of bending a thin metal ruler to trace  $y_t$ : the bending cost corresponds to the second-difference penalty. Endpoints are fragile (the *endpoint problem*); be cautious interpreting the last few observations.

**Practice tips.** Common conventions are: quarterly data  $\lambda = 1600$ , annual  $\lambda \approx 6.25$ , monthly  $\lambda \approx 129,600$ . Treat these as useful defaults, not laws; always check robustness.

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<sup>1</sup>Notes from slides:  $y_t = \tau_t + x_t$  and polynomial  $\tau_t$  variants are presented; unit-root tests (“Dickey–Fuller tests”) are mentioned as tools to assess stationarity; the rule of thumb that differencing targets the stochastic trend appears alongside examples for GDP and employment.

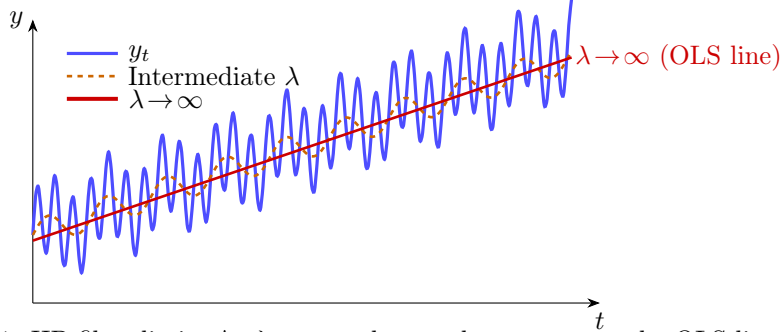


Figure 1: HP filter limit. As  $\lambda$  grows, the trend converges to the OLS linear trend.

### Limit cases.

- $\lambda \rightarrow 0$ : the penalty vanishes and  $\tau_t \rightarrow y_t$  (virtually no smoothing).
- $\lambda \rightarrow \infty$ : the curvature penalty dominates and the trend becomes *affine in  $t$* , i.e. a straight line  $\tau_t = a + bt$ —specifically the OLS linear trend.

The case where  $\lambda \rightarrow \infty$  is interesting to understand the logic. The HP objective is a tradeoff: fit the data (small  $\sum (y_t - \tau_t)^2$ ) but avoid bending (small  $\sum (\Delta^2 \tau_t)^2$ ). When  $\lambda$  is huge, bending becomes infinitely costly, so we must have

$$\Delta^2 \tau_t = \tau_{t+1} - 2\tau_t + \tau_{t-1} = 0 \quad \text{for all interior } t,$$

which implies a constant slope and therefore a straight line:

$$\tau_t = a + bt.$$

*Proof.* Set  $d_t := \tau_t - \tau_{t-1}$ . From  $\Delta^2 \tau_t = \tau_{t+1} - 2\tau_t + \tau_{t-1} = 0$  we get  $d_{t+1} - d_t = 0$ , hence

$$d_{t+1} = d_t = b \forall t.$$

Summing,  $\tau_t - \tau_1 = \sum_{k=2}^t d_k = (t-1)b$ , so

$$\tau_t = \tau_1 + (t-1)b = (\tau_1 - b) + bt := a + bt.$$

To pin down  $a, b$ , choose the least-squares fit to  $\{(t, y_t)\}$ :

$$(a, b) = \arg \min_{a, b} \sum_{t=1}^T (y_t - (a + bt))^2,$$

whose solution is the usual OLS line with  $b = \frac{\sum (t-\bar{t})(y_t - \bar{y})}{\sum (t-\bar{t})^2}$ ,  $a = \bar{y} - b\bar{t}$ . □

## 1.5 Linear filters: time and frequency

### Definition (time-domain view)

A *linear time-invariant* (LTI) filter transforms a series  $\{y_t\}$  as

$$\hat{y}_t = \sum_{k=-K}^K a_k y_{t-k},$$

where  $\{a_k\}$  are the *weights* (the impulse response) and  $K$  is the window half-width.

- **Two-sided (non-causal):** uses lags and leads ( $k < 0$  and  $k > 0$ ); best for retrospective analysis.
- **One-sided (causal / “real time”):** uses only lags ( $k \geq 0$ ); avoids looking ahead but typically adds delay and ripple.
- **Level preservation:** if  $\sum_k a_k = 1$ , a constant mean passes through undistorted (unit gain at zero frequency).

This operation is a *convolution*:  $\hat{y} = a * y$ . The filter’s response to a unit impulse is exactly  $\{a_k\}$ .

### Definition (frequency-domain view)

Represent the series as a superposition of angular frequencies  $\omega \in [-\pi, \pi]$  ( $\omega = 2\pi/P$  for period  $P$ ). Any LTI filter has a *frequency response*

$$\rho(\omega) = \sum_{k=-K}^K a_k e^{-i\omega k},$$

whose *magnitude*  $|\rho(\omega)|$  is the **gain** (how much each frequency passes) and whose *argument*  $\arg \rho(\omega)$  is the **phase** (horizontal shift). If  $x_t$  has spectral density  $f_x(\omega)$ , then

$$\text{var}(\hat{x}_t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\rho(\omega)|^2 f_x(\omega) d\omega \quad (\text{standard convention}).$$

*Mental rule:* set  $|\rho(\omega)| \approx 1$  where you care (the “pass band”), and near 0 where you want attenuation (very low-frequency trend or very high-frequency noise).

**Symmetry and phase.** If the filter is *symmetric* ( $a_{-k} = a_k$ ), then  $\rho(\omega)$  is real and the phase is 0 or  $\pi$  (*zero phase*): the filter does not shift timing; it only rescales frequencies. Causal filters typically induce nonzero phase (delay).

### Core examples

(i)  **$M$ -term moving average (MA).**

$$a_k = \begin{cases} 1/M, & k = 0, \dots, M-1 \quad (\text{one-sided}) \\ 1/M, & k = -(M-1)/2, \dots, (M-1)/2 \quad (\text{two-sided, } M \text{ odd}) \end{cases}$$

$$\rho(\omega) = \frac{1}{M} \sum_{k=0}^{M-1} e^{-i\omega k} = e^{-i\omega(M-1)/2} \frac{\sin(M\omega/2)}{M \sin(\omega/2)}.$$

*Quick read:* a **low-pass** smoother. It attenuates rapid wiggles (high frequencies) and preserves slow drift. The symmetric version avoids phase delay.

(ii) **First difference (high-pass).**

$$\hat{y}_t = y_t - y_{t-1}, \quad a_0 = 1, \quad a_1 = -1.$$

$$\rho(\omega) = 1 - e^{-i\omega} = e^{-i\omega/2} \cdot 2i \sin(\omega/2), \quad |\rho(\omega)| = 2 |\sin(\omega/2)|.$$

*Quick read:* **kills** zero frequency (the level/trend) and passes cycle; within  $[0, \pi]$ , higher  $\omega$  gets larger gain.

(iii) **Band-pass filter (ideal vs. approximations).** The *ideal* keeps  $|\rho(\omega)| = 1$  for  $\omega \in [\omega_L, \omega_H]$  and 0 outside; it is not finitely implementable. Two popular approximations:

- **Baxter–King (BK):** two-sided, symmetric, finite window (removes frequencies outside a chosen range; sacrifices endpoints).
- **Christiano–Fitzgerald (CF):** band-pass approximation that can be implemented *almost* one-sided (better for real time, at the cost of phase and border behavior).

*Business-cycle convention (quarterly):* pass periods  $P \in [6, 32]$  quarters, i.e.  $\omega_H = 2\pi/6$  and  $\omega_L = 2\pi/32$ .

### Useful connections

- **HP as a (near) linear filter:** HP acts like a *low-pass* LTI filter in the interior of the sample; it attenuates very low frequencies (trend) and passes cycle. At the endpoints it is no longer strictly LTI (the *endpoint* issue).
- **Sum of weights:**  $\sum_k a_k = 1$  preserves the mean;  $\sum_k a_k = 0$  (e.g., the difference) removes it.
- **Effect on second moments:** filtering  $x_t$  by  $\{a_k\}$  multiplies its spectrum by  $|\rho(\omega)|^2$ . For ARMA data, you can combine the filter polynomial with the process polynomial.

### Design and diagnostics

#### How to choose a filter (quick checklist).

1. **Goal:** growth (log-differences), cycle (band-pass), or smooth trend (low-pass)?
2. **Causality:** for *real time*, use one-sided; for historical analysis, use symmetric (zero phase).
3. **Frequency band:** set  $[\omega_L, \omega_H]$  using periods of interest (e.g., 6–32 quarters).
4. **Robustness:** compare at least two methods (e.g., log-diffs vs. HP/BK/CF).

#### Common pitfalls and fixes.

- **Leakage:** finite filters don't separate bands perfectly; avoid claims driven by razor-thin peaks; inspect  $|\rho(\omega)|$ .
- **Gibbs ripples:** sharper band edges create more ringing; smoother windows reduce ripples at the cost of gentler transitions.
- **Endpoints:** two-sided filters lose data at the borders or become biased; document how many points you drop or use extensions/nowcasts.
- **Phase delay:** causal filters shift timing; report and, if needed, *correct* the delay when comparing peaks/troughs.

#### Mini-propositions (handy in class)

- **Symmetry  $\Rightarrow$  zero phase:** if  $a_{-k} = a_k$ , then  $\rho(\omega) \in \mathbb{R}$  and the filter does not shift time (it only rescales each frequency).
- **Difference removes the mean:** with  $a_0 = 1$ ,  $a_1 = -1$ , we have  $\sum_k a_k = 0$  and  $|\rho(0)| = 0$ ; hence it eliminates the level trend.

- **Moving average preserves the mean:**  $\sum_k a_k = 1$  and  $|\rho(0)| = 1$ ; thus it does not remove a perfectly constant level.

**Definition (VAR).** Suppose we have  $n$  stationary time series variables and we want to track their relationships over time. An order- $p$  *vector autoregression* (VAR( $p$ )) is a statistical model of the form

$$\mathbf{y}_t = \mathbf{c} + \sum_{s=1}^p \mathbf{\Phi}_s \mathbf{y}_{t-s} + \boldsymbol{\epsilon}_t \quad (1.1)$$

where

- $\mathbf{y}_t \in \mathbb{R}^{n \times 1}$  is a vector giving the observations of our  $n$  variables at time  $t$ ;
- $\mathbf{c} \in \mathbb{R}^{n \times 1}$  is a constant;
- $\mathbf{\Phi}_s \in \mathbb{R}^{n \times n}$  captures the relationships between our time series variables at different lags, in the sense that  $\Phi_s^{(i,j)}$  captures the impact of the variable  $j$  from  $s$  periods ago on variable  $i$  today; and
- $\boldsymbol{\epsilon}_t \in \mathbb{R}^{n \times 1}$  is vector white noise:  $\mathbb{E}[\boldsymbol{\epsilon}_t] = \mathbf{0}$ ,  $\mathbb{E}[\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_t^T] = \boldsymbol{\Omega} \in \mathbb{R}^{n \times n}$  for all  $t$ , where  $\boldsymbol{\Omega} = [\sigma_{i,j}]_{j,i=1}^n$  is the variance-covariance matrix for  $\boldsymbol{\epsilon}_t$ , and  $\mathbb{E}[\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_j] = \mathbf{0}_{n \times n}$  for all  $i \neq j$ .

**Remark** (Estimating a VAR using OLS). If a set of  $n$  variables follow a VAR( $p$ ), and we have data  $\{\mathbf{y}_{-p+1}, \mathbf{y}_{-p+2}, \dots, \mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_T\}$ , then we can estimate Equation (1.1) using equation-by-equation OLS. In particular, for each  $i = 1, \dots, n$ , row  $i$  of Equation (1.1) gives the value of variable  $i$  at time  $t$  as

$$y_{t,i} = c_i + \sum_{s=1}^p \sum_{j=1}^n \Phi_s^{(i,j)} y_{t-s,j} + \epsilon_{t,i} \quad (1.2)$$

with  $\mathbb{E}[\epsilon_{t,i}] = 0$  and  $\mathbb{E}[\epsilon_{t,i}^2] = \sigma_{i,i}^2$ . We can use the sample  $\{y_{1,i}, y_{2,i}, \dots, y_{T,i}\}$  to estimate this equation, obtaining OLS estimates  $\hat{c}_i$ , and  $\hat{\Phi}_s^{(i,j)}$  for each  $s = 1, \dots, p$  and  $j = 1, \dots, n$ :

$$y_{t,i} = \hat{c}_i + \sum_{s=1}^p \sum_{j=1}^n \hat{\Phi}_s^{(i,j)} y_{t-s,j} + \hat{\epsilon}_{t,i} \quad (1.3)$$

After doing this for each variable, we can concatenate the intercept estimates to get  $\hat{\mathbf{c}}$ , and the coefficient estimates to get matrices  $\hat{\mathbf{\Phi}}_s$  for each  $s = 1, \dots, p$ . We can estimate  $\hat{\boldsymbol{\Omega}} = [\hat{\sigma}_{i,j}]_{i,j=1}^n$  by setting  $\hat{\sigma}_{i,i} = \frac{1}{T} \sum_{t=1}^T \hat{\epsilon}_{t,i}^2$  for each  $i = 1, \dots, n$ , and  $\hat{\sigma}_{i,j} = \frac{1}{T} \sum_{t=1}^T \hat{\epsilon}_{t,i} \hat{\epsilon}_{t,j}$  for  $i \neq j$ .

**Remark** (Presenting a VAR without a constant). Any VAR can be presented without a constant term by demeaning. Thus, without loss of generality, we can present any VAR without the constant term  $\mathbf{c}$ . To see this, suppose we have a VAR( $p$ ) as in Equation (1.1):

$$\mathbf{y}_t = \mathbf{c} + \sum_{s=1}^p \mathbf{\Phi}_s \mathbf{y}_{t-s} + \boldsymbol{\epsilon}_t.$$

Since each of the  $n$  underlying variables are assumed stationary, we can define  $\boldsymbol{\mu} := \mathbb{E}[\mathbf{y}_t]$ , which does not depend on time. Taking the expectation of both sides Equation (1.1), we then get

$$\boldsymbol{\mu} = \mathbf{c} + \sum_{s=1}^p \mathbf{\Phi}_s \boldsymbol{\mu}. \quad (1.4)$$

Thus,

$$\mathbf{c} = \boldsymbol{\mu} - \sum_{s=1}^p \boldsymbol{\Phi}_s \boldsymbol{\mu} = \left( \mathbf{I}_n - \sum_{s=1}^p \boldsymbol{\Phi}_s \right) \boldsymbol{\mu}$$

and so

$$\boldsymbol{\mu} = \left( \mathbf{I}_n - \sum_{s=1}^p \boldsymbol{\Phi}_s \right)^{-1} \mathbf{c}.$$

Now subtract  $\boldsymbol{\mu}$  from both sides of Equation 1.1, and add and subtract  $(\sum_{s=1}^p \boldsymbol{\Phi}_s) \boldsymbol{\mu}$  from the right-hand side:

$$\begin{aligned} \mathbf{y}_t - \boldsymbol{\mu} &= \mathbf{c} - \boldsymbol{\mu} + \left( \sum_{s=1}^p \boldsymbol{\Phi}_s \right) \boldsymbol{\mu} + \sum_{s=1}^p \boldsymbol{\Phi}_s \mathbf{y}_{t-s} - \left( \sum_{s=1}^p \boldsymbol{\Phi}_s \right) \boldsymbol{\mu} + \boldsymbol{\epsilon}_t \\ &= \mathbf{c} - \left( \mathbf{I}_n - \sum_{s=1}^p \boldsymbol{\Phi}_s \right) \boldsymbol{\mu} + \sum_{s=1}^p \boldsymbol{\Phi}_s (\mathbf{y}_{t-s} - \boldsymbol{\mu}) + \boldsymbol{\epsilon}_t \\ &= \sum_{s=1}^p \boldsymbol{\Phi}_s (\mathbf{y}_{t-s} - \boldsymbol{\mu}) + \boldsymbol{\epsilon}_t. \end{aligned}$$

Thus, defining  $\mathbf{x}_t \equiv \mathbf{y}_t - \boldsymbol{\mu}$ , we have

$$\mathbf{x}_t = \sum_{s=1}^p \boldsymbol{\Phi}_s \mathbf{x}_{t-s} + \boldsymbol{\epsilon}_t, \quad (1.5)$$

which is a VAR( $p$ ) with no constant term that has the same innovations and coefficient matrices as the VAR( $p$ ) we began with.

**Remark** (Representing VAR( $p$ ) as VAR(1)). Any VAR( $p$ ) (for finite  $p$ ) can be represented as a VAR(1). Consider the VAR( $p$ ) on  $n$  variables (sin constante):

$$\mathbf{y}_t = \sum_{s=1}^p \boldsymbol{\Phi}_s \mathbf{y}_{t-s} + \boldsymbol{\epsilon}_t. \quad (1.6)$$

Para cada  $t$ , definí

$$\boldsymbol{\xi}_t = \begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} \in \mathbb{R}^{np \times 1}, \quad \mathbf{v}_t = \begin{bmatrix} \boldsymbol{\epsilon}_t \\ \mathbf{0}_{n \times 1} \\ \vdots \\ \mathbf{0}_{n \times 1} \end{bmatrix} \in \mathbb{R}^{np \times 1},$$

y definí la matriz compañera:

$$\mathbf{F} = \begin{bmatrix} \boldsymbol{\Phi}_1 & \boldsymbol{\Phi}_2 & \cdots & \boldsymbol{\Phi}_{p-1} & \boldsymbol{\Phi}_p \\ \mathbf{I}_n & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{np \times np}.$$

Notá que  $\mathbf{v}_t$  es ruido blanco vectorial: para todo  $t$ ,

$$\mathbb{E}[\mathbf{v}_t] = \mathbf{0}_{np \times 1}, \quad \mathbb{E}[\mathbf{v}_t \mathbf{v}_t^\top] = \begin{bmatrix} \mathbf{\Omega} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix},$$

y para  $t \neq \tau$ ,

$$\mathbb{E}[\mathbf{v}_t \mathbf{v}_\tau^\top] = \mathbf{0}_{np \times np}.$$

Entonces, el VAR(1) en  $np$  variables

$$\boldsymbol{\xi}_t = \mathbf{F} \boldsymbol{\xi}_{t-1} + \mathbf{v}_t$$

representa esencialmente el mismo proceso que el VAR( $p$ ) original.

**Remark** (Representing a VAR( $p$ ) as a MA( $\infty$ )). Under certain conditions, we can give a VAR( $p$ ) a representation as an MA( $\infty$ ) process. Without loss of generality, we consider a VAR(1) on  $n$  variables with no constant term:

$$\mathbf{y}_t = \boldsymbol{\Phi} \mathbf{y}_{t-1} + \boldsymbol{\epsilon}_t. \quad (1.7)$$

We can make recursive substitutions in Equation (1.7) to get

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\Phi} (\boldsymbol{\Phi} \mathbf{y}_{t-2} + \boldsymbol{\epsilon}_{t-1}) + \boldsymbol{\epsilon}_t \\ &= \boldsymbol{\Phi}^2 \mathbf{y}_{t-2} + \boldsymbol{\Phi} \boldsymbol{\epsilon}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \boldsymbol{\Phi}^2 (\boldsymbol{\Phi} \mathbf{y}_{t-3} + \boldsymbol{\epsilon}_{t-2}) + \boldsymbol{\Phi} \boldsymbol{\epsilon}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \boldsymbol{\Phi}^3 \mathbf{y}_{t-3} + \boldsymbol{\Phi}^2 \boldsymbol{\epsilon}_{t-2} + \boldsymbol{\Phi} \boldsymbol{\epsilon}_{t-1} + \boldsymbol{\epsilon}_t \\ &\vdots \\ &= \boldsymbol{\Phi}^{k+1} \mathbf{y}_{t-k-1} + \boldsymbol{\epsilon}_t + \sum_{j=1}^k \boldsymbol{\Phi}^j \boldsymbol{\epsilon}_{t-j}. \end{aligned}$$

If all the eigenvalues of  $\boldsymbol{\Phi}$  lie strictly within the unit circle, then  $\lim_{k \rightarrow \infty} \boldsymbol{\Phi}^{k+1} = \mathbf{0}_{n \times n}$ , the process is stationary, and we can take the limit of the recursion to get the following representation as an MA( $\infty$ ):

$$\mathbf{y}_t = \boldsymbol{\epsilon}_t + \sum_{j=1}^{\infty} \boldsymbol{\Phi}^j \boldsymbol{\epsilon}_{t-j}. \quad (1.8)$$

This expresses today's observation as an infinite sum of all past innovations, with the influence of past innovations decreasing as we go further and further back in the past).

**Remark** (Impulse response functions using the MA( $\infty$ ) representation). Suppose we have the following VAR( $p$ ) on  $n$  variables (w.l.o.g. no constant):

$$\mathbf{y}_t = \sum_{s=1}^p \boldsymbol{\Phi}_s \mathbf{y}_{t-s} + \boldsymbol{\epsilon}_t. \quad (1.9)$$

Assume it admits an MA( $\infty$ ) representation

$$\mathbf{y}_t = \boldsymbol{\epsilon}_t + \sum_{j=1}^{\infty} \boldsymbol{\Psi}_j \boldsymbol{\epsilon}_{t-j}. \quad (1.10)$$

Define the matrix polynomial  $\Psi(x) := \mathbf{I}_n + \sum_{j=1}^{\infty} \Psi_j x^j$ . Then (1.10) is

$$\mathbf{y}_t = \Psi(L) \boldsymbol{\epsilon}_t, \quad (1.11)$$

where  $L$  is the lag operator,  $L\mathbf{x}_t \equiv \mathbf{x}_{t-1}$ . Likewise, defining  $\Phi(x) := \mathbf{I}_n - \sum_{s=1}^p \Phi_s x^s$ , we can rewrite (1.9) as

$$\boldsymbol{\epsilon}_t = \Phi(L) \mathbf{y}_t. \quad (1.12)$$

Combining (1.11) and (1.12),

$$\begin{aligned} \mathbf{y}_t &= \Psi(L) \boldsymbol{\epsilon}_t \\ &= \Psi(L) \Phi(L) \mathbf{y}_t, \end{aligned}$$

hence  $\Psi(L) \Phi(L) = \mathbf{I}_n$ . Expanding,

$$\begin{aligned} \Psi(L) \Phi(L) &= \left( \mathbf{I}_n + \sum_{j=1}^{\infty} \Psi_j L^j \right) \left( \mathbf{I}_n - \sum_{s=1}^p \Phi_s L^s \right) \\ &= \mathbf{I}_n - \sum_{s=1}^p \Phi_s L^s + \sum_{j=1}^{\infty} \Psi_j L^j - \sum_{j=1}^{\infty} \sum_{s=1}^p \Psi_j \Phi_s L^{j+s} \\ &= \mathbf{I}_n + \sum_{j=1}^{\infty} \left( \Psi_j - \sum_{i=1}^p \Psi_{j-i} \Phi_i \right) L^j, \end{aligned}$$

where we set  $\Psi_0 = \mathbf{I}_n$  and  $\Psi_k = \mathbf{0}_{n \times n}$  for  $k < 0$ . Since the product equals  $\mathbf{I}_n$ , the coefficient of each power of  $L^j$  ( $j \geq 1$ ) must be  $\mathbf{0}_{n \times n}$ , yielding the recursion

$$\Psi_j = \sum_{i=1}^p \Psi_{j-i} \Phi_i, \quad j = 1, 2, \dots$$

(equivalent a escribir la suma hasta  $i \leq \min\{p, j\}$ ).

Why is this useful? From  $\mathbf{y}_t = \boldsymbol{\epsilon}_t + \sum_{j=1}^{\infty} \Psi_j \boldsymbol{\epsilon}_{t-j}$ , the moving-average matrices  $\{\Psi_j\}$  describen cómo shocks reducidos se transmiten en el tiempo. En particular, el efecto contemporáneo en la variable  $i$  de un shock unitario en la componente  $j$  ocurrido  $s$  periodos atrás es la entrada  $(i, j)$  de  $\Psi_s$ :

$$\frac{\partial y_{i,t+s}}{\partial \epsilon_{j,t}} = (\Psi_s)_{ij}.$$

Graficar estos coeficientes en distintos leads/lags da la *impulse response function* (IRF). La información de  $\{\Phi_s\}$  y la de  $\{\Psi_j\}$  es equivalente (vía la relación anterior). **Importante:** la IRF de forma reducida *no* tiene interpretación causal sin una identificación estructural adicional (p.ej., restricciones contemporáneas, long-run, signos, etc.).

**Definition.** (Granger causality) Consider the following bivariate VAR( $p$ ):

$$\begin{bmatrix} y_{t,1} \\ y_{t,2} \end{bmatrix} = \sum_{s=1}^p \begin{bmatrix} \phi_s^{(1,1)} & \phi_s^{(1,2)} & \phi_s^{(2,1)} & \phi_s^{(2,2)} \end{bmatrix} \begin{bmatrix} y_{t-s,1} \\ y_{t-s,2} \end{bmatrix} + \begin{bmatrix} \epsilon_{t,1} \\ \epsilon_{t,2} \end{bmatrix}. \quad (1.13)$$

We say that  $y_2$  does not *Granger cause*  $y_1$  if each matrix  $\phi_s$  is lower triangular; that is,  $\phi_s^{(1,2)} = 0$  for all  $s = 1, \dots, p$ , implying that past values of  $y_2$  play no role in determining today's value of  $y_1$ . We test for Granger causality as follows. First, we estimate the following *unrestricted equation* using OLS (a

direct reading off of the first line of Equation (1.13)):

$$y_{t,1} = \sum_{s=1}^p \phi_s^{(1,1)} y_{t-s,1} + \sum_{s=1}^p \phi_s^{(1,2)} y_{t-s,2} + \epsilon_{t,1}. \quad (1.14)$$

Save the sum of squares of the residuals from estimating Equation (1.14) as  $SSR_1$ . Next, estimate with OLS the following “restricted equation” postulating that past values of  $y_2$  have no impact on contemporary  $y_1$ :

$$y_{t,1} = \sum_{s=1}^p \phi_s^{(1,1)} y_{t-s,1} + \epsilon_{t,1}. \quad (1.15)$$

Save the sum of squares of the residuals from estimate Equation (1.15) as  $SSR_0$ . It will always be the case that  $SSR_1 \geq SSR_0$ , since are adding more covariates to otherwise the same equation. The question is whether the difference is significant enough for us to reject lack of Granger causality. We conduct statistical tests on the difference  $SSR_1 - SSR_0$  (either  $p$  or  $F$  tests).

**Remark.** (Limitations of reduced-form VARs) Without a minimum amount of theory, we cannot make causal statements using VARs. The innovations  $\epsilon_t$  cannot be interpreted as shocks with economic meaning, because they are all correlated with each other. As an example, consider the following structural model:

$$\begin{aligned} y_{1,t} &= b_1 y_{2,t} + k_1 + b_{1,1} y_{1,t-1} + b_{1,2} y_{2,t-1} + u_{1,t}, \\ y_{2,t} &= b_2 y_{1,t} + k_2 + b_{2,1} y_{1,t-1} + b_{2,2} y_{2,t-1} + u_{2,t}, \end{aligned}$$

where  $\begin{bmatrix} u_{1,t} & u_{2,t} \end{bmatrix}^T$  is vector white noise. (This is not yet a VAR, because the variables are in terms of each others’ contemporaneous values). Concretely, thing of  $y_1$  as GDP and  $y_2$  as governmetn spending. If  $1 - b_1 b_2 \neq 0$ , we can use substitution to obtain the following representation of the model as a VAR (let’s take it on faith):

$$\begin{aligned} y_{1,t} &= \frac{k_1 + b_1 k_2}{1 - b_1 b_2} + \frac{b_{11} + b_1 b_{21}}{1 - b_1 b_2} y_{1,t-1} + \frac{b_{12} + b_1 b_{22}}{1 - b_1 b_2} y_{2,t-1} + \frac{u_{1,t} + b_1 u_{2,t}}{1 - b_1 b_2}, \\ y_{2,t} &= \frac{k_2 + b_2 k_1}{1 - b_1 b_2} + \frac{b_{21} + b_2 b_{21}}{1 - b_1 b_2} y_{1,t-1} + \frac{b_{22} + b_2 b_{12}}{1 - b_1 b_2} y_{2,t-1} + \frac{u_{2,t} + b_2 u_{1,t}}{1 - b_1 b_2}. \end{aligned}$$

or

$$\mathbf{y}_t = \mathbf{c} + \mathbf{\Phi} \mathbf{y}_{t-1} + \boldsymbol{\epsilon}_t, \quad (1.16)$$

where

$$\mathbf{c} = \begin{bmatrix} \frac{k_1 + b_1 k_2}{1 - b_1 b_2} & \frac{k_2 + b_2 k_1}{1 - b_1 b_2} \end{bmatrix},$$

$$\mathbf{\Phi} = \begin{bmatrix} \frac{b_{11} + b_1 b_{21}}{1 - b_1 b_2} & \frac{b_{12} + b_1 b_{22}}{1 - b_1 b_2} & \frac{b_{21} + b_2 b_{21}}{1 - b_1 b_2} & \frac{b_{22} + b_2 b_{12}}{1 - b_1 b_2} \end{bmatrix},$$

and

$$\boldsymbol{\epsilon}_t = \begin{bmatrix} \frac{u_{1,t} + b_1 u_{2,t}}{1 - b_1 b_2} & \frac{u_{2,t} + b_2 u_{1,t}}{1 - b_1 b_2} \end{bmatrix}.$$

We can estimate these VAR parameters using OLS. But we are not able to recover the structural parameters  $b_1$  and  $b_2$ . Note that if we knew  $b_1$  and  $b_2$ , we could use our OLS estimates to estimate the other structural parameters by just solving some simple linear systems of equations. What if we don’t know  $b_1$  and  $b_2$ ? We know that  $\text{Var}(\epsilon_{1,t}) = \frac{\sigma_1^2 + b_1^2 \sigma_2^2}{(1 - b_1 b_2)^2}$ ,  $\text{Var}(\epsilon_{2,t}) = \frac{\sigma_2^2 + b_2^2 \sigma_1^2}{(1 - b_1 b_2)^2}$ , and  $\text{Cov}(\epsilon_{1,t}, \epsilon_{2,t}) = \frac{b_2}{(1 - b_1 b_2)^2} \sigma_1^2 + \frac{b_1}{(1 - b_1 b_2)^2} \sigma_2^2$ . This is three equations in four unknowns ( $\sigma_1$ ,  $\sigma_2$ ,  $b_1$ , and  $b_2$ ). Without some

more assumptions, we're screwed in terms of recovering structural parameters. If we assume that today's GDP has no contemporaneous effect on government spending (so  $b_1 = 0$ ), we can make progress on estimating the other parameters. Indeed, our equations reduce to

$$\text{Var}(\epsilon_{1,t}) = \sigma_1^2, \text{Cov}(\epsilon_{1,t}, \epsilon_{2,t}) = b_2 \sigma_1^2, \text{Var}(\epsilon_{2,t}) = \sigma_2^2 + b_2^2 \sigma_1^2$$

from we can estimate structural parameters, utilizing our OLS estimates:

$$\begin{aligned}\hat{\sigma}_1^2 &= \text{Var}(\hat{\epsilon}_{1,t}), \\ \hat{b}_2 &= \frac{\text{Cov}(\hat{\epsilon}_{1,t}, \hat{\epsilon}_{2,t})}{\hat{\sigma}_1^2} = \frac{\text{Cov}(\hat{\epsilon}_{1,t}, \hat{\epsilon}_{2,t})}{\text{Var}(\hat{\epsilon}_{1,t})}, \\ \hat{\sigma}_2^2 &= \text{Var}(\hat{\epsilon}_{2,t}) - \hat{b}_2^2 \hat{\sigma}_1^2 = \text{Var}(\hat{\epsilon}_{1,t}) - \frac{(\text{Cov}(\hat{\epsilon}_{1,t}, \hat{\epsilon}_{2,t}))^2}{\text{Var}(\hat{\epsilon}_{1,t})}.\end{aligned}$$

**Remark.** Let's more formally exhibit the identification problem inherent in structural VARs. Suppose the true, underlying linear structural model is given by

$$\mathbf{B}_0 \mathbf{y}_t = \sum_{s=1}^p \mathbf{B}_s \mathbf{y}_{t-s} + \mathbf{u}_t. \quad (1.17)$$

where

- $\mathbf{y}_t \in \mathbb{R}^{n \times 1}$  is our collection of  $n$  variables at time  $t$ ,
- $\mathbf{u}_t \in \mathbb{R}^{n \times 1}$  is the vector of *structural shocks* at time  $t$ . This is vector white noise, and by definition its components are uncorrelated. By rescaling, we may assume  $\text{Var}(u_{i,t}) = 1$  for each  $i$ . Thus, for all  $t$ ,

$$\mathbb{E}[\mathbf{u}_t] = \mathbf{0}, \quad \mathbb{E}[\mathbf{u}_t \mathbf{u}_t^\top] = \mathbf{I}_n,$$

and for all  $i \neq j$ ,

$$\mathbb{E}[\mathbf{u}_i \mathbf{u}_j^\top] = \mathbf{0}_{n \times n}.$$

- $\mathbf{B}_s \in \mathbb{R}^{n \times n}$ ,  $s = 0, 1, \dots, p$ , are coefficient matrices. For  $s \geq 1$ , the  $(i, j)$  entry of  $\mathbf{B}_s$  gives the impact of variable  $j$  lagged  $s$  periods on variable  $i$  today.  $\mathbf{B}_0$  is the *contemporaneous effects matrix*. (Note: after rescaling  $\mathbf{u}_t$ , the diagonal of  $\mathbf{B}_0$  need not be 1.)

If  $\mathbf{B}_0$  is not invertible, the model cannot be inverted. Suppose instead  $\mathbf{B}_0$  is invertible. Then multiplying (1.17) by  $\mathbf{B}_0^{-1}$  yields the VAR( $p$ ) representation:

$$\begin{aligned}\mathbf{y}_t &= \sum_{s=1}^p \mathbf{B}_0^{-1} \mathbf{B}_s \mathbf{y}_{t-s} + \mathbf{B}_0^{-1} \mathbf{u}_t \\ &= \sum_{s=1}^p \Phi_s \mathbf{y}_{t-s} + \epsilon_t,\end{aligned}$$

where  $\Phi_s := \mathbf{B}_0^{-1} \mathbf{B}_s$  and  $\epsilon_t := \mathbf{B}_0^{-1} \mathbf{u}_t$  are the reduced-form VAR innovations.

Note that for all  $t$ ,

$$\begin{aligned}\mathbb{E}[\epsilon_t] &= \mathbb{E}[\mathbf{B}_0^{-1} \mathbf{u}_t] \\ &= \mathbf{B}_0^{-1} \mathbb{E}[\mathbf{u}_t] \\ &= \mathbf{B}_0^{-1} \mathbf{0}_{n \times 1} \\ &= \mathbf{0}_{n \times 1},\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[\epsilon_t \epsilon_t^\top] &= \mathbb{E}[(\mathbf{B}_0^{-1} \mathbf{u}_t)(\mathbf{B}_0^{-1} \mathbf{u}_t)^\top] \\ &= \mathbf{B}_0^{-1} \mathbb{E}[\mathbf{u}_t \mathbf{u}_t^\top] \mathbf{B}_0^{-T} \\ &= \mathbf{B}_0^{-1} \mathbf{I}_n \mathbf{B}_0^{-T} \\ &= \mathbf{B}_0^{-1} \mathbf{B}_0^{-T} \\ &:= \mathbf{\Omega}.\end{aligned}$$

For  $i \neq j$ ,

$$\begin{aligned}\mathbb{E}[\epsilon_i \epsilon_j^\top] &= \mathbb{E}[(\mathbf{B}_0^{-1} \mathbf{u}_i)(\mathbf{B}_0^{-1} \mathbf{u}_j)^\top] \\ &= \mathbf{B}_0^{-1} \mathbb{E}[\mathbf{u}_i \mathbf{u}_j^\top] \mathbf{B}_0^{-T} \\ &= \mathbf{B}_0^{-1} \mathbf{0}_{n \times n} \mathbf{B}_0^{-T} \\ &= \mathbf{0}_{n \times n}.\end{aligned}$$

So from  $\{\mathbf{y}_t\}$  we can estimate  $\hat{\mathbf{\Phi}}_s$ ,  $\hat{\epsilon}_t$ , and  $\hat{\mathbf{\Omega}}$ . Ideally we would like to recover  $\mathbf{u}_t$  and  $\mathbf{B}_0$ , but absent additional assumptions, this is impossible:  $\mathbf{\Omega} = \mathbf{B}_0^{-1} \mathbf{B}_0^{-T}$  does not uniquely pin down  $\mathbf{B}_0$ . In fact, infinitely many  $\mathbf{A} \in \mathbb{R}^{n \times n}$  satisfy  $\mathbf{\Omega} = \mathbf{A}^{-1} \mathbf{A}^{-T}$ .

Indeed, let  $\mathbf{R}$  be any orthogonal matrix ( $\mathbf{R}\mathbf{R}^\top = \mathbf{I}_n$ ). Define fake shocks  $\tilde{\mathbf{u}}_t := \mathbf{R}\mathbf{u}_t$ . Then

$$\begin{aligned}\mathbb{E}[\tilde{\mathbf{u}}_t \tilde{\mathbf{u}}_t^\top] &= \mathbb{E}[(\mathbf{R}\mathbf{u}_t)(\mathbf{R}\mathbf{u}_t)^\top] \\ &= \mathbb{E}[\mathbf{R}\mathbf{u}_t \mathbf{u}_t^\top \mathbf{R}^\top] \\ &= \mathbf{R} \mathbb{E}[\mathbf{u}_t \mathbf{u}_t^\top] \mathbf{R}^\top \\ &= \mathbf{R} \mathbf{I}_n \mathbf{R}^\top \\ &= \mathbf{I}_n.\end{aligned}$$

Thus,

$$\mathbf{\Omega} = (\mathbf{R}\mathbf{B}_0)^{-1} (\mathbf{R}\mathbf{B}_0)^{-T}.$$

So  $\mathbf{R}\mathbf{B}_0$  is observationally equivalent to  $\mathbf{B}_0$ , and  $\tilde{\mathbf{u}}_t$  indistinguishable from the true structural shocks.

**Remark** (Recursive Cholesky decomposition). Pick up where we left off in the last remark. Recall that a Cholesky decomposition matrix of a real matrix  $\mathbf{A}$  is an equation  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$  where  $\mathbf{L}$  is a real, lower-triangular matrix with nonnegative diagonal. It is a fact that every positive semidefinite matrix has a unique Cholesky decomposition. Also, we should know that the inverse of a lower triangular matrix is also a lower triangular matrix. Returning to  $\mathbf{\Omega} = \mathbf{B}_0^{-1} \mathbf{B}_0^{-T}$ . Note that  $\mathbf{\Omega}$  and  $\hat{\mathbf{\Omega}}$  are positive semidefinite. If we assume the assumption that  $\mathbf{B}_0$  (and hence  $\mathbf{B}_0^{-1}$ ) is lower diagonal, then  $\mathbf{\Omega} = \mathbf{B}_0^{-1} \mathbf{B}_0^{-T}$  is the unique Cholesky decomposition of  $\mathbf{\Omega}$ . Hence we can uniquely identify  $\mathbf{B}_0$  under the assumption it is lower

diagonal (amounting to an economic reordering of variables). With  $\mathbf{B}_0$  in hand, we can compute impulse response functions to structural shocks. Practically, given our data, we estimate  $\widehat{\boldsymbol{\Omega}}$ . Using MATLAB, find the Cholesky decomposition of  $\widehat{\boldsymbol{\Omega}}$ , which will turn out to be  $\widehat{\mathbf{B}}_0^{-1}$ . Then, if we want to simulate the response to a structural shock to  $\boldsymbol{\epsilon}_t$ , we can use our VAR via the innovation  $\mathbf{u}_t = \widehat{\mathbf{B}}_0^{-1} \boldsymbol{\epsilon}_t$ .

**Remark** (Other canonical identification ideas). We present an alternative identification assumption to assuming that  $\mathbf{B}_0$  is lower triangular. Recall the MA( $\infty$ ) representation of the reduced-form VAR:

$$\mathbf{y}_t = \boldsymbol{\epsilon}_t + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s \boldsymbol{\epsilon}_{t-s} = \boldsymbol{\Psi}(L) \boldsymbol{\epsilon}_t,$$

where  $\boldsymbol{\Psi}(L)$  is the infinite polynomial  $\boldsymbol{\Psi}(x) = \mathbf{I}_n + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s x^s$ . Plugging in that  $\boldsymbol{\epsilon}_t = \mathbf{B}_0^{-1} \mathbf{u}_t$ , we get that

$$\mathbf{y}_t = \mathbf{B}_0^{-1} \mathbf{u}_t + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s \mathbf{B}_0^{-1} \mathbf{u}_{t-s} = \boldsymbol{\Psi}(L) (\mathbf{B}_0^{-1} \mathbf{u}_t) = \mathbf{A}(L) \mathbf{u}_t,$$

where  $\mathbf{A}(L)$  is the infinite polynomial  $\mathbf{A}(x) = \mathbf{B}_0^{-1} + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s \mathbf{B}_0^{-1} x^s$ . Note that  $\mathbf{A}(0) = \mathbf{B}_0^{-1}$ . So, the Cholesky identification strategy amounts to placing restrictions on  $\mathbf{A}(0)$ . Consider instead  $\mathbf{A}(1)$ :

$$\mathbf{A}(1) = \mathbf{B}_0^{-1} + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s \mathbf{B}_0^{-1} = \left( \mathbf{I}_n + \sum_{s=1}^{\infty} \boldsymbol{\Psi}_s \right) \mathbf{B}_0^{-1} = \boldsymbol{\Psi}(1) \mathbf{B}_0^{-1}.$$

Note that  $\mathbf{A}(1)$  gives the cumulative sum of all responses of VAR variables to a structural shock  $\mathbf{u}_t$ . Hence if our variables are in differences to begin with, then  $\mathbf{A}(1)$  gives the sum of the total response of differences to the shock—ie, the long-run change in the level. Therefore, by making assumptions on  $\mathbf{A}(1)$ , such as assuming certain long-run relationships between variables hold, or saying we don't want certain structural shocks to effect certain variables, etc, we can aid identification. Eg, assume  $\mathbf{A}(1)$  is lower triangular, etc.

In practice, we estimate the variance-covariance matrix  $\widehat{\boldsymbol{\Omega}}$  from our data for the reduced-form VAR. We estimate a candidate  $\widehat{\mathbf{B}}_0$  as the inverse of the Cholesky decomposition  $\widehat{\mathbf{A}}$  of  $\widehat{\boldsymbol{\Omega}}$ . Then, we draw orthogonal matrices  $\mathbf{R}$  and consider a new contemporaneous effects matrix  $\mathbf{R}\mathbf{B}_0$ . We continue searching until the matrix  $\mathbf{A}(1) = \boldsymbol{\Psi}(1) (\mathbf{R}\mathbf{B}_0)^{-1} = \boldsymbol{\Psi}(1) (\mathbf{A}\mathbf{R}^{-1})$  is zero in the elements for which we want to assume zero long-run effects.

## 2 Consumption

**Remark.** (Simple Model of Consumption Smoothing) We consider a simple model of consumption smoothing.

- Time is discrete, indexed by  $t$ . We begin at time  $t = 1$ .
- Each period  $t$ , agents choose consumption  $C_t \geq 0$  and saving/borrowing.. The agent's current stock of wealth is denoted by  $A_t$ .
- Agents possess perfect foresight and certainty about:
  - End of life: finite and known. The agent dies at time  $T$ .
  - Labor income/endowment stream  $(Y_1, Y_2, \dots, Y_T)$  received each period, with  $Y_t \geq 0$  for all  $t = 1, \dots, T$ .
  - Interest rates are constant across time, and zero:  $r_t = r = 0$  for all  $t = 1, \dots, T$ . This is a “pure storage technology.”
- No government.
- Perfect capital markets (same interest rate for borrowing and lending). The resources placed in the capital markets are non-perishable.
- Default is forbidden: the agent must end life with positive wealth.  $A_T \geq 0$ .

The agent derives present utility from consumption through a per-period *felicity* (or flow-utility) function,  $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ . We assume  $u$  to be:

- Strictly increasing.
- Strictly concave.
- We assume  $u$  to be (once or twice) differentiable as needed. If it is once differentiable, then the assumption that  $u$  is strictly increasing implies  $u'(C) > 0$  for all  $C \in \mathbb{R}_+$ . If it is twice differentiable, then the strict concavity assumption implies  $u''(C) < 0$  for all  $C \in \mathbb{R}_+$ .
- Also, we assume that the felicity functions are such that zero consumption will never be chosen in any period. If it is differentiable, this is accomplished by requiring  $\lim_{C \rightarrow 0^+} u'(C) = +\infty$ .

The consumption good  $C_t$  is non-durable—it disappears after giving flow utility to the consumer.

*Intertemporal* (or *lifetime*) utility is given by a discounted sum of per-period flow utilities. Specifically, for a *discount factor*  $\beta \in \mathbb{R}_+$ , lifetime utility is given by the function  $U : \mathbb{R}_+^T \rightarrow \mathbb{R}$  defined by  $U(C_1, C_2, \dots, C_T) = \sum_{t=1}^T \beta^{t-1} u(C_t)$ . Note that this form of intertemporal utility implies that more or less consumption yesterday or tomorrow has no impact on the marginal utility of more consumption today: utility is *additively separable* in consumption from different periods. From the corresponding assumptions on  $u$ , we see that  $U$  is strictly increasing in each argument and strictly concave. The *discount rate*  $\gamma$  is derived implicitly from the discount factor  $\beta$  via  $\beta := \frac{1}{1+\gamma}$ . For now, assume  $\beta = 1$ . Then,  $\gamma = 0$ —the agent discounts the future subjectively at the same rate as the capital market does (interest rate of zero). (Note that in infinite horizon models, we need  $\beta < 1$  so that intertemporal utility is finite). The agent is born with resources  $A_0$ . Each period, she enters with wealth  $A_{t-1}$ , obtains income  $A_t$  and chooses her consumption  $C_t$ , implicitly defining saving/borrowing  $A_t$  as the difference  $A_t = A_{t-1} + Y_t - C_t$ . Recall that agents must die with nonnegative networth:  $A_T \geq 0$ . Moreover, we assume there is no bequest motive, so agents will never want to die with nonnegative wealth, as they

can get flow utility from consuming whatever is left with no future consequences:  $A_T \leq 0$ . Together, these conditions imply  $A_T = 0$ . The agent's choices for  $(C_t)_{t=1}^T$  must satisfy the *intertemporal budget constraint* (IBC):

$$\sum_{t=1}^T C_t \leq A_0 + \sum_{t=1}^T Y_t.$$

In words, her lifetime consumption cannot exceed her lifetime wealth. (We assume that the exogenous variables  $A_0$  and  $Y_t$  are such that  $A_0 + \sum_{t=1}^T Y_t \geq 0$ , or else the problem is not sensible). Since intertemporal utility  $U$  is strictly increasing, it is clear that the IBC will always hold with equality. Where does the IBC come from? In fact, the IBC is derived from the institutional per-period *flow-budget constraints* (FBC), which say that each period's consumption, minus borrowing or plus saving, cannot exceed the resources available to the consumer at the beginning of the period:

$$C_1 + A_1 \leq A_0 + Y_1, C_2 + A_2 \leq A_1 + Y_2, \dots, C_t + A_t \leq A_{t-1} + Y_t, \dots, C_T \leq A_{T-1} + Y_T.$$

(For the period- $T$  FBC, recall that the agent always chooses  $A_T = 0$ ). Rearrange each flow budget constraint to get a bound on each period's consumption:

$$C_1 \leq A_0 - A_1 + Y_1, C_2 \leq A_1 - A_2 + Y_2, \dots, C_t \leq A_{t-1} - A_t + Y_t, \dots, C_T \leq A_{T-1} + Y_T.$$

These constraints state that each period's consumption cannot exceed that period's income less the change in wealth. Summing these  $T$  equations gives the IBC:

$$\begin{aligned} \sum_{t=1}^T C_t &\leq \sum_{t=1}^{T-1} ((A_{t-1} - A_t) + Y_t) + A_{T-1} + Y_T \\ &= A_0 + \sum_{t=1}^T Y_t. \end{aligned}$$

Alternatively, we could start with the IBC and derive the flow-budget constraints: **TODOTDO**

We proceed to formally state the agent's decision problem: given initial endowment  $A_0 \in \mathbb{R}$  and a sequence of labor endowments  $(Y_1, Y_2, \dots, Y_T) \in \mathbb{R}_+^T$  satisfying  $A_0 + \sum_{t=1}^T Y_t$ ,

$$\max_{(C_1, \dots, C_T) \in \mathbb{R}_+^T} U(C_1, \dots, C_T) = \sum_{t=1}^T C_t \text{ s.t. } \sum_{t=1}^T C_t \leq A_0 + \sum_{t=1}^T Y_t.$$

We see that the seemingly dynamic problem of choosing consumption each period is mathematically equivalent to a static decision. We solve the problem. Recall that zero consumption will never be chosen in any period. Moreover, recall that the IBC will always be satisfied with equality. Thus we can write define the Lagrangian  $\mathcal{L} : \mathbb{R}_{++}^T \times \mathbb{R}_+ \rightarrow \mathbb{R}$  via

$$\mathcal{L}(C_1, C_2, \dots, C_T, \lambda) = \sum_{t=1}^T u(C_t) - \lambda \left[ \sum_{t=1}^T C_t - A_0 - \sum_{t=1}^T Y_t \right],$$

and solve the unconstrained maximization problem

$$\max_{(C_1, C_2, \dots, C_T) \in \mathbb{R}_+^T, \lambda \in \mathbb{R}_+} \mathcal{L}(C_1, C_2, \dots, C_T, \lambda).$$

Since  $U$  is concave, any critical points of  $\mathcal{L}$  will solve this problem. For each  $t = 1, \dots, T$ , the first-order

condition yields

$$u'(C_t^*) - \lambda^* = 0 \Rightarrow u'(C_t^*) = \lambda^*.$$

In particular, this tells us that  $u'(C_1^*) = u'(C_2^*) = \dots = u'(C_T^*)$ . Since  $u''(C) < 0$  for all  $C$ , then  $u'$  is strictly monotone, and hence it is invertible. Therefore, we have that  $C_1^* = C_2^* = \dots = C_T^*$ . Denote by  $\bar{C}$  the value  $C_t^*$  for all  $t = 1, \dots, T$ . The first-order conditions have told us about the growth path of  $C_t$  (namely, that it stays constant), but we still do not know the level. To find this, we take the first-order condition for  $\lambda$ :

$$\sum_{t=1}^T C_t^* - A_0 - \sum_{t=1}^T Y_t = 0 \Rightarrow \sum_{t=1}^T \bar{C} = A_0 + \sum_{t=1}^T Y_t \Rightarrow \bar{C} = \frac{1}{T} [A_0 + \sum_{t=1}^T Y_t].$$

Thus, we see that the agent chooses the same consumption every period, equal to the average of total lifetime income. The marginal propensity to consume out of current income is the same each period, at  $\frac{1}{T}$ . What does this result imply about saving? Defining per-period saving as  $S_t = Y_t - C_t$  for  $t = 1, \dots, T$ , we find

$$S_t = Y_t - C_t = Y_t - \frac{1}{T} [A_0 + \sum_{t=1}^T Y_t].$$

As an alternative frame for saving, note

$$\begin{aligned} A_t - A_{t-1} &= (A_{t-1} + Y_t - C_t) - A_{t-1} \\ &= Y_t - C_t \\ &= S_t \end{aligned}$$

so saving/dissaving is the transitory change in net asset position. So,  $S_t > 0$  precisely when the period's transitory income is greater than average lifetime income, and  $S_t < 0$  when transitory income is less than average lifetime income. Punchline: even under total certainty and no risk, the agent will save and borrow to smooth consumption!

#### Cookie-Cutter: FBC $\rightarrow$ IBC (telescoping trick)

Consider the flow budget constraint (no labor income):

$$A_{t+1} = (A_t - C_t)(1+r) = (1+r)A_t - (1+r)C_t.$$

Divide by  $(1+r)^{t+1}$  to obtain the *discounted-assets form*:

$$\frac{A_{t+1}}{(1+r)^{t+1}} = \frac{A_t}{(1+r)^t} - \frac{C_t}{(1+r)^t}. \quad (\star)$$

Sum  $(\star)$  from  $t = 0$  to  $T$ ; the  $A$ -terms telescope:

$$\frac{A_{T+1}}{(1+r)^{T+1}} - \frac{A_0}{1} = - \sum_{t=0}^T \frac{C_t}{(1+r)^t}.$$

Rearranging,

$$\sum_{t=0}^T \frac{C_t}{(1+r)^t} = A_0 - \frac{A_{T+1}}{(1+r)^{T+1}}.$$

Impose the *No-Ponzi / transversality condition*  $\lim_{T \rightarrow \infty} \frac{A_{T+1}}{(1+r)^{T+1}} = 0$  to get the **intertemporal budget constraint (IBC)**:

$$\sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} = A_0.$$

Mnemonic for exams: **D-S-S-L** = *Discount* by  $(1+r)^{t+1}$ , *Shift* indices, *Sum* (telescopes), take the *Limit*.<sup>a</sup>

<sup>a</sup>Useful variants:

- With income  $Y_t$ :  $A_{t+1} = (A_t + Y_t - C_t)(1+r) \Rightarrow \sum_{t \geq 0} \frac{C_t}{(1+r)^t} = A_0 + \sum_{t \geq 0} \frac{Y_t}{(1+r)^t}$ .
- Time-varying safe rate  $r_t$ : discount by  $R_{0,t} \equiv \prod_{j=0}^{t-1} (1+r_j)$ ; IBC becomes  $\sum_{t \geq 0} \frac{C_t}{R_{0,t}} = A_0 + \sum_{t \geq 0} \frac{Y_t}{R_{0,t}}$  with  $\lim_{T \rightarrow \infty} \frac{A_{T+1}}{R_{0,T+1}} = 0$ .
- Using  $\beta \equiv \frac{1}{1+r}$ :  $A_{t+1}\beta^{t+1} = A_t\beta^t - C_t\beta^t \Rightarrow \sum_{t \geq 0} \beta^t C_t = A_0$  (and similarly with  $Y_t$ ).
- Risky/uncertain returns: use the stochastic discount factor  $M_{t+1}$ ; discounted assets are  $M_{0,t}A_t$  and the IBC is  $\mathbb{E}_0 \left[ \sum_{t \geq 0} M_{0,t} C_t \right] = A_0 + \mathbb{E}_0 \left[ \sum_{t \geq 0} M_{0,t} Y_t \right]$  with  $\lim_{T \rightarrow \infty} \mathbb{E}_0[M_{0,T+1} A_{T+1}] = 0$ .
- Gross return notation  $R=1+r$ : replace  $(1+r)^t$  by  $R^t$  throughout.
- Growing measurement units (inflation or population growth  $g$ ): work in real/per-capita terms or discount additionally by the growth factor so the transversality condition holds.
- Finite horizon  $T$ : no limit; instead  $\sum_{t=0}^T \frac{C_t}{R_{0,t}} = A_0 + \sum_{t=0}^T \frac{Y_t}{R_{0,t}} - \frac{A_{T+1}}{R_{0,T+1}}$ .

**Remark** (Two timing conventions yield the same intertemporal budget). *Notation*:  $R = 1+r$ ,  $q = 1/R$ .

(A) **Pay-tomorrow (payoff units)**.

$$A_{t+1} = R(A_t + Y_t - C_t) \quad \Longleftrightarrow \quad C_t + qA_{t+1} = A_t + Y_t.$$

Divide by  $R^{t+1}$ , sum, and impose No-Ponzi:

$$\sum_{t \geq 0} \frac{C_t}{R^t} = A_0 + \sum_{t \geq 0} \frac{Y_t}{R^t}.$$

(B) **Buy-today (present-value units)**.

$$C_t + \frac{W_{t+1}}{R} = W_t + Y_t \quad \Longleftrightarrow \quad W_{t+1} = R(W_t + Y_t - C_t).$$

Divide by  $R^t$ , sum, and impose No-Ponzi:

$$\sum_{t \geq 0} \frac{C_t}{R^t} = W_0 + \sum_{t \geq 0} \frac{Y_t}{R^t}.$$

*Dictionary.* At each date  $t$ ,  $W_t = A_t$  (same “today-pesos”). The difference is whether  $R$  multiplies savings ( $A_{t+1} = R(\cdot)$ ) or appears in the purchase price of assets ( $W_{t+1}/R$ ).

**Remark.** (Simple model of retirement savings) Let us make Model into a very basic life cycle model. Let  $T_j$  be the (known) *retirement date*, and set  $Y_t = \bar{Y}$  for  $t = 1, \dots, T_j$  for fixed  $\bar{Y} \geq 0$  and  $Y_t = 0$  for  $t \geq T_j + 1$ . Assume  $A_0 = 0$  for simplicity. Application of the formula  $C_t = \bar{C} = \frac{1}{T} \left[ A_0 + \sum_{t=1}^T Y_t \right]$  for all  $t \in \{1, \dots, T\}$  to our specific circumstances yields

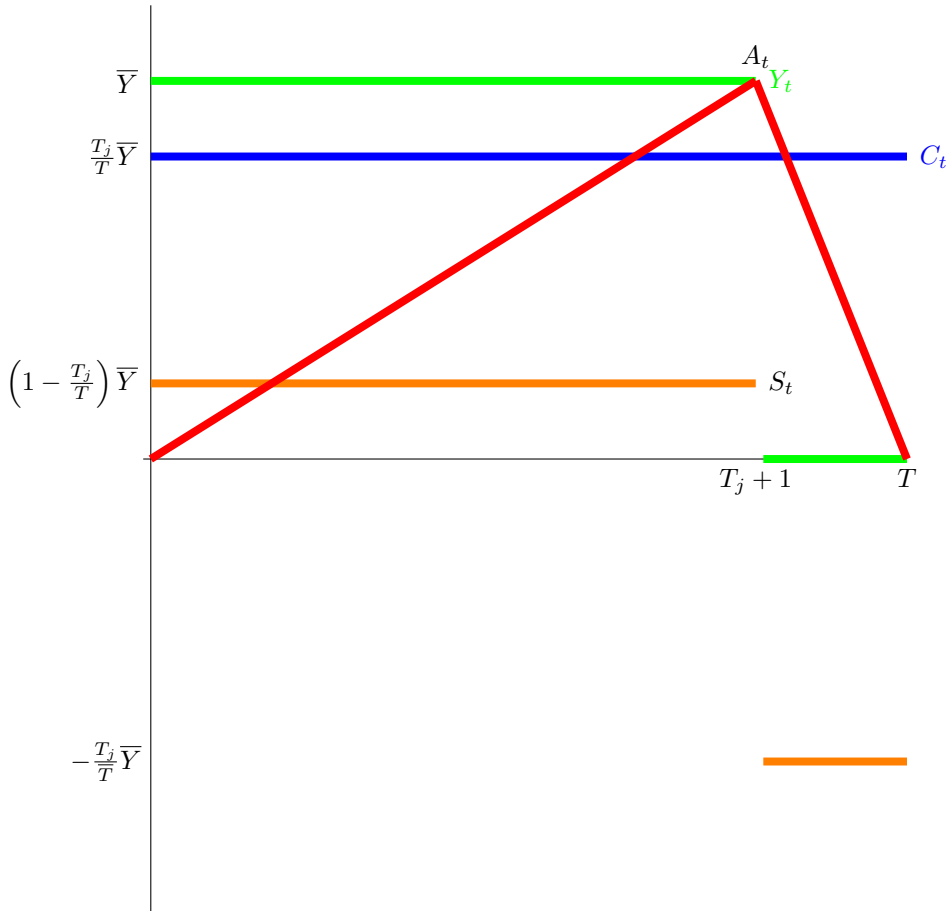
$$C_t = \bar{C} = \frac{T_j}{T} \bar{Y}.$$

Thus, savings  $S_t = Y_t - C_t$  is given by

$$S_t = \begin{cases} \left(1 - \frac{T_j}{T}\right) \bar{Y}, & t = 1, \dots, T_j, \\ -\frac{T_j}{T} \bar{Y}, & t = T_j + 1, \dots, T. \end{cases}$$

So during the working life,  $S_t > 0$ , and during retirement,  $S_t < 0$ . So, the lifecycle paths of the variables are as follows:

- Consumption stays constant every period at  $\bar{C} = \frac{T_j}{T} \bar{Y}$ .
- Income: constant at  $\bar{Y}$  for  $t = 1, \dots, T_j$ , then constant at zero for  $t = T_j + 1, \dots, T$ .
- Savings: the difference between income and consumption.
- Assets: triangular shape: increase until  $T_j$ , then declining to zero until  $T$ .



**Remark.** (Relative risk aversion) For a twice differentiable, strictly increasing, strictly concave felicity function  $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ , we define the *relative risk aversion* at  $C \in \mathbb{R}_{++}$  by  $-\frac{u''(C)C}{u'(C)}$ . (Recall  $u''(C) < 0$  for all  $C$ , so relative risk aversion is positive). Intuitively,  $-u''(C)$  governs the degree of concavity in the consumer's preferences, thereby governing their risk aversion—how much they prefer a sure bet in the middle versus a risky bet with the same expected utility. To get this into a form that is invariant across affine transformations of a utility function (which do not affect preferences), divide by  $u'(C)$ . (Indeed, if  $v : \mathbb{R}_+ \rightarrow \mathbb{R}$  is another utility function given by  $v(C) = au(C) + b$ , then  $-\frac{v''(C)}{v'(C)} = -\frac{au''(C)}{au'(C)} = -\frac{u''(C)}{u'(C)}$ .) Finally, note that the expression  $-\frac{u''(C)}{u'(C)}$  is currently in units of inverse consumption; to make it unitless, we multiply by  $C$  to obtain  $-\frac{u''(C)C}{u'(C)}$ , the coefficient of relative risk aversion. In general, the coefficient

of relative risk aversion varies depending on  $C$ . But a special class of utility functions have a constant relative risk aversion at all levels of consumption. For any  $\sigma \geq 0$ , we define the CRRA (constant relative risk aversion) utility function  $u : \mathbb{R}_{++} \rightarrow \mathbb{R}$  by

$$\begin{aligned} u(C) &= \frac{C^{1-\sigma} - 1}{1-\sigma}, \quad \sigma \neq 1, \\ u(C) &= \ln(C), \quad \sigma = 1 \end{aligned}$$

which has constant relative risk aversion of  $\sigma$  at each level of consumption (stay tuned). In fact, this is the *only* class of utility functions with constant relative risk aversion (**prove**). (Often, the minus 1 in the numerator is omitted because it does not change preferences, but it is useful theoretically). Note that for  $\sigma = 0$ , we simply have  $u(C) = C - 1$ , so utility is simply linear. We will justify below that  $\ln(C)$  is the limiting case of  $u(C)$  as  $\sigma \rightarrow 1$ . For  $1 \neq \sigma \geq 0$ , we show that the CRRA utility function with parameter  $\sigma$  has constant coefficient of relative risk aversion  $\sigma$ . Indeed, for all  $C \in \mathbb{R}_{++}$ ,

$$u'(C) = C^{-\sigma}, u''(C) = -\sigma C^{-\sigma-1}$$

and so

$$\text{Relative risk aversion at } C = -\frac{u''(C)C}{u'(C)} = -\frac{-\sigma C^{-\sigma-1}C}{C^{-\sigma}} = \sigma$$

so relative risk aversion is the same at all consumption levels, and is the parameter  $\sigma$ . We show that  $u(C) \rightarrow \ln(C)$  as  $\sigma \rightarrow 1$ : using L'Hôpital's rule,

$$\lim_{\sigma \rightarrow 1} \frac{C^{1-\sigma} - 1}{1-\sigma} = \lim_{\sigma \rightarrow 1} \frac{-\ln(C) C^{1-\sigma}}{-1} = \ln(C).$$

And the utility function  $u(C) = \ln(C)$  has constant relative risk aversion of  $\sigma = 1$ :

$$u'(C) = \frac{1}{C}, u''(C) = -\frac{1}{C^2}$$

and so

$$\text{Relative risk aversion at } C = -\frac{u''(C)C}{u'(C)} = -\frac{-\frac{1}{C^2}C}{\frac{1}{C}} = 1.$$

**Remark.** (General model of consumption and saving under perfect foresight) We generalize Model as follows:

1. Discount factor  $\beta < 1$ , implying a discount rate  $\gamma = \frac{1-\beta}{\beta} > 0$ . The agent (subjectively) discounts the future.
2. Strictly positive interest rate  $r > 0$ . Still constant over time and same for borrowing and saving.
3. Government levying proportional, time-varying consumption taxes  $\tau_t^c$  and  $\tau_t^y$ ,  $t = 1, \dots, T$ . Government revenues are thrown in the ocean.

Otherwise, we keep the same general setting. The intertemporal utility function is now

$$U(C_1, C_2, \dots, C_T) = \sum_{t=1}^T \beta^{t-1} u(C_t).$$

The flow budget-constraints:

$$\begin{aligned}
C_1(1 + \tau_1^c) + A_1 &= Y_1(1 - \tau_1^y) + A_0(1 + r), \\
C_2(1 + \tau_2^c) + A_2 &= Y_2(1 - \tau_2^y) + A_1(1 + r), \\
&\vdots \\
C_t(1 + \tau_t^c) + A_t &= Y_t(1 - \tau_t^y) + A_{t-1}(1 + r), \\
&\vdots \\
C_T(1 + \tau_T^c) &= Y_T(1 - \tau_T^y) + A_{T-1}(1 + r).
\end{aligned}$$

(Recall that we assume  $A_T = 0$ —no bequest motive and no default). To derive the intertemporal budget constraint, we multiply the period- $t$  flow budget constraint by  $\frac{1}{(1+r)^{t-1}}$  for  $t = 1, \dots, T$ , to obtain

$$\frac{1}{(1+r)^{t-1}} [C_t(1 + \tau_t^c) + A_t] = \frac{1}{(1+r)^{t-1}} [Y_t(1 - \tau_t^y)] + \frac{1}{(1+r)^{t-2}} A_{t-1}, \quad t = 1, \dots, T.$$

Rearrange to

$$\frac{C_t(1 + \tau_t^c)}{(1+r)^{t-1}} = \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + \frac{A_{t-1}}{(1+r)^{t-2}} - \frac{A_t}{(1+r)^{t-1}}, \quad t = 1, \dots, T.$$

Finally, sum up these  $T$  equations to get the intertemporal budget constraint:

$$\begin{aligned}
\sum_{t=1}^T \frac{C_t(1 + \tau_t^c)}{(1+r)^{t-1}} &= \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + \sum_{t=1}^T \frac{A_{t-1}}{(1+r)^{t-2}} - \sum_{t=1}^T \frac{A_t}{(1+r)^{t-1}} \\
&= \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + \sum_{t=0}^{T-1} \frac{A_t}{(1+r)^{t-1}} - \sum_{t=1}^T \frac{A_t}{(1+r)^{t-1}} \\
&= \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + \frac{A_0}{(1+r)^{0-1}} + \frac{A_T}{(1+r)^{T-1}} \\
&= \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + (1+r) A_0.
\end{aligned}$$

That is,

$$\sum_{t=1}^T \frac{C_t(1 + \tau_t^c)}{(1+r)^{t-1}} = \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + (1+r) A_0,$$

that is, the present value of (consumption tax-inclusive) lifetime consumption equals the present value of lifetime income. We now formally state the problem:

$$\begin{aligned}
\max_{C_1, C_2, \dots, C_T \in \mathbb{R}_+^T} \quad & U(C_1, C_2, \dots, C_T) = \sum_{t=1}^T \beta^{t-1} u(C_t), \\
\text{s.t.} \quad & \sum_{t=1}^T \frac{C_t(1 + \tau_t^c)}{(1+r)^{t-1}} \leq \sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + (1+r) A_0
\end{aligned}$$

with  $A_0 \in \mathbb{R}$  given,  $A_T = 0$ ,  $(Y_1, \dots, Y_T) \in \mathbb{R}_+^T$  given,  $\beta < 1$  given,  $r > 0$  given, and  $(\tau_t^c, \tau_t^y)_{t=1}^T$  given, with  $\sum_{t=1}^T \frac{Y_t(1 - \tau_t^y)}{(1+r)^{t-1}} + (1+r) A_0 \geq 0$ . As before, our assumptions on felicity imply that each  $C_t$  chosen will be strictly positive, and the intertemporal budget constraint will hold with equality. Hence our decision

is equivalent to:

$$\begin{aligned} & \max_{(C_1, \dots, C_T) \in \mathbb{R}_{++}^T, \lambda \in \mathbb{R}_{++}} \mathcal{L}(C_1, \dots, C_T, \lambda) \\ &= \sum_{t=1}^T \beta^{t-1} u(C_t) + \lambda \left[ \sum_{t=1}^T \frac{Y_t (1 - \tau_t^y)}{(1+r)^{t-1}} + (1+r) A_0 - \sum_{t=1}^T \frac{C_t (1 + \tau_t^c)}{(1+r)^{t-1}} \right]. \end{aligned}$$

This problem is concave, so it suffices to find the critical points. For each  $t = 1, \dots, T-1$ , the FOC for  $C_t$  is

$$\beta^{t-1} u'(C_t^*) = \lambda^* \frac{1 + \tau_t^c}{(1+r)^{t-1}}$$

and the FOC for  $C_{t+1}$  is

$$\beta^t u'(C_{t+1}^*) = \lambda^* \frac{1 + \tau_{t+1}^c}{(1+r)^t}.$$

Dividing these two conditions gives the *consumption Euler equation*:

$$\beta \frac{u'(C_{t+1}^*)}{u'(C_t^*)} = \frac{1 + \tau_{t+1}^c}{(1+r)(1 + \tau_t^c)}, \quad t = 1, \dots, T-1.$$

Notice that income  $Y_t$  and labor income taxes  $\tau_t^y$  do not factor into this equation at all. Do a first-order Taylor approximation of  $u'$  about  $C_t^*$ , evaluated at  $C_{t+1}^*$ :

$$u'(C_{t+1}^*) = u'(C_t^*) + u''(C_t^*) (C_{t+1}^* - C_t^*) + O\left((C_{t+1}^* - C_t^*)^2\right), \quad t = 1, \dots, T-1.$$

Divide each of these equations through by  $u'(C_t^*)$  to get

$$\frac{u'(C_{t+1}^*)}{u'(C_t^*)} = 1 + \frac{u''(C_t^*) \cdot C_t^*}{u'(C_t^*)} \cdot \frac{C_{t+1}^* - C_t^*}{C_t^*} + O\left((C_{t+1}^* - C_t^*)^2\right).$$

(Drop stars from now on). We then see that to a first-order,

$$\frac{u'(C_{t+1})}{u'(C_t)} = 1 - \text{coefficient of relative risk aversion} * \text{growth rate of consumption}$$

so  $\frac{u'(C_{t+1})}{u'(C_t)}$  is inversely related to the growth rate of consumption.

Recall that the *elasticity of intertemporal substitution* at time  $t$  is given by

$$-\frac{d\left(\frac{C_{t+1}}{C_t}\right)}{d\left(\frac{u'(C_{t+1})}{u'(C_t)}\right)} \cdot \frac{\frac{u'(C_{t+1})}{u'(C_t)}}{\frac{C_{t+1}}{C_t}}$$

This captures the degree to which the growth rate of consumption is sensitive to changes in the marginal rate of substitution ( $u'(C_{t+1})/u'(C_t)$ ). If it is high, then a small increase in the intertemporal rate of substitution is associated with a large decrease in the growth rate of consumption. Note we can write

the elasticity of intertemporal substitution as

$$- \left( \frac{d \left( \ln \left( \frac{C_{t+1}}{C_t} \right) \right)}{d \left( \ln \left( \frac{u'(C_{t+1})}{u'(C_t)} \right) \right)} \right).$$

If our utility function is CRRA with parameter  $-\sigma$ , then this further reduces to

$$- \left( \frac{d \left( \ln \left( \frac{C_{t+1}}{C_t} \right) \right)}{d \left( \ln \left( \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \right) \right)} \right) = \frac{1}{\sigma} \left( \frac{d \left( \ln \left( \frac{C_{t+1}}{C_t} \right) \right)}{d \left( \ln \left( \frac{C_{t+1}}{C_t} \right) \right)} \right) = \frac{1}{\sigma}.$$

So under CRRA utility, the elasticity of intertemporal substitution is equal to the inverse of the coefficient of relative risk aversion. That is weird! Back to the model. Denote by  $\sigma_t = -\frac{u''(C_t)C_t}{u'(C_t)}$  the coefficient of relative risk aversion at time  $t$ . (If utility is CRRA, then  $\sigma_t = \sigma$  for all  $t$ ). Then, we found that for  $t = 1, \dots, T_1$ ,

$$\beta \frac{u'(C_{t+1})}{u'(C_t)} = \frac{1 + \tau_{t+1}^c}{(1+r)(1+\tau_t^c)}$$

and up to the first order,

$$\frac{u'(C_{t+1})}{u'(C_t)} = 1 - \sigma_t \cdot \frac{\Delta C_{t+1}}{C_t}.$$

Equating these expressions gives that, up to a first order, for each  $t = 1, \dots, T-1$ , we have the following approximation for the growth rate of consumption:

$$\frac{u'(C_{t+1})}{u'(C_t)} = \frac{1 + \tau_{t+1}^c}{\beta(1+r)(1+\tau_t^c)} = 1 - \sigma_t \cdot \frac{\Delta C_{t+1}}{C_t} \Rightarrow \frac{\Delta C_{t+1}}{C_t} = \frac{1}{\sigma_t} \left[ 1 - \frac{1 + \tau_{t+1}^c}{\beta(1+r)(1+\tau_t^c)} \right].$$

This is called a *Keynes-Ramsey* formula: the Euler equation has been transformed from an expression involving marginal utilities into an expression about the growth rate of consumption. Some remarks on the above Keynes-Ramsey formula:

- Income  $Y_t$ , income taxes  $\tau_t^y$ , and initial wealth  $A_0$  are all irrelevant for consumption growth.
- Suppose consumption taxes will increase tomorrow:  $\tau_t^c < \tau_{t+1}^c$ . This will cause a decrease in the growth rate of consumption (and if  $\beta(1+r) = 1$ , it will certainly cause negative consumption growth rate). The level of consumption today depends on the balance between the intertemporal substitution effect and the intertemporal wealth effect from future tax increases. Could potentially cause a consumption boom this period depending.

From now on, we assume all consumption taxes are constant:  $\tau_t^c = \tau$  for all  $t$ . We made our point about unconventional fiscal policy. Then, the Keynes-Ramsey formula reduces to

$$\begin{aligned} \frac{\Delta C_{t+1}}{C_t} &= \frac{1}{\sigma_t} \left[ 1 - \frac{1}{\beta(1+r)} \right] \\ &= \frac{1}{\sigma_t} \left[ \frac{\beta(1+r) - 1}{\beta(1+r)} \right] \\ &= \frac{1}{\sigma_t} \left[ \frac{\beta(1+r) - \beta(1+\gamma)}{\beta(1+r)} \right] \\ &= \frac{1}{\sigma_t} \left[ \frac{\beta((1+r) - (1+\gamma))}{\beta(1+r)} \right] \end{aligned}$$

where, recall that  $\gamma$  is the discount rate defined such that  $\beta = \frac{1}{1+\gamma}$ . Finally, we see that, again up to a first order, and assuming  $\sigma_t$  is roughly constant at some  $\sigma$  (literally true if felicity is CRRA),

$$\frac{\Delta C_{t+1}}{C_t} = \frac{r - \gamma}{\sigma(1+r)}.$$

From this new Keynes-Ramsey formula, we get some more insights:

1. If relative risk aversion  $\sigma$  is high, so that  $\frac{1}{\sigma}$  is low, then any change in interest or discount rates translates into less change on the consumption growth rate than would a lower  $\sigma$ .
2. If  $r = \gamma$ , implying  $r = \frac{1-\beta}{\beta} \Rightarrow \beta r = 1 - \beta \Rightarrow \beta(1+r) = 1$ , then the market interest rate equals the discount rate, and consumption growth is zero. Intuitively, the market compensates the agent just enough for her impatience that she does not tilt consumption.
3. If  $r > \gamma$ , then the market discounts the future more than the agent does. The agent is patient and defers consumption—consumption growth is positive.
4. If  $r < \gamma$ , then the market discounts the future less than does the agent. The agent is too impatient to reap the rewards of waiting, and consumption growth is negative.

We can do some comparative statics on what happens if exogenous parameters change:

1. Suppose that the discount rate  $\gamma$  fall, that is,  $\beta$  increases. We see that consumption growth will fall. Intuitively, the agent discounts the future more, is more impatient, and shifts relatively more consumption to today.
2. The derivative of the Keynes-Ramsey formula with respect to  $r$  is

$$\frac{\sigma(1+r) - (r-\gamma)\sigma}{\sigma^2(1+r)^2} = \frac{1+\gamma}{\sigma(1+r)^2} > 0$$

implying that an increase in the interest rate will cause consumption growth to increase.

3. Again, none of this has to do with levels.

Punchline: up to a first order, we have that consumption will be constant over time (as in Model ) if consumption taxes are constant over time and the consumers are as patient as the market. In general, however, consumption will not be constant, but grows and shrinks over time. Even in this perfect foresight world, there is *consumption tilting* when consumption taxes are changing or consumers have differing patience from the market.

In continuous time (need  $\beta < 1$ ), the annoying  $1+r$  in the denominator of the Keynes-Ramsey formula drops out. Assuming consumption taxes are constant over time, and that discounting is proportional to time. We can write (some of this math is a little iffy)

$$\beta \frac{u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+r} \Rightarrow \frac{u'(C_{t+1})}{u'(C_t)} = \frac{1+\gamma}{1+r}$$

so for small increments of time  $\Delta t$ ,

$$\frac{u'(C_{t+\Delta t})}{u'(C_t)} = \frac{1+\gamma\Delta t}{1+r\Delta t}.$$

We can approximate  $C_{t+\Delta t}$  as  $C_t + \Delta t \dot{C}_t$ , where  $\dot{C}_t = \frac{\partial C}{\partial t}$  is the instantaneous rate of change of  $C$ . Then we use a Taylor expansion of  $u'$  about  $C_t$  to get the approximation  $u'(C_t + \Delta t \dot{C}_t) = u'(C_t) +$

$u''(C_t) \Delta t \dot{C}_t$ . Thus,

$$\begin{aligned}
& \frac{u'(C_t) + u''(C_t) \Delta t \dot{C}_t}{u'(C_t)} = \frac{1 + \gamma \Delta t}{1 + r \Delta t} \\
\Rightarrow & 1 + \frac{u''(C_t) \Delta t \dot{C}_t}{u'(C_t)} = \frac{1 + \gamma \Delta t}{1 + r \Delta t} \\
\Rightarrow & \frac{u''(C_t) \Delta t \dot{C}_t}{u'(C_t)} = \frac{1 + \gamma \Delta t - 1 - r \Delta t}{1 + r \Delta t} = \frac{\Delta t (\gamma - r)}{1 + r \Delta t} \\
\Rightarrow & \frac{u''(C_t) \dot{C}_t}{u'(C_t)} = \frac{\gamma - r}{1 + r \Delta t}.
\end{aligned}$$

Multiply the left-hand side by  $\frac{C_t}{C_t}$ . Assume the elasticity of intertemporal substitution is roughly constant at  $\frac{1}{\sigma} = -\frac{u'(C)}{u''(C)C}$ . Then,

$$\frac{u''(C_t) C_t}{u'(C_t)} \cdot \frac{\dot{C}_t}{C_t} = \frac{\gamma - r}{1 + r \Delta t} \Rightarrow -\sigma \cdot \frac{\dot{C}_t}{C_t} = \frac{\gamma - r}{1 + r \Delta t} \Rightarrow \frac{\dot{C}_t}{C_t} = \frac{1}{\sigma} \cdot \frac{r - \gamma}{1 + r \Delta t}.$$

Taking  $\Delta t \rightarrow 0$ :

$$\frac{\dot{C}_t}{C_t} = \frac{r - \gamma}{\sigma}$$

saying that the rate of change of consumption is the difference between the market and subjective interest rates, modulated by the intertemporal elasticity of substitution. Return now to the case of discrete time. Assume that consumption taxes are constant. Then, the Euler equation reads

$$\beta (1 + r) u'(C_{t+1}) = u'(C_t).$$

We can interpret this intuitively with a marginal benefit-marginal cost analysis. If the consumer is behaving optimally, then any infinitesimal change should not result in a change in utility. So suppose she saves an infinitesimal amount  $dC$  more today and consumes it tomorrow. How does that impact her utility? She loses approximately  $u'(C_t) dC$  utils from the loss of consumption today, but gains  $\beta (1 + r) u'(C_{t+1}) dC$  tomorrow. Thus these quantities should be the same if she is at the optimum. We can solve the model in closed-form in the case of  $\beta (1 + r) = 1$ , that is,  $\gamma = r$ . Assume also that consumption taxes are zero. In all periods. Then the Euler equation reads

$$u'(C_{t+1}) = u'(C_t)$$

so since  $u'$  is invertible, we have  $C_1 = C_2 = \dots = C_T = \bar{C}$ . Denote lifetime income by  $W = A_0 (1 + r) +$

$\sum_{t=1}^T \frac{1-\tau_t^y}{(1+r)^{t-1}} Y_t$ . Plug these into the intertemporal budget constraint to get

$$\begin{aligned}
& \sum_{t=1}^T \frac{C_t}{(1+r)^{t-1}} = W \\
& \Rightarrow \bar{C} \sum_{t=1}^T \frac{1}{(1+r)^{t-1}} = W \\
& \Rightarrow \bar{C} \frac{\left(\frac{1}{1+r}\right)^T - 1}{\left(\frac{1}{1+r}\right) - 1} = W \\
& \Rightarrow \bar{C} \frac{(1+r) \left( \left(\frac{1}{1+r}\right)^T - 1 \right)}{-r} = W \\
& \Rightarrow \bar{C} = \frac{r}{(1+r) \left( 1 - \left(\frac{1}{1+r}\right)^T \right)} W.
\end{aligned}$$

As  $T \rightarrow \infty$ , then each period's consumption approaches the "annuity value" of lifetime wealth (with the pesky  $1+r$  in the denominator, an artifact of discrete time):

$$\bar{C} = \frac{r}{1+r} W.$$

Note that the marginal propensity to consume out of current income is  $r \frac{1}{1+r} = \left(\frac{1-\beta}{\beta}\right) \beta = 1 - \beta$ . Since  $\beta$  is thought of as close to one, this means the marginal propensity to consume is small. In the general case of  $\beta(1+r) \neq 1$  (that is,  $r \neq \gamma$ ), we solve numerically using a shooting algorithm. We only need to solve for one thing,  $C_1$ , since perfect foresight and the Euler equations imply everything else. We guess a value for  $C_1$ , use the Euler equations to compute the implied  $C_2, C_3, \dots, C_T$ , and check how close the present value of lifetime consumption is to  $W$ . Iterate until you are arbitrarily close to equality in the IBC. This type of algorithm can accommodate all sorts of time-varying stuff (interest rates, discount factors, taxes, etc.) so long as it is deterministic and known.

**Remark.** (Model of consumption and savings under income risk, with quadratic felicity) We now present a model of consumption and savings where we drop the perfect foresight assumption. Specifically, we allow per-period income to be random. We utilize the *quadratic felicity function*

$$u(C_t) = C_t - \frac{a}{2} C_t^2$$

for a constant  $a > 0$ . Note that marginal utility is linear:  $u'(C_t) = 1 - aC_t$ . This is convenient analytically, but quadratic felicity has some drawbacks:

1. Since  $u'(0) = 1$ , the condition that  $\lim_{C \rightarrow 0^+} u'(C) = \infty$  does not hold. Thus we could get stuck at a boundary solution where someone finds it optimal to starve to death.
2. Note that marginal utility is negative for  $C > \frac{1}{a}$ . This means that quadratic felicity violates the assumption of strictly increasing marginal utility. In practice, we can often get around this issue by setting  $a$  to be as small as we need.

For simplicity, assume that all taxes are zero:  $\tau_t^c = \tau_t^y = 0$  for all  $t$ . Assume as well that  $\gamma = r = 0$ , so  $\beta = 1$ . None of these assumptions are fundamental, but just for expositional clarity so we can abstract away from consumption tilting. The income stream  $\{Y_t\}_{t=1}^T$  is a stochastic process. At  $t = 1$ , the consumer only knows  $Y_1$ , and can only immediately control  $C_1$ . The consumer's problem is to choose

decision rules  $C_1(Y_1)$ ,  $C_2(Y_1, Y_2)$ , ...,  $C_T(Y_1, \dots, Y_T)$ , where  $C_t(Y_1, \dots, Y_t)$  is a mapping from the set of all possible histories  $Y_1, \dots, Y_t$  into a nonnegative consumption level, subject to the intertemporal budget constraint. Formally, the problem is

$$\max_{C_1 \geq 0, \dots, C_T \geq 0} \mathbb{E}_1 \left[ \sum_{t=1}^T \left( C_t - \frac{a}{2} C_t^2 \right) \right], \quad \text{s.t.} \quad \sum_{t=1}^T C_t \leq A_0 + \sum_{t=1}^T Y_t$$

where again,  $C_t \geq 0$  is not a single number, but rather a mapping from the space of histories  $(Y_1, \dots, Y_t)$  into  $\mathbb{R}_+$ . The budget constraint must hold with in *every* state of the world, not just in expectation. We require that  $A_0 + \sum_{t=1}^T Y_t \geq 0$  in every state of the world, so that  $A_T < 0$  does not happen from now fault of the consumer. We solve for now using a variational argument. We assume equality in the intertemporal budget constraint (even though inequality could occur with quadratic felicity, since agents don't always want to consume everything.). Suppose we are at a solution, and that we defer an infinitesimal amount of consumption from period 1 to period  $t$ . Note that the utility cost of giving up an infinitesimal amount of consumption today is  $u'(C_1) dC = (1 - aC_1) dC$ . The expected utility gain from the future consumption is  $\mathbb{E}_1[u'(C_t) dC] = \mathbb{E}_1[(1 - aC_t) dC] = (1 - a\mathbb{E}_1(C_t)) dC$ . Equating the benefit and cost gives  $(1 - aC_1) dC = (1 - a\mathbb{E}_1(C_t)) dC$ , which yields the Euler equation

$$C_1 = \mathbb{E}_1[C_t] \text{ for all } t.$$

So consumption is constant in expectation. Hence we will smooth consumption in expectation. Now, assuming equality in the IBC, we take the expectation of both sides:

$$\begin{aligned} \sum_{t=1}^T C_t &= A_0 + \sum_{t=1}^T Y_t \\ \Rightarrow \mathbb{E}_1 \left[ \sum_{t=1}^T C_t \right] &= \mathbb{E}_1 \left[ A_0 + \sum_{t=1}^T Y_t \right] \\ \Rightarrow C_1 + \sum_{t=2}^T \mathbb{E}_1[C_t] &= A_0 + \sum_{t=1}^T \mathbb{E}_1[Y_t] \\ \Rightarrow TC_1 &= A_0 + \sum_{t=1}^T \mathbb{E}_1[Y_t] \\ \Rightarrow C_1 &= \frac{1}{T} \left( A_0 + \sum_{t=1}^T \mathbb{E}_1[Y_t] \right) \end{aligned}$$

so period-1 consumption is the time average of *expected* lifetime income, given the knowledge we have at period 1. This behavior is called *certainty equivalence*: the consumer behaves almost as under certainty, just with the known income stream being replaced by the expected income stream. There is no precautionary savings present. It turns out that certainty equivalence is unique to the quadratic felicity function. The result ultimately stems from the fact that for quadratic felicity,  $\mathbb{E}[u'(C_t)] = u'(\mathbb{E}[C_t])$ , which does not hold in general due to Jensen's inequality. Under quadratic felicity, the derivation of  $\mathbb{E}_1[C_t] = C_1$  for all  $t$  can clearly be generalized:  $\mathbb{E}_t[C_{t+s}] = C_t$  for all  $t < s \leq T$ , using an identical variational argument. In particular, we have that *random walk result*:

$$\mathbb{E}_t[C_{t+1}] = C_t \text{ for all } t.$$

This says that individual consumption follows a martingale: today's consumption is the best predictor of tomorrow's consumption. We reformulate the random-walk result in terms of changes in consumption.

By the definition of the expectation operator, we can write

$$C_t = \mathbb{E}_{t-1} [C_t] + e_t,$$

where  $e_t$  is a random variable satisfying  $\mathbb{E} [e_t] = 0$ . Using the random walk result, we can rewrite this as

$$C_t = C_{t-1} + e_t \Rightarrow \Delta C_t = e_t.$$

Therefore, the random walk result implies that changes in individual consumption are unpredictable from information in the previous period. We discuss two tests of the random walk result in the literature. Using aggregate consumption data, Hall (1978) regressed aggregate consumption change  $\Delta C_t$  on variables known at time  $t - 1$ , such as  $C_{t-1}$ ,  $Y_{t-1}$ , and a stock market index. Aside from the stock market index, none of them were predictive. Some issues: how do we know we are capturing the whole information set at  $t - 1$ ? There could be other relevant variables there that we don't observe. Moreover, since we are using aggregate data but the result was derived at an individual level, Hall is implicitly invoking a representative agent assumption. Campbell and Mankiw (1989) tested the random walk result against a specific alternative. Namely, the assume that a fraction  $1 - \lambda$  of the population follows the random walk in their consumption, while a fraction  $\lambda$  is hand-to-mouth:  $\Delta C_t = \Delta Y_t$ . Then, aggregate consumption is  $\Delta C_t = \lambda \Delta Y_t + (1 - \lambda) e_t$ . Since  $(1 - \lambda) e_t$  is a regression error term, they estimated this equation to estimate  $\lambda$ . One issue with OLS is that  $\Delta Y_t$  and  $e_t$  are likely correlated, since they represent news about current income and news about changes in permanent income. As an IV for  $\Delta Y_t$ , they use lagged  $\Delta Y_t$  and lagged  $\Delta C_t$ , and estimate  $\lambda \approx 0.5$ , a significant departure from the random walk result, which implies  $\lambda = 0$ . Still, there is the issue that they are using aggregate consumption data, and things are in partial equilibrium. Newer papers try to microfound things more rigorously. Back to the model. Recall that

$$C_1 = \frac{1}{T} \left( A_0 + \sum_{t=1}^T \mathbb{E}_1 [Y_t] \right).$$

Let us find period 2 consumption. Taking the expectation of the IBC in period 2, we have

$$\begin{aligned} \mathbb{E}_2 \left[ \sum_{t=1}^T C_t \right] &= \mathbb{E}_2 \left[ A_0 + \sum_{t=1}^T Y_t \right] \\ \Rightarrow C_1 + C_2 + \sum_{t=3}^T \mathbb{E}_2 [C_t] &= A_0 + Y_1 + \sum_{t=2}^T \mathbb{E}_2 [Y_t] \\ \Rightarrow C_1 + (T - 1) C_2 &= A_0 + Y_1 + \sum_{t=2}^T \mathbb{E}_2 [Y_t] \\ \Rightarrow (T - 1) C_2 &= (A_0 + Y_1 - C_1) + \sum_{t=2}^T \mathbb{E}_2 [Y_t] \\ \Rightarrow C_2 &= \frac{1}{T - 1} \left( A_1 + \sum_{t=2}^T \mathbb{E}_2 [Y_t] \right). \end{aligned}$$

So  $C_2$  is the average of the expected remaining lifetime resources. We can further massage this

expression to

$$\begin{aligned}
C_2 &= \frac{1}{T-1} \left( A_0 + Y_1 - C_1 + \sum_{t=2}^T \mathbb{E}_1[Y_t] + \sum_{t=2}^T \mathbb{E}_2[Y_t] - \sum_{t=2}^T \mathbb{E}_1[Y_t] \right) \\
&= \frac{1}{T-1} \left( A_0 - C_1 + \sum_{t=1}^T \mathbb{E}_1[Y_t] + \sum_{t=2}^T \mathbb{E}_2[Y_t] - \sum_{t=2}^T \mathbb{E}_1[Y_t] \right) \\
&= \frac{1}{T-1} \left( TC_1 - C_1 + \sum_{t=2}^T \mathbb{E}_2[Y_t] - \sum_{t=2}^T \mathbb{E}_1[Y_t] \right) \\
&= C_1 + \frac{1}{T-1} \left( \sum_{t=2}^T \mathbb{E}_2[Y_t] - \sum_{t=2}^T \mathbb{E}_1[Y_t] \right).
\end{aligned}$$

Therefore, the change in consumption is

$$\begin{aligned}
\Delta C_2 &= C_2 - C_1 \\
&= \frac{1}{T-1} \left( \sum_{t=2}^T \mathbb{E}_2[Y_t] - \sum_{t=2}^T \mathbb{E}_1[Y_t] \right);
\end{aligned}$$

that is, the change in consumption is the change in expected lifetime income that occurs between period 1 and period 2. So consumption will only change in response to previously unexpected changes in income.

**Remark.** (Model with income risk, generalized) We consider the same model of income risk with quadratic felicity as in Model , but we drop the assumption of  $r = \gamma = 0$ . Specifically, we still impose that  $r = \gamma$ , but we allow it to take values greater than zero:  $r = \gamma > 0$ , so  $\beta < 1$ . The new optimization problem is

$$\begin{aligned}
&\max_{C_1, \dots, C_T} \mathbb{E}_1 \left[ \sum_{t=1}^T \beta \left( C_t - \frac{a}{2} C_t^2 \right) \right] \\
&\text{s.t. } \sum_{t=1}^T \frac{C_t}{(1+r)^{t-1}} \leq A_0(1+r) + \sum_{t=1}^T \frac{Y_t}{(1+r)^{t-1}},
\end{aligned}$$

where as before, we are not choosing single values for  $C_t$ , but rather contingency plans (mappings from the realized history up to period  $t$  into  $C_t$ ), and the  $Y_t$ 's are random variables, so the budget constraint holds in every possible history. We use a variational argument to show that consumption follows a martingale. The utility cost of giving up an infinitesimal amount of period-1 consumption is  $u'(C_1) = 1 - aC_1$ . If it is invested until period  $t$  and then consumed, the expected utility gain as of period 1 is  $\mathbb{E}_1 \left[ \beta^{t-1} (1+r)^{t-1} u'(C_t) \right] = 1 - a\mathbb{E}_1[C_t]$  (recall that  $r = \gamma$ , so  $\beta(1+r) = \beta(1+\gamma) = 1$ ). Equating the benefits and costs gives

$$\mathbb{E}_1[C_t] = C_1.$$

Take the period-1 expectation of the IBC to obtain

$$\begin{aligned}
\mathbb{E}_1 \left[ \sum_{t=1}^T \frac{C_t}{(1+r)^{t-1}} \right] &= \mathbb{E}_1 \left[ (1+r) A_0 + \sum_{t=1}^T \frac{Y_t}{(1+r)^{t-1}} \right] \\
\Rightarrow \sum_{t=1}^T \frac{\mathbb{E}_1[C_t]}{(1+r)^{t-1}} &= (1+r) A_0 + \sum_{t=1}^T \frac{\mathbb{E}_1[Y_t]}{(1+r)^{t-1}} \\
\Rightarrow TC_1 \sum_{t=1}^T \frac{1}{(1+r)^{t-1}} &= (1+r) A_0 + \sum_{t=1}^T \frac{\mathbb{E}_1[Y_t]}{(1+r)^{t-1}} \\
\Rightarrow C_1 &= \frac{r}{(1+r) \left( 1 - \left( \frac{1}{1+r} \right)^T \right)} \left[ (1+r) A_0 + \sum_{t=1}^T \frac{\mathbb{E}_1[Y_t]}{(1+r)^{t-1}} \right].
\end{aligned}$$

As  $T \rightarrow \infty$ , this expression approaches

$$C_1 = \frac{r}{1+r} \left[ (1+r) A_0 + \sum_{t=1}^{\infty} \frac{\mathbb{E}_1[Y_t]}{(1+r)^{t-1}} \right].$$

This shows that we have certainty equivalence once again, where we behave almost as in the perfect foresight case, but just replacing actual lifetime income with expected lifetime income. In general, we have that as  $T \rightarrow \infty$

$$C_t = \frac{r}{1+r} \left[ A_{t-1} (1+r) + \sum_{s=0}^{\infty} \frac{\mathbb{E}_t[Y_{t+s}]}{(1+r)^s} \right].$$

Let's derive that. Fix  $t$ . We have the martingale result that  $\mathbb{E}_t[Y_s] = Y_t$  for  $s > t$ , by a variational argument:  $u'(C_t) = \beta^{t-s} (1+r)^{t-s} \mathbb{E}_t[u'(C_s)]$ . With this in hand, take the period- $t$  expectation of the IBC to obtain

$$\begin{aligned}
\mathbb{E}_t \left[ \sum_{s=1}^T \frac{C_s}{(1+r)^{s-1}} \right] &= \mathbb{E}_t \left[ (1+r) A_0 + \sum_{s=1}^T \frac{Y_s}{(1+r)^{s-1}} \right] \\
\Rightarrow \sum_{s=1}^{t-1} \frac{C_s}{(1+r)^{s-1}} + \frac{C_t}{(1+r)^{t-1}} + \sum_{s=t+1}^T \frac{\mathbb{E}_t[C_s]}{(1+r)^{s-1}} &= (1+r) A_0 + \sum_{s=1}^{t-1} \frac{Y_s}{(1+r)^{s-1}} + \sum_{s=t}^T \frac{\mathbb{E}_t[Y_s]}{(1+r)^{s-1}} \\
\Rightarrow C_t \sum_{s=t}^T \frac{1}{(1+r)^{s-1}} &= (1+r) A_0 + \sum_{s=1}^{t-1} \frac{Y_s - C_s}{(1+r)^{s-1}} + \sum_{s=t}^T \frac{\mathbb{E}_t[Y_s]}{(1+r)^{s-1}} \\
\Rightarrow C_t \sum_{s=0}^{T-s} \frac{1}{(1+r)^{s+t-1}} &= (1+r) A_0 + \sum_{s=1}^{t-1} \frac{A_s - (1+r) A_{s-1}}{(1+r)^{s-1}} + \sum_{s=0}^{T-s} \frac{\mathbb{E}_t[Y_{t+s}]}{(1+r)^{s+t-1}}.
\end{aligned}$$

Now, note that

$$\begin{aligned}
(1+r) A_0 + \sum_{s=1}^{t-1} \frac{A_s - (1+r) A_{s-1}}{(1+r)^{s-1}} &= (1+r) A_0 + \sum_{s=1}^{t-1} \frac{A_s}{(1+r)^{s-1}} - \sum_{s=1}^{t-1} \frac{A_{s-1}}{(1+r)^{s-2}} \\
&= \sum_{s=0}^{t-1} \frac{A_s}{(1+r)^{s-1}} - \sum_{s=0}^{t-2} \frac{A_s}{(1+r)^{s-1}} \\
&= \frac{A_{t-1}}{(1+r)^{t-2}}.
\end{aligned}$$

Plugging this back into our expression, we get

$$\begin{aligned}
C_t \sum_{s=0}^{T-t} \frac{1}{(1+r)^{s+t-1}} &= \frac{A_{t-1}}{(1+r)^{t-2}} + \sum_{s=0}^{T-t} \frac{\mathbb{E}_t[Y_{t+s}]}{(1+r)^{s+t-1}} \\
\Rightarrow C_t \sum_{s=0}^{T-t} \frac{1}{(1+r)^s} &= (1+r) A_{t-1} + \sum_{s=0}^{T-t} \frac{\mathbb{E}_t[Y_{t+s}]}{(1+r)^s} \\
\Rightarrow C_t &= \frac{r}{(1+r) \left(1 - \left(\frac{1}{1+r}\right)^{T+t-1}\right)} \left[ (1+r) A_{t-1} + \sum_{s=0}^{T-t} \frac{\mathbb{E}_t[Y_{t+s}]}{(1+r)^s} \right].
\end{aligned}$$

Taking  $T \rightarrow \infty$  gives the desired result:

$$C_t = \frac{r}{1+r} \left[ A_{t-1} (1+r) + \sum_{s \geq 0} \frac{1}{(1+r)^s} \mathbb{E}_t[Y_{t+s}] \right].$$

It can be then derived ([todo on PS2](#)) that the change in consumption depends only on updates in news. This holds under certainty equivalence and without motives for consumption tilting ( $r = \gamma$ ). Indeed, we have

$$\begin{aligned}
C_t &= \frac{r}{1+r} \left( A_{t-1} (1+r) + \sum_{s=0}^{\infty} \frac{1}{(1+r)^s} E_t[Y_{t+s}] \right) \\
&= \frac{r}{1+r} \left[ (1+r) (Y_{t-1} + (1+r) A_{t-2} - C_{t-1}) + \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} + \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right] \\
&= \frac{r}{1+r} \left[ (1+r)^2 A_{t-2} + (1+r) Y_{t-1} - (1+r) C_{t-1} + (1+r) \sum_{s=1}^{\infty} \frac{E_t[Y_{t-1+s}]}{(1+r)^s} + \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right] \\
&= (1+r) \frac{r}{1+r} \left[ (1+r) A_{t-2} + \sum_{s=0}^{\infty} \frac{E_t[Y_{t-1+s}]}{(1+r)^s} \right] - r C_{t-1} + \frac{r}{1+r} \left[ \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right] \\
&= C_{t-1} + \frac{r}{1+r} \left[ \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right]
\end{aligned}$$

and so

$$\Delta C_t = \frac{r}{1+r} \left[ \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right].$$

If we make assumptions on the process generating  $Y_t$ , we can get nicer results for how consumption changes in response to shocks. For instance, first suppose that  $Y_t$  follows a process  $Y_t = \mu + e_t$ , where  $e_t$  is iid white noise. Then,

$$\begin{aligned}
C_t - C_{t-1} &= \frac{r}{1+r} \left[ \sum_{s=0}^{\infty} \frac{E_t[Y_{t+s}]}{(1+r)^s} - \sum_{s=0}^{\infty} \frac{E_{t-1}[Y_{t+s}]}{(1+r)^s} \right] \\
&= \frac{r}{1+r} \left[ E_t[\mu + e_t] - E_{t-1}[\mu + e_t] + \sum_{s=1}^{\infty} \frac{\mathbb{E}_t[\mu + e_{t+s}] - E_{t-1}[\mu + e_{t+s}]}{(1+r)^s} \right] \\
&= \frac{r}{1+r} \left[ \mu + e_t - \mu - 0 + \sum_{s=1}^{\infty} \frac{\mu + 0 - \mu - 0}{(1+r)^s} \right] \\
&= \frac{r}{1+r} e_t.
\end{aligned}$$

Therefore, we see that consumption barely changes in response to a transitory income shock  $e_t$  (since

$\frac{r}{1+r}$  is likely much less than one).

If  $Y_t$  is a stationary AR( $p$ )  $Y_t = \sum_{i=1}^p \rho_i Y_{t-i} + e_t$ , with  $\sum_{i=1}^p \rho_i \leq 1$ , then (prove)

$$C_t - C_{t-1} = \frac{r}{1+r} \cdot \frac{1}{1 - \sum_{i=1}^p \frac{\rho_i}{(1+r)^i}} \cdot e_t.$$

In the case of an AR(1), this simplifies to (prove)

$$C_t - C_{t-1} = \frac{r}{1+r-\rho} e_t.$$

So as  $\rho \rightarrow 1$ ,  $C_t$  increasingly approaches following a random walk, where consumption changes one-for-one with a transitory income shock:

$$C_t - C_{t-1} = e_t.$$

**Remark** (Income risk with precautionary savings). We turn to the case of income risk where we move away from quadratic felicity, so that certainty equivalence does not hold. Under CARA utility and a normal income process, we will show that consumption now follows a random walk with drift. For simplicity, assume that  $r = \gamma = 0$ , so  $\beta = 1$ . We make use of the *constant absolute risk aversion (CARA)* utility function

$$u(C) = -\frac{1}{\alpha} e^{-\alpha C}.$$

The CARA function, fittingly enough, has constant absolute risk aversion  $-\frac{u''(C)}{u'(C)} = \alpha$ . We assume labor income follows a process

$$Y_t = Y_{t-1} + e_t,$$

with  $e_t \sim N(0, \sigma^2)$  iid. As before, we state the consumer's problem:

$$\max_{C_1, \dots, C_T} \sum_{t=1}^T u(C_t) \quad \text{s.t.} \quad \sum_{t=1}^T C_t \leq A_0 + \sum_{t=1}^T Y_t.$$

We derive the FOC using a variational argument. If we give up one unit of consumption today, the utility cost is  $u'(C_t) = e^{-\alpha C_t}$ . If we save this amount to tomorrow and consume it then, the utility benefit is  $\beta(1+r) E_t[u'(C_{t+1})] = E_t[e^{-\alpha C_{t+1}}]$ . Hence we have the first-order condition

$$e^{-\alpha C_t} = E_t[e^{-\alpha C_{t+1}}].$$

We conjecture that consumption obeys the following random walk with drift, and then confirm that it satisfies that first-order condition:

$$C_{t+1} = C_t + \mu + e_{t+1}.$$

Note that the shocks to the consumption process are the same as the shocks to the income process. It is a fact that for a normal random variable  $X$ , we have  $E[e^{\alpha X}] = e^{\alpha E[X]} \cdot e^{\frac{\alpha^2 \text{Var}(X)}{2}}$  (prove). Plugging this into the right-hand side of the first-order condition gives

$$\begin{aligned} E_t[e^{-\alpha C_{t+1}}] &= E_t[e^{-\alpha(C_t + \mu + e_{t+1})}] \\ &= E_t[e^{-\alpha C_t} \cdot e^{-\alpha \mu} \cdot e^{-\alpha e_{t+1}}] \\ &= e^{-\alpha C_t} \cdot e^{-\alpha \mu} \cdot E_t[e^{-\alpha e_{t+1}}] \\ &= e^{-\alpha C_t} \cdot e^{-\alpha \mu} \cdot e^{\frac{\alpha^2 \sigma^2}{2}}. \end{aligned}$$

Equating this to the left-hand side of the first-order condition yields

$$e^{-\alpha C_t} = e^{-\alpha C_t} \cdot e^{-\alpha \mu} \cdot e^{\frac{\alpha^2 \sigma^2}{2}} \Rightarrow e^{\alpha \mu} = e^{\frac{\alpha^2 \sigma^2}{2}} \Rightarrow \alpha \mu = \frac{\alpha^2 \sigma^2}{2} \Rightarrow \mu = \frac{\alpha \sigma^2}{2}.$$

Hence, the solution is

$$C_{t+1} = C_t + \frac{\alpha \sigma^2}{2} + e_{t+1}.$$

Therefore, consumption is a random walk with drift, where the drift is increasing in absolute risk aversion and the variance of the income process.

Note that we can write

$$C_2 = C_1 + \frac{\alpha \sigma^2}{2} + e_2 = C_1 + \frac{\alpha \sigma^2}{2} + (Y_2 - Y_1)$$

and

$$C_3 = C_2 + \frac{\alpha \sigma^2}{2} + (Y_3 - Y_2)$$

and so

$$C_3 = C_1 + 2 \cdot \frac{\alpha \sigma^2}{2} + (Y_3 - Y_1).$$

Iterating this forward, we get that for any  $t$ ,

$$C_t = C_1 + (t-1) \frac{\alpha \sigma^2}{2} + (Y_t - Y_1).$$

Plug this into the IBC:

$$\begin{aligned} \sum_{t=1}^T \left( C_1 + (t-1) \frac{\alpha \sigma^2}{2} + Y_t - Y_1 \right) &= A_0 + \sum_{t=1}^T Y_t \\ \Rightarrow TC_1 + \frac{\alpha \sigma^2}{2} \sum_{t=1}^{T-1} t + \sum_{t=1}^T Y_t - TY_1 &= A_0 + \sum_{t=1}^T Y_t \\ \Rightarrow TC_1 &= A_0 + TY_1 - \frac{\alpha \sigma^2 \cdot T(T-1)}{4} \\ \Rightarrow C_1 &= \frac{A_0}{T} + Y_1 - \frac{\alpha \sigma^2 (T-1)}{4}. \end{aligned}$$

Recall that in the certainty-equivalence case, with quadratic felicity and  $\gamma = r = 0$ , we had  $C'_1 = \frac{A_0}{T} + \frac{1}{T} \sum_{t=1}^T E_1[Y_t]$ . If we made the random-walk hypothesis on labor income, then we would have  $C'_1 = \frac{A_0}{T} + Y_1$ . Hence, under CARA-normal assumptions, we have that  $C_1 = C'_1 - \frac{\alpha \sigma^2 (T-1)}{4}$ . Consumption is lower because of *precautionary saving*. Note that precautionary savings is increasing in absolute risk aversion  $\alpha$ , the volatility of the income process  $\sigma$ , and lifespan  $T$ . In general for any  $t$ , we have that for  $s \geq t$  ([show](#)),

$$C_s = C_t + (s-t) \frac{\alpha \sigma^2}{2} + Y_s - Y_t.$$

Therefore, manipulating the budget constraint, we find that

$$\begin{aligned}
\sum_{s=1}^T C_s &= A_0 + \sum_{s=1}^T Y_s \\
\Rightarrow \sum_{s=1}^{t-1} C_s + \sum_{s=t}^T \left( C_t + (s-t) \frac{\alpha\sigma^2}{2} + Y_s - Y_t \right) &= A_0 + \sum_{s=1}^T Y_s \\
\Rightarrow (T-t+1)C_t + \frac{\alpha\sigma^2}{2} \sum_{s=0}^{T-s} s + \sum_{s=t}^T Y_s - (T-t+1)Y_t &= A_0 + \sum_{s=1}^{t-1} (Y_s - C_s) + \sum_{s=t}^T Y_s \\
\Rightarrow (T-t+1)C_t + \frac{\alpha\sigma^2(T-s)(T-t+1)}{4} - (T-t+1)Y_t &= A_0 + \sum_{s=1}^{t-1} (A_s - A_{s-1}) = A_{t-1} \\
\Rightarrow C_t &= \frac{A_{t-1}}{T-t+1} + Y_t - \frac{\alpha\sigma^2(T-t)}{4}.
\end{aligned}$$

Note that this formula gives  $C_T = A_{T-1} + Y_T$ , as it should be. Finally, we must show that the process does indeed follow the random walk with drift. First, note that for any  $t$ ,

$$\begin{aligned}
A_{t-1} &= A_{t-2} + Y_{t-1} - C_{t-1} \\
&= A_{t-2} + Y_{t-1} - \frac{A_{t-2}}{T-s+2} - Y_{t-1} + \frac{\alpha\sigma^2(T-s+1)}{4} \\
&= A_{t-2} \left( 1 - \frac{1}{T-s+2} \right) + \frac{\alpha\sigma^2(T-t+1)}{4} \\
&= A_{t-2} \left( \frac{T-t+1}{T-t+2} \right) + \frac{\alpha\sigma^2(T-t+1)}{4}.
\end{aligned}$$

Then, we compute the change in consumption as

$$\begin{aligned}
C_t - C_{t-1} &= \frac{A_{t-1}}{T-t+1} + Y_t - \frac{\alpha\sigma^2(T-t)}{4} - \frac{A_{t-2}}{T-t+2} - Y_{t-1} + \frac{\alpha\sigma^2(T-t+1)}{4} \\
&= \frac{A_{t-2}}{T-t+2} + \frac{\alpha\sigma^2}{4} + \frac{\alpha\sigma^2((T-t+1) - (T-t))}{4} - \frac{A_{t-2}}{T-t+2} + Y_t - Y_{t-1} \\
&= \frac{\alpha\sigma^2}{2} + \underbrace{Y_t - Y_{t-1}}_{=e_t} \\
&= \frac{\alpha\sigma^2}{2} + e_t.
\end{aligned}$$

## Prudence, mean-preserving spreads, and precautionary saving

We have shown that, for a particular combination of a felicity function (CARA) and income process (Gaussian random walk), we can derive a *particular* consumption process (random walk with drift, with innovations the same as in the income process) as the outcome. When directly compared to the certainty-equivalent case (where the consumption outcome is a random walk), we obtain *precautionary saving* (captured by the drift of the random walk).

In what follows, we show what causes precautionary saving *more generally*, independently of the particular function forms assumed above: the condition  $u'''(\cdot) > 0$ , i.e., strict convexity of marginal felicity.

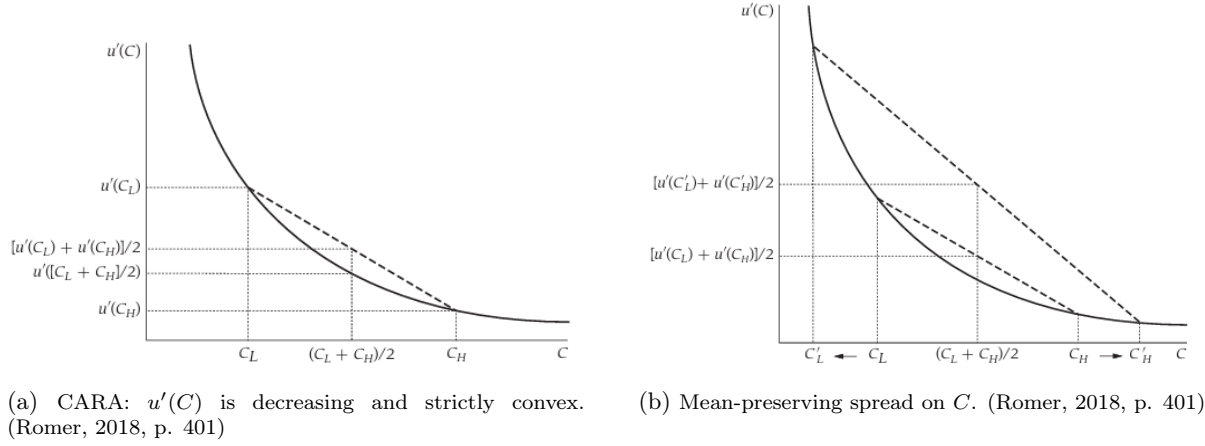
### The role of $u'''(\cdot) > 0$ under CARA felicity

For CARA,

$$u(C_t) = -\frac{1}{\alpha}e^{-\alpha C_t}, \quad u'(C_t) = e^{-\alpha C_t} > 0 \text{ (strict monotonicity),}$$

$$u''(C_t) = -\alpha e^{-\alpha C_t} < 0 \text{ (strict concavity),} \quad u'''(C_t) = \alpha^2 e^{-\alpha C_t} > 0 \text{ (strict convexity of } u'(C_t))$$

Looking at the third derivative we know that the marginal felicity  $u'(\cdot)$  in the CARA case is *strictly convex*.



To make the argument cleanest, consider  $r = \gamma$  (so that  $\beta(1+r) = 1$ ). Consider a *mean-preserving spread* (MPS) applied to  $C_{t+1}$ , which by definition preserves the conditional mean:

$$\underbrace{\mathbb{E}_t \left[ C_{t+1}^{\text{pre MPS}} \right] = \mathbb{E}_t \left[ C_{t+1}^{\text{post MPS}} \right]}_{\text{mean preservation}}$$

Using the Euler equation (with  $r = \gamma$ ): **I do not totally understand the math here.**

$$\underbrace{u'(C_t) = \mathbb{E}_t \left[ u' \left( C_{t+1}^{\text{post MPS}} \right) \right]}_{\text{Euler}} > \underbrace{\mathbb{E}_t \left[ u' \left( C_{t+1}^{\text{pre MPS}} \right) \right]}_{\text{MPS} + u'''(\cdot) > 0} > \underbrace{u' \left( \mathbb{E}_t \left[ C_{t+1}^{\text{pre MPS}} \right] \right)}_{\text{Jensen} + u'''(\cdot) > 0}.$$

From  $u'(C_t) > u' \left( \mathbb{E}_t [C_{t+1}^{\text{pre MPS}}] \right)$  and since  $u''(\cdot) < 0$  (i.e.,  $u'(\cdot)$  is strictly decreasing), it follows that

$$C_t < \mathbb{E}_t \left[ C_{t+1}^{\text{pre MPS}} \right].$$

By the definition of a mean-preserving spread,

$$C_t < \mathbb{E}_t \left[ C_{t+1}^{\text{post MPS}} \right].$$

The same chain of arguments for the certainty-equivalent (quadratic felicity, i.e. linear  $u'(\cdot)$ ) case yields

$$C_t^{\text{c.e.}} = \mathbb{E}_t \left[ C_{t+1}^{\text{c.e., post MPS}} \right].$$

Choosing the *same* MPS across felicity specifications (so that the comparison is sensible) implies

$$\mathbb{E}_t \left[ C_{t+1}^{\text{post MPS}} \right] = \mathbb{E}_t \left[ C_{t+1}^{\text{c.e., post MPS}} \right].$$

Taken together, we conclude:

$$C_t^{\text{c.e.}} > C_t, \quad \text{or equivalently} \quad S_t^{\text{precautionary}} \equiv C_t^{\text{c.e.}} - C_t > 0.$$

**Intuition.** Consider two people, A with *linear* marginal felicity (quadratic utility;  $u''' = 0$ ) and B with *strictly convex* marginal felicity ( $u''' > 0$ ). Start both with a consumption lottery for tomorrow with the same expectation (and, for ease of comparison, the same variance). Expose both to the same mean-preserving spread (riskier lottery while the expected consumption tomorrow remains the same). Person A will not change her consumption today. Person B will reduce her consumption today and save more (precautionary saving).

Income risk is a case where the *third* derivative matters in Economics. Economists call the property  $u'''(\cdot) > 0$  *prudence*. As we have seen, prudence (a property of the felicity function—a primitive) and precautionary saving (an *outcome*) are tightly connected:  $u'''(\cdot) > 0 \Rightarrow$  marginal felicity is strictly convex  $\Rightarrow$  mean-preserving spreads raise  $\mathbb{E}[u'(\cdot)]$  and induce lower  $C_t$  today.

**Addendum.** Risk aversion ( $u''(C) < 0$ ) and prudence ( $u'''(C) > 0$ ) play different roles. Risk aversion governs dislike of risk *in levels*; prudence governs the *precautionary* response of current consumption to future risk. Under  $\beta(1+r) = 1$  and a mean-preserving spread (MPS) in  $C_{t+1}$ :

- If  $u''(C) < 0$  and  $u'''(C) = 0$  (e.g., quadratic utility), the MPS leaves  $C_t$  *unchanged*  $\Rightarrow$  certainty equivalence.
- If  $u''(C) < 0$  and  $u'''(C) > 0$  (e.g., CARA/CRRA), the MPS raises  $\mathbb{E}_t[u'(C_{t+1})]$ , so by Euler  $u'(C_t)$  must rise  $\Rightarrow C_t$  falls  $\Rightarrow$  higher saving (precautionary saving).
- If  $u''(C) < 0$  and  $u'''(C) < 0$  (“imprudence”), the MPS lowers  $\mathbb{E}_t[u'(C_{t+1})]$   $\Rightarrow C_t$  rises  $\Rightarrow$  lower saving.

## Dynamic Programming

**An Initial Remark** This subsection is methodological. We will first apply dynamic programming to the consumption problem, but the techniques are broadly applicable beyond that context. The discussion is slightly more abstract than the substance-oriented material elsewhere in the lecture; the goal is to equip you with a general tool.

### Statement of the Problem

We work in discrete time, indexed by  $t = 0, 1, 2, \dots$  (starting at  $t = 0$  is a convention). Intertemporal aggregation is *additively separable*: lifetime value is the discounted sum of per-period flow payoffs.<sup>2</sup> The horizon is infinite for most of what follows, and—for now—we assume perfect foresight/certainty.

Given an initial state  $x_0 \in \mathbb{R}^n$ , the planner's value function is

$$V(x_0) \equiv \max_{\{u_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(x_t, u_t)$$

subject to the law of motion

$$x_{t+1} = g(x_t, u_t), \quad t = 0, 1, 2, \dots$$

For ease of notation we suppress boldface; it should be clear that  $x_t$  and  $u_t$  can be vectors:

- $x_t \in \mathbb{R}^n$  for all  $t$  (the *state*).
- $u_t \in \mathbb{R}^k$  for all  $t$  (the *control*).
- The discount factor is  $\beta \in (0, 1)$  in the infinite-horizon case (so that the objective is well-defined).
- The per-period flow payoff is a real-valued mapping

$$r : \mathbb{R}^n \times \mathbb{R}^k \longrightarrow \mathbb{R}, \quad (x, u) \mapsto r(x, u).$$

- $V(x_0)$  is measured in the same “units” as  $r$  and represents the maximal discounted value starting at  $x_0$ .

**What does  $r(x_t, u_t)$  capture?** In general  $r$  can depend on both states and controls. In the baseline consumption application we will take

$$r(x_t, u_t) = u(C_t),$$

i.e., felicity depends only on the choice  $C_t$  (do *not* confuse the function  $u(\cdot)$  here with the control vector notation). One could also allow  $r$  to depend on components of the state (e.g., preference shocks), in which case  $r = r(x_t, u_t)$  truly uses both arguments.

**Example (time-varying CRRA).** A simple specification with time-varying risk aversion  $\sigma_t$  is

$$u_t(C_t) = \frac{C_t^{1-\sigma_t}}{1-\sigma_t},$$

where the subscript on  $u_t$  reminds us that felicity parameters may change over time.

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<sup>2</sup>Additive separability means the objective contains no cross-time interaction terms such as  $h(x_t, x_{t+1})$ ; instead it is  $\sum_{t=0}^{\infty} \beta^t r(x_t, u_t)$ .

## States vs. Controls: Formulation Dependence

What one labels as a *state* or a *control* is formulation-dependent. In the baseline consumption formulation, today's consumption  $C_t$  is the control and beginning-of-period assets are a component of the state, e.g.  $x_t = (A_{t-1}, Y_t, \dots)$ . An equivalent formulation replaces  $C_t$  with a flow budget constraint and takes next period's assets  $A_t$  as the control; then  $A_t$  becomes part of tomorrow's state  $x_{t+1}$ . Thus, today's control can become tomorrow's state via the transition  $g$ .

**Minimal example.** Budget constraint with zero interest for simplicity:

$$A_t = A_{t-1} + Y_t - C_t, \quad C_t \geq 0,$$

and per-period payoff  $r(x_t, u_t) = u(C_t)$ . The stochastic object for next period is  $Y_{t+1}$ .

**Formulation A (choose consumption).** State today:  $x_t = (A_{t-1}, Y_t)$ . Control today:  $u_t = C_t \in [0, A_{t-1} + Y_t]$ . Transition (law of motion):

$$x_{t+1} = (A_t, Y_{t+1}) \quad \text{with} \quad A_t \equiv A_{t-1} + Y_t - C_t.$$

Bellman equation:

$$V(A_{t-1}, Y_t) = \max_{0 \leq C \leq A_{t-1} + Y_t} \left\{ u(C) + \beta \mathbb{E}[V(A_{t-1} + Y_t - C, Y_{t+1}) \mid Y_t] \right\}.$$

**Formulation B (choose next period's assets).** Same state today:  $x_t = (A_{t-1}, Y_t)$ . Control today:  $u_t = A_t \in [\underline{A}, A_{t-1} + Y_t]$  (an occasionally imposed borrowing limit  $\underline{A}$ , here assume  $\underline{A} = 0$  for homogeneity with Formulation A). Consumption is now *implied* by the choice of  $A_t$ :

$$C_t \equiv A_{t-1} + Y_t - A_t.$$

Transition:

$$x_{t+1} = (A_t, Y_{t+1}).$$

Bellman equation:

$$V(A_{t-1}, Y_t) = \max_{\underline{A} \leq A' \leq A_{t-1} + Y_t} \left\{ u(A_{t-1} + Y_t - A') + \beta \mathbb{E}[V(A', Y_{t+1}) \mid Y_t] \right\}.$$

**Why this is the same problem.** Formulation A optimizes directly over  $C$  and then *induces*  $A' = A_{t-1} + Y_t - C$ . Formulation B optimizes directly over  $A'$  and then *induces*  $C = A_{t-1} + Y_t - A'$ . Substituting  $A' \leftrightarrow C$  turns one Bellman equation into the other. Hence, “today's control becomes part of tomorrow's state” simply because the control feeds the transition map  $g$ :

$$g(A_{t-1}, Y_t; C_t) = (A_{t-1} + Y_t - C_t, Y_{t+1}) \quad \text{or} \quad g(A_{t-1}, Y_t; A_t) = (A_t, Y_{t+1}).$$

**Mnemonic.** *State* = what you need to carry into the Bellman problem to evaluate feasibility and continuation value next period. *Control* = what you pick today; once picked, it is recorded in the next state via  $x_{t+1} = g(x_t, u_t)$ .

**Regularity: Existence, Boundedness, and Concavity** TB Revised For existence of an optimal policy and well-behaved value functions it is standard to assume: (i) the feasible correspondence  $U(x)$

is nonempty, compact-valued, and (jointly) continuous in  $x$ ; (ii)  $g(x, u)$  is continuous; (iii)  $r(x, u)$  is continuous and bounded above (or  $\beta < 1$  with growth conditions). Under these conditions standard fixed-point arguments (e.g., Blackwell's sufficient conditions) deliver a unique bounded solution to the Bellman equation.

If, in addition, the *feasible set is convex* (for each  $x$ , the set  $\{(x', u) : x' = g(x, u), u \in U(x)\}$  is convex) and the flow payoff  $r$  is *concave* in  $(x, u)$ , then the Bellman operator preserves concavity and the value function  $V$  is concave.

### The Sequence Problem

Given  $x_0 \in \mathbb{R}^n$ ,  $\beta \in (0, 1)$ , a flow payoff  $r : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}$ , and a transition map  $g : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ , the decision maker solves

$$V(x_0) \equiv \max_{\{u_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \quad \text{s.t.} \quad x_{t+1} = g(x_t, u_t), \quad x_0 \text{ given.}$$

This is the *sequence problem*. The objective is *additively separable over time*.

### Dynamic Programming and the Recursive Problem

The idea is to replace the infinite sequence  $\{u_t\}_{t \geq 0}$  with a (time-invariant, in infinite horizon) *policy function*  $u_t = h(x_t)$  so that the path generated by  $x_{t+1} = g(x_t, h(x_t))$  solves the sequence problem. Many macro problems admit such *recursive solutions* (state  $\rightarrow$  choice  $\rightarrow$  next state); with finite memory one can pack lags into  $x_t$ .

**Bellman's optimality principle.** Whatever the current state and decision, the remaining decisions must be optimal for the state induced by today's choice.

**Bellman equation.** Dropping time subscripts and denoting next period with a prime,

$$V(x) = \max_{u \in U(x)} \left\{ r(x, u) + \beta V(g(x, u)) \right\}, \quad x' = g(x, u).$$

Here  $r(x, u)$  is the *flow value* and  $V(g(x, u))$  the (discounted) *continuation value*. A maximizer defines  $u = h(x)$  and then

$$V(x) = r(x, h(x)) + \beta V(g(x, h(x))).$$

Both displays are *functional equations* in the two unknowns  $h(\cdot)$  and  $V(\cdot)$ .

**Existence, Uniqueness, and Iteration** Under standard regularity (nonempty compact feasible sets  $U(x)$ , continuity of  $r$  and  $g$ , and boundedness or  $\beta < 1$  with growth control), the Bellman operator has a unique bounded fixed point  $V$ . With concave  $r$  and a convex feasible set,  $V$  is concave. Value iteration  $V_{j+1} = \mathcal{T}V_j$  with

$$(\mathcal{T}V)(x) = \max_u \{ r(x, u) + \beta V(g(x, u)) \}$$

converges to  $V$  from any bounded continuous  $V_0$ .

### First-Order and Envelope Conditions

Assume interior solutions and differentiability of  $r$  and  $g$ . A necessary condition for  $u = h(x)$  is

$$\frac{\partial r(x, u)}{\partial u} + \beta \frac{\partial V(x')}{\partial x'} \frac{\partial g(x, u)}{\partial u} = 0, \quad x' = g(x, u).$$

Differentiating the Bellman equation w.r.t.  $x$  at the optimum  $u = h(x)$  yields

$$\frac{\partial V(x)}{\partial x} = \frac{\partial r(x, u)}{\partial x} + \frac{\partial r(x, u)}{\partial u} \frac{\partial h(x)}{\partial x} + \beta \frac{\partial V(x')}{\partial x'} \left[ \frac{\partial g(x, u)}{\partial x} + \frac{\partial g(x, u)}{\partial u} \frac{\partial h(x)}{\partial x} \right].$$

If, in addition,  $g$  can be written as  $g(u)$  (or more generally  $\partial g / \partial x = 0$  after re-formulation), the *Benveniste–Scheinkman envelope condition* simplifies to

$$\boxed{\frac{\partial V(x)}{\partial x} = \frac{\partial r(x, u)}{\partial x} \quad \text{at } u = h(x).}$$

Substituting the envelope condition back into the FOC gives the standard *Euler equation* that contains  $h(\cdot)$  but no  $V(\cdot)$ :

$$\frac{\partial r(x, u)}{\partial u} + \beta \frac{\partial r(x', u')}{\partial x'} \frac{\partial g(x, u)}{\partial u} = 0, \quad x' = g(x, u), \quad u' = h(x').$$

This functional equation in the policy  $h$  can be attacked with analytical or numerical methods.

**Remark.** Before an example:

- The FOC will almost always exist, as long as we have differentiability of the functions involved.
- However, the FOC in a useful way, both economically and for numerical purposes, i.e., as an Euler equation without the value function, does not always exist; only when the Benveniste-Scheinkman formula holds.
- If this is not the case, the solution has to come directly from the Bellman equation (which is also a function equation, but in  $V(x)$ ).

## Example I: Consumption–Saving

We revisit the standard consumption–saving problem under perfect foresight. Let assets be  $A$ , income  $Y$ , interest rate  $r$ , and period felicity  $u(\cdot)$ .

$$r(x, u) = u(C), \quad g(x, u) : A' = (1 + r)A + Y - C,$$

with  $Y$  exogenous, control variable  $C$ , and state variables  $(A, Y)$ . State the problem at  $t = 0$  as

$$\max_{\{C_t, A_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t) \quad \text{s.t.} \quad A_{t+1} = (1 + r)A_t + Y_t - C_t, \quad A_0 \text{ given.}$$

Equivalently, using forward substitution,

$$\sum_{t=0}^{\infty} \frac{1}{(1 + r)^t} C_t = A_0(1 + r) + \sum_{t=0}^{\infty} \frac{1}{(1 + r)^t} Y_t.$$

With state  $(A, Y)$  and control  $C$ ,

$$V(A, Y) = \max_{C, A'} \{u(C) + \beta V(A', Y')\} \quad \text{s.t.} \quad A' = (1 + r)A + Y - C.$$

Plugging the constraint,

$$V(A, Y) = \max_C u(C) + \beta V((1 + r)A + Y - C, Y').$$

The FOC w.r.t.  $C$  is

$$u'(C) - \beta V_A(A', Y') = 0 \iff u'(C) = \beta V_A(A', Y'),$$

which is *not yet* an Euler equation because it still contains  $V_A(\cdot)$ . Because  $A' = (1+r)A + Y - C(A)$  depends on  $A$ , the Benveniste–Scheinkman envelope in its simple form does not apply directly. Differentiating the Bellman equation at the optimum  $C = C(A, Y)$  gives

$$\frac{\partial V(A, Y)}{\partial A} = u'(C) \frac{\partial C}{\partial A} - \beta V_A(A', Y') \frac{\partial C}{\partial A} + \underbrace{\beta V_A(A', Y') (1+r)}_{=u'(C)} = u'(C)(1+r).$$

Using the FOC  $u'(C) = \beta V_A(A', Y')$  to cancel the bracketed terms yields

$$\frac{\partial V(A, Y)}{\partial A} = \underbrace{\beta V_A(A', Y') (1+r)}_{=u'(C)} \implies V_A(A', Y') = u'(C')(1+r) \text{ (holds tomorrow).}$$

Substituting back into the FOC produces the familiar Euler equation:

$$\boxed{u'(C) = \beta(1+r) u'(C')}.$$

### Equivalent Formulation Making Envelope Immediate

We can instead take next period's assets as the control,  $u = A'$ , and write

$$V(A, Y) = \max_{A'} u(A(1+r) + Y - A') + \beta V(A', Y').$$

Here  $g(x, u)$  is the identity in  $u$ <sup>3</sup>

so the Benveniste–Scheinkman envelope gives

$$V_A(A, Y) = u'(C) (1+r).$$

The FOC  $-u'(C) + \beta V_A(A', Y') = 0$  then implies

$$V_A(A', Y') = \frac{u'(C)}{\beta} \implies u'(C) = \beta(1+r) u'(C'),$$

the same Euler condition as above.

**Remarks** Three important points:

- $Y$  is technically part of the state; under perfect foresight it can be suppressed in notation without loss.
- Both derivations highlight the “formulation dependence” of states/controls: taking  $C$  or  $A'$  as the control leads to the same optimality condition, but different algebra.

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<sup>3</sup>“ $g(x, u)$  is the identity at  $u$ ” means that, when taking  $A'$  as the control, the transition is  $x' = g(x, u) = (A', Y')$ : the first component of the future state is exactly the control, so that  $\partial A' / \partial u = 1$  and, crucially for the envelope with respect to  $A$ ,  $\partial x' / \partial A = (\partial A' / \partial A, \partial Y' / \partial A) = (0, 0)$  (because  $A' = u$  and  $Y'$  is exogenous). Intuition: if I choose  $C$  as the control, the constraint  $A' = (1+r)A + Y - C$  makes  $A'$  *dependent* on  $A$ ; then when deriving  $V$  with respect to  $A$ , a “cross” term appears via  $A'$  (and must be canceled using the FOC). On the other hand, if I choose  $A'$  as the control variable, the future state no longer changes when I vary  $A$  in that derivative: there is no cross effect and the envelope condition is immediate,  $V_A(A, Y) = u'(C)(1+r)$ .

- Under the usual concavity/compactness conditions, the Bellman problem admits a unique time-invariant value function and policy.

## Statement of the Problem II

### Sequence problem

Given an initial state  $x_0$ , consider

$$V(x_0) \equiv \max_{\{u_t\}_{t \geq 0}} E_0 \sum_{t=0}^{\infty} \beta^t r(x_t, u_t)$$

subject to

$$x_{t+1} = g(x_t, u_t, \varepsilon_{t+1}), \quad \{\varepsilon_t\} \text{ i.i.d. with distribution } F, \quad x_0 \in \mathbb{R}^n \text{ given.}$$

### Recursive formulation (Bellman equation)

$$V(x) = \max_u \left\{ r(x, u) + \beta \mathbb{E} [V(g(x, u, \varepsilon)) | x] \right\},$$

where the conditional expectation is

$$\mathbb{E} [V(g(x, u, \varepsilon)) | x] \equiv \int V(g(x, u, \varepsilon)) dF(\varepsilon).$$

**First-order condition** For an interior optimum,

$$\frac{\partial r(x, u)}{\partial u} + \beta E \left[ \frac{\partial V(x')}{\partial x'} \frac{\partial g(x, u, \varepsilon)}{\partial u} \middle| x \right] = 0.$$

**Envelope (Benveniste–Scheinkman).** If  $\partial g(x, u, \varepsilon)/\partial x = 0$ , the usual reasoning yields

$$\frac{\partial V(x)}{\partial x} = \frac{\partial r(x, u)}{\partial x}.$$

And you get

$$\frac{\partial r(x, u)}{\partial u} + \beta E \left[ \frac{\partial r(x', u')}{\partial x'} \frac{\partial g(x, u, \varepsilon)}{\partial u} \middle| x \right] = 0.$$

## Example II: Consumption–Savings with Stochastic Income

Let current assets  $A$  and current income  $Y$  be the state. The Bellman problem (taking next period's assets  $A'$  as the control) is

$$V(A, Y) = \max_{A'} u(A(1+r) + Y - A') + \beta \mathbb{E} [V(A', Y') | Y]. \quad (\text{Bellman})$$

**Why is  $Y$  a state?** It carries information about  $Y'$  (e.g., if  $Y$  follows a Markov process), so it matters for expectations even though today it only enters the budget through resources.

**Envelope (identity in the control).** With  $A'$  as the control, the transition for the asset component is the identity: the choice  $A'$  is next period's asset state. Hence, when differentiating with respect to  $A$ , we hold  $A'$  fixed and obtain the envelope directly:

$$V_A(A, Y) = u_C(C)(1+r), \quad C \equiv A(1+r) + Y - A'. \quad (\text{Envelope})$$

Note: If instead  $C$  were the control,  $A' = (1+r)A + Y - C$  depends on  $A$  and a cross term would appear; it cancels with the FOC below.

**FOC.**

$$-u_C(C) + \beta \mathbb{E}[V_A(A', Y') | Y] = 0. \quad (\text{FOC})$$

Using the envelope *one period ahead*,  $V_A(A', Y') = (1+r)u_C(C')$ , we obtain the familiar stochastic Euler equation:

$$u_C(C) = \beta(1+r) \mathbb{E}[u_C(C') | Y]. \quad (\text{Euler})$$

**What not to confuse (derivative bookkeeping).**

- $\frac{\partial}{\partial A'} \{u(A(1+r) + Y - A')\} = -u_C(C)$ .
- $\frac{\partial}{\partial A} \{u(A(1+r) + Y - A')\} = u_C(C)(1+r)$  is the derivative wrt the *current* state  $A$ , not wrt the *control*  $A'$ . Do not mix the role of  $A'$  as control with the role of  $A$  as (future) state variable.
- Hence  $\frac{\partial r(x', u')}{\partial x'}$  in the slide's abstract notation corresponds here to  $(1+r)u_C(C')$ .
- If  $g$  denotes the law of motion for the asset state, then with  $A'$  as control we have  $g(A', \cdot) = A'$  (identity), so  $\partial g / \partial A' = 1$ .

Combining the FOC with the one-step-ahead envelope gives

$$-u_C(C) + \beta(1+r) \mathbb{E}[u_C(C') | Y] = 0 \iff u_C(C) = \beta(1+r) \mathbb{E}[u_C(C') | Y].$$

To avoid any confusion about what to plug into the general Euler equation, start from the FOC

$$-u_C(C) + \beta \mathbb{E}[V_A(A', Y') | Y] = 0,$$

then apply Benveniste–Scheinkman to get  $V_A(A, Y) = u_C(C)(1+r)$  and shift it forward to  $(A', Y')$  to obtain  $V_A(A', Y') = (1+r)u_C(C')$ . Substituting back delivers the Euler equation.

If  $Y$  is i.i.d., define *cash on hand*

$$Z \equiv A(1+r) + Y.$$

Now  $Y$  does not forecast  $Y'$  and can be dropped from the state. The Bellman equation collapses to a single state variable:

$$V(Z) = \max_{A'} u(Z - A') + \beta \mathbb{E}[V(Z')], \quad Z' = A'(1+r) + Y'.$$

**Economic takeaway.**  $Y$  entered the state only because it predicted  $Y'$ . When that forecasting role disappears (i.i.d. case), current resources are fully summarized by  $Z$ . In general, look for transformations that compress forecasting information into fewer state variables (e.g., cash-on-hand with i.i.d. shocks; companion-form stacking when only finitely many lags matter).

## Summary so far

So far, Dynamic Programming has delivered (via Bellman + Benveniste–Scheinkman) the Euler equation. The next step is more substantive: can we compute/characterize the value function  $V(x)$  and the policy  $h(x)$ ? To push in that direction, we will need a bit more mathematical background.

# Contraction Mapping for the Bellman Operator

## Bellman operator as a map on functions

Let  $X$  be the state space and  $U$  the set of feasible controls at each  $x \in X$ . Given:

- a bounded one-period return (flow utility)  $r : X \times U \rightarrow \mathbb{R}$ ,
- a transition  $g : X \times U \times E \rightarrow X$  with shock  $\varepsilon \in E$ ,
- a discount factor  $\beta \in (0, 1)$ ,
- and a conditional expectation operator  $\mathbb{E}[\cdot \mid x]$  over  $\varepsilon$ ,

define the *Bellman operator*  $T : \mathcal{S} \rightarrow \mathcal{S}$  by

$$[T(V)](x) \equiv \max_{u \in U(x)} \left\{ r(x, u) + \beta \mathbb{E} [ V(g(x, u, \varepsilon)) \mid x ] \right\}.$$

Here  $\mathcal{S}$  is a set of real-valued functions on  $X$  (specified below).

**Interpretation.**  $T$  takes a candidate continuation value  $V$  and produces a new function: (i) forecast next period's value through  $g$  and expectation, (ii) discount, (iii) add current flow  $r$ , (iv) maximize over  $u$ . The Bellman equation is the fixed-point condition

$$T(V) = V.$$

Let  $\mathcal{S}$  be the space of bounded real functions on  $X$ , and endow it with the sup norm

$$\|V\|_{\infty} \equiv \sup_{x \in X} |V(x)|, \quad d(V, \tilde{V}) = \|V - \tilde{V}\|_{\infty}.$$

Then  $(\mathcal{S}, d)$  is a complete metric space (the space of bounded functions is complete under the sup norm).

## Contraction Mapping Theorem (Banach)

**Theorem 1 (CMT).** Suppose  $T : \mathcal{S} \rightarrow \mathcal{S}$  is a contraction with modulus  $\beta \in (0, 1)$  under the metric  $d$ , i.e.

$$d(TV, T\tilde{V}) \leq \beta d(V, \tilde{V}) \quad \forall V, \tilde{V} \in \mathcal{S}.$$

Then:

1. **Existence and Uniqueness.**  $T$  has exactly one fixed point  $V^* \in \mathcal{S}$  with  $T(V^*) = V^*$ .
2. **Convergence of Value Iteration.** For any  $V_0 \in \mathcal{S}$ , the sequence  $V_{n+1} = T(V_n)$  satisfies  $V_n \rightarrow V^*$ .
3. **Geometric error bound.** For all  $n \geq 0$ ,

$$d(V_n, V^*) \leq \beta^n d(V_0, V^*).$$

4. **Operational bound (no  $V^*$ ).** For all  $n \geq 0$ ,

$$d(V_n, V^*) \leq \frac{1}{1 - \beta} d(V_n, V_{n+1}).$$

This bound is convenient because it only involves iterates you can compute.

**Intuition** A contraction shrinks distances by a fixed factor  $\beta < 1$ . Iterating  $T$  keeps shrinking the gap to the (unique) fixed point. The “geometry” is that  $T$  pulls every candidate value function closer to  $V^*$  by at least a factor  $\beta$  each step.

## When is the Bellman operator a contraction?

In our setting with  $\|\cdot\|_\infty$ ,

$$\|TV - T\tilde{V}\|_\infty = \sup_x \left| \max_u \{r(x, u) + \beta \mathbb{E}[V(x')|x]\} - \max_u \{r(x, u) + \beta \mathbb{E}[\tilde{V}(x')|x]\} \right| \leq \beta \|V - \tilde{V}\|_\infty,$$

because the  $r$  cancels and the max is 1-Lipschitz:  $|\max_i a_i - \max_i b_i| \leq \max_i |a_i - b_i|$ . Moreover, conditional expectation is a contraction in sup norm:  $\sup_x |\mathbb{E}[V(x') - \tilde{V}(x') | x]| \leq \|V - \tilde{V}\|_\infty$ . Hence with  $\beta \in (0, 1)$ ,  $T$  is a contraction on  $(\mathcal{S}, \|\cdot\|_\infty)$ .

**Remark** (Quick checklist). To argue  $T$  is a contraction under  $\|\cdot\|_\infty$  it suffices that: (i)  $r$  is bounded and  $U(x)$  nonempty; (ii)  $\mathbb{E}[\cdot | x]$  maps bounded functions to bounded functions and is 1-Lipschitz in sup norm; (iii)  $\beta \in (0, 1)$ . In practice this is exactly the standard consumption–savings setup: bounded (or suitably normalized) returns, discounting, and well-behaved expectations.

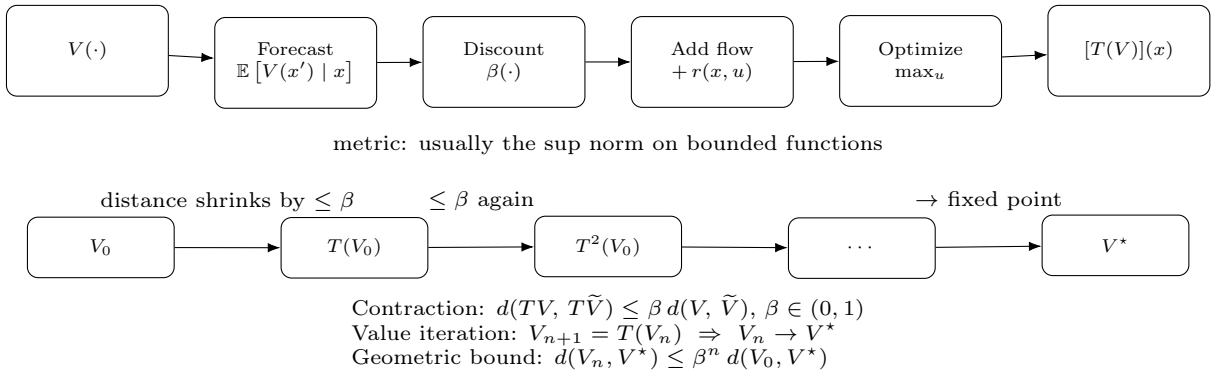
## Why this is useful

If  $T$  is a contraction with modulus  $\beta$ :

- **Existence** of a solution to the Bellman equation.
- **Uniqueness** of that solution.
- **Computation:** value iteration  $V_{n+1} = T(V_n)$  converges to  $V^*$  from any  $V_0$ , with a known rate ( $\beta$ ) and practical error bounds.

In applications we often just iterate and see whether  $V_n$  stabilizes. Convergence suggests (and under contraction guarantees) existence and uniqueness. A simple robustness check is to try different  $V_0$  and verify they converge to the same limit; monitor  $d(V_n, V_{n+1})$  to certify accuracy via the operational bound.

Figure 3: Bellman operator as a pipeline (top) and as a contraction driving value iteration to the unique fixed point  $V^*$  (bottom).



## Blackwell's sufficient conditions

Let  $B(X)$  be the set of bounded, real-valued functions on a state space  $X$  and

$$d(V, \tilde{V}) = \|V - \tilde{V}\|_\infty \equiv \sup_{x \in X} |V(x) - \tilde{V}(x)|$$

be the sup-norm metric. For  $V, \tilde{V} \in B(X)$ , write  $V \leq \tilde{V}$  to mean  $V(x) \leq \tilde{V}(x)$  for all  $x$ ; and for  $a \in \mathbb{R}$ ,  $(V + a)(x) \equiv V(x) + a$ .

**Theorem 2** (Blackwell). *Let  $T : B(X) \rightarrow B(X)$  satisfy:*

1. **Monotonicity:** *if  $V \leq \tilde{V}$  then  $TV \leq T\tilde{V}$ .*
2. **Discounting:** *there exists  $\beta \in (0, 1)$  such that for all  $V \in B(X)$  and  $a \geq 0$ ,*

$$T(V + a) \leq T(V) + \beta a.$$

*Then  $T$  is a contraction on  $(B(X), \|\cdot\|_\infty)$  with modulus  $\beta$ :*

$$d(TV, T\tilde{V}) \leq \beta d(V, \tilde{V}) \quad \forall V, \tilde{V} \in B(X).$$

*Consequently,  $T$  has a unique fixed point  $V^*$ , and for any  $V_0$  the iterates  $V_{n+1} = T(V_n)$  converge to  $V^*$  with geometric rate  $\beta$ .*

*Proof (three short steps). Step 1.* For any  $V, \tilde{V}$ ,

$$V(x) = \tilde{V}(x) + (V(x) - \tilde{V}(x)) \leq \tilde{V}(x) + \sup_{y \in X} |V(y) - \tilde{V}(y)| = \tilde{V}(x) + d(V, \tilde{V}),$$

so  $V \leq \tilde{V} + d(V, \tilde{V})$ .

*Step 2.* By monotonicity and discounting,

$$TV \leq T(\tilde{V} + d(V, \tilde{V})) \leq T\tilde{V} + \beta d(V, \tilde{V}).$$

Swapping  $V, \tilde{V}$  gives  $T\tilde{V} \leq TV + \beta d(V, \tilde{V})$ .

*Step 3.* The two inequalities imply, for all  $x$ ,

$$|TV(x) - T\tilde{V}(x)| \leq \beta d(V, \tilde{V}),$$

hence  $d(TV, T\tilde{V}) \leq \beta d(V, \tilde{V})$ . □

**Intuition.** Monotonicity: improving continuation values cannot hurt today's evaluation. Discounting: adding a constant  $a$  to continuation values moves  $T$  by at most  $\beta a$ . Together they bound  $T$ 's "sensitivity" to changes in  $V$  by the factor  $\beta < 1$ , which is exactly the contraction property.

## Garbage In—Garbage Out (GIGO) for Bellman fixed points

Let  $(\mathcal{S}, d)$  be a complete metric space of functions (e.g.  $B(X)$  with the sup norm). A subset  $\mathcal{S}' \subset \mathcal{S}$  is *closed* if every convergent sequence in  $\mathcal{S}'$  has its limit in  $\mathcal{S}'$ .

**Closedness facts (with  $\|\cdot\|_\infty$ ).** (i) Bounded and continuous functions  $X \rightarrow \mathbb{R}$  form a closed set in  $B(X)$ . (ii) On an interval  $[a, b]$ , bounded and increasing functions  $[a, b] \rightarrow \mathbb{R}$  form a closed set. (iii)

On  $[a, b]$ , bounded and convex functions form a closed set. (Each follows since uniform limits preserve continuity, monotonicity, and convexity.)

**Lemma (GIGO).** *Let  $T : \mathcal{S} \rightarrow \mathcal{S}$  be a contraction with modulus  $\beta \in (0, 1)$  and let  $V^*$  be its unique fixed point. If  $\emptyset \neq \mathcal{S}' \subset \mathcal{S}$  is closed and invariant under  $T$  (i.e.  $T(\mathcal{S}') \subset \mathcal{S}'$ ), then  $V^* \in \mathcal{S}'$ . More generally, if  $T(\mathcal{S}') \subset \mathcal{S}'' \subset \mathcal{S}'$  with  $\mathcal{S}''$  closed, then  $V^* \in \mathcal{S}''$ .*

*Proof.* Pick any  $V_0 \in \mathcal{S}'$ . By invariance,  $V_n \equiv T^n(V_0) \in \mathcal{S}'$  for all  $n$ . By contraction,  $V_n \rightarrow V^*$  in  $d$ . Closedness of  $\mathcal{S}'$  gives  $V^* \in \mathcal{S}'$  (and similarly for  $\mathcal{S}''$ ).  $\square$

**Applications.** Let  $T$  be the Bellman operator on  $(B(X), \|\cdot\|_\infty)$ .

- *Continuity:* If  $T$  maps bounded continuous functions into bounded continuous functions, then  $V^*$  is continuous.
- *Monotonicity:* On  $[a, b]$ , if  $T$  maps bounded increasing functions into increasing functions, then  $V^*$  is increasing.
- *Convexity/concavity:* On  $[a, b]$ , if  $T$  maps bounded convex (resp. concave) functions into convex (resp. concave) functions, then  $V^*$  is convex (resp. concave).

**Strict versions.** If, for every increasing  $V$ , the image  $T(V)$  is *strictly* increasing, then starting from any increasing  $V_0$  the sequence  $T^n(V_0)$  is strictly increasing for all  $n$  and the limit  $V^*$  is strictly increasing. (Analogous statements hold for strict convexity/concavity.)

**Intuition.** Start with a “good” guess  $V_0$  (continuous, increasing, convex,  $\dots$ ). If  $T$  never leaves that class (invariance) and value iteration converges, the limit inherits the good property by closedness. Garbage in  $\Rightarrow$  garbage out; good in  $\Rightarrow$  good out.

## Equivalence of the sequence and the recursive problems

Let states evolve by  $x_{t+1} = g(x_t, u_t, \varepsilon_{t+1})$  and per-period payoff be  $r(x_t, u_t)$  with discount  $\beta \in (0, 1)$ .

**Finite horizon (perfect foresight for simplicity).** Fix a terminal value  $V_{T+1}(\cdot)$ . Define recursively

$$V_t(x) = \max_u \left\{ r(x, u) + \beta V_{t+1}(g(x, u)) \right\}, \quad t = T, \dots, 0.$$

By successive substitution,

$$V_0(x_0) = \max_{\{u_0, \dots, u_T\}} \sum_{t=0}^T \beta^t r(x_t, u_t) + \beta^{T+1} V_{T+1}(x_{T+1}),$$

subject to  $x_{t+1} = g(x_t, u_t)$ . Hence the recursive and sequence problems are the *same* object written differently; there is no fixed point (values are time-indexed).

**Infinite horizon.** Let  $V$  solve the Bellman equation

$$V(x) = \max_u \left\{ r(x, u) + \beta \mathbb{E} [V(g(x, u, \varepsilon')) \mid x] \right\}.$$

Iterating the recursion  $T+1$  times along any feasible policy  $\{u_t\}$  yields the telescoping representation

$$V(x_0) = \sup_{\{u_t\}} \mathbb{E} \left[ \sum_{t=0}^T \beta^t r(x_t, u_t) + \beta^{T+1} V(x_{T+1}) \mid x_0 \right].$$

If the *tail term* vanishes,

$$\lim_{T \rightarrow \infty} \beta^{T+1} \mathbb{E}[|V(x_{T+1})| \mid x_0] = 0,$$

then

$$V(x_0) = \sup_{\{u_t\}} \mathbb{E} \left[ \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \mid x_0 \right],$$

so the sequence and recursive formulations are equivalent.

**Easy sufficient condition (practical).** If  $|r(x, u)| \leq \bar{r} < \infty$  and  $\beta \in (0, 1)$ , the fixed point satisfies

$$\|V\|_{\infty} \leq \frac{\bar{r}}{1 - \beta},$$

hence  $\beta^{T+1} \mathbb{E}[|V(x_{T+1})|] \leq \beta^{T+1} \|V\|_{\infty} \rightarrow 0$  and equivalence follows. (Variants: any bound of the form  $|V(x)| \leq K$  suffices; in growth models one uses no-Ponzi/borrowing constraints or weighted norms to ensure the tail goes to zero.)

**Intuition.** Backward induction (finite  $T$ ) or repeated substitution (infinite horizon) “unwraps” the Bellman equation into the sequence objective plus a single terminal piece. Under discounting, the terminal piece dies out; what remains is exactly the sequence problem.

## Numerical solution: value function iteration

Given the Bellman operator

$$[T(V)](x) \equiv \max_{u \in U(x)} \left\{ r(x, u) + \beta \mathbb{E}[V(g(x, u, \varepsilon)) \mid x] \right\},$$

value function iteration (VFI) computes  $V$  by fixed-point iteration.

**Algorithm.** Choose any  $V_0$  (e.g.,  $V_0 \equiv 0$ ) and for  $j = 0, 1, 2, \dots$  update

$$V_{j+1}(x) = \max_{u \in U(x)} \left\{ r(x, u) + \beta \mathbb{E}[V_j(g(x, u, \varepsilon)) \mid x] \right\}.$$

Stop when  $\|V_{j+1} - V_j\|_{\infty} < \text{tol}$ . Under the contraction conditions,  $V_j \rightarrow V^*$  and the *operational* error bound holds:

$$\|V^* - V_j\|_{\infty} \leq \frac{\beta}{1 - \beta} \|V_j - V_{j-1}\|_{\infty}.$$

**Practical remarks.** (i) With a finite state space, the update is a finite-dimensional fixed point in  $\mathbb{R}^n$ ; contraction carries over and convergence is guaranteed. (ii) With continuous states, one works on a grid and interpolates (e.g., linear, monotone cubic). Contraction applies to the *true* Bellman operator; the discretized/interpolated operator need not be a contraction—if convergence is slow/unstable, refine the grid or use structure-preserving interpolation (monotonicity/concavity). (iii) Expectations: use quadrature or simulation consistent with the shock process and the interpolation scheme.

**Howard policy improvement / modified policy iteration.** Let  $h_j$  be the maximizer (policy) associated with  $V_j$ . Rather than re-maximizing every step, *evaluate* the current policy several times before the next maximization:

$$V_{j,\ell+1}(x) = r(x, h_j(x)) + \beta \mathbb{E}[V_{j,\ell}(g(x, h_j(x), \varepsilon)) \mid x], \quad \ell = 0, \dots, L-1,$$

starting from  $V_{j,0} = V_j$ ; then set  $V_{j+1} = V_{j,L}$  and update  $h_{j+1}$ . This typically yields large speedups because costly maximizations are done only every  $L$  iterations and values improve monotonically.

## Euler-equation methods (policy approximation)

When the first-order approach is valid, the optimal policy  $u = h(x)$  satisfies the functional equation

$$R(x, h(x)) \equiv \left. \frac{\partial r(x, u)}{\partial u} \right|_{u=h(x)} + \beta \mathbb{E} \left[ \left. \frac{\partial r(x', u')}{\partial x'} \frac{\partial g(x, u, \varepsilon)}{\partial u} \right|_{u=h(x), u'=h(x')} \middle| x \right] = 0.$$

Approximate  $h(\cdot)$  in a parametric family  $h(x; \theta)$  and choose  $\theta$  so that  $R(x, h(x; \theta)) \approx 0$ :

- *Collocation*: enforce  $R(\xi_k, h(\xi_k; \theta)) = 0$  at nodes  $\{\xi_k\}$ .
- *Residual minimization*: choose  $\theta$  to minimize  $\sum_k w_k R(\xi_k, h(\xi_k; \theta))^2$  (deterministic or simulation-based).

This targets the *policy* directly. It is fast but requires (a) valid FOCs (concavity, interior solutions), (b) careful treatment of constraints (e.g., occasionally binding), and (c) accurate expectations consistent with the parameterization.

## Summary and guidance

- VFI solves for  $V(\cdot)$ ; the policy follows from the argmax. It is robust (works with kinks/non-differentiabilities) but can be slow.
- Euler-equation methods solve for  $h(\cdot)$ ; very fast when FOCs are valid and constraints are well handled.
- Once you have either  $V$  or  $h$ , the other is essentially implied.
- For speed with reliability: start with VFI on a coarse grid to get a good initial policy, then switch to policy improvement and/or an Euler residual refinement.

## Closed-form solutions: when and why

Only a handful of dynamic programs admit paper-and-pencil solutions. Two canonical ones: (i) Brock–Mirman (optimal growth, Cobb–Douglas output,  $\delta = 1$ , log utility); (ii) consumption–saving with HARA preferences under special income processes. In both, the strategy is *guess and verify*: posit a parametric form for  $V$  (and often for the policy) and check optimality.

**Why useful?** (i) Solving by hand builds intuition for numerical algorithms (what objects the Bellman update is pushing). (ii) These are limit cases of richer models; their closed forms are excellent initial guesses for value/policy iteration.

## Brock–Mirman (deterministic, capital only)

Planner chooses  $\{C_t, K_{t+1}\}_{t \geq 0}$  to maximize

$$V(K_0) \equiv \max_{\{C_t, K_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \log C_t \quad \text{s.t.} \quad K_{t+1} = K_t^\alpha - C_t, \quad C_t \geq 0, \quad K_{t+1} \geq 0,$$

with  $\beta \in (0, 1)$  and  $\alpha \in (0, 1)$ . Resource feasibility implies  $K_{t+1} \in [0, K_t^\alpha]$ .

**Recursive form.**

$$V(K) = \max_{K' \in [0, K^\alpha]} \left\{ \log(K^\alpha - K') + \beta V(K') \right\}.$$

**Guess.** Try  $V(K) = A + B \log K$  with constants  $(A, B)$  to be determined.

**Verification.** FOC of the Bellman problem under the guess:

$$-\frac{1}{K^\alpha - K'} + \beta \frac{B}{K'} = 0 \quad \implies \quad K' = \frac{\beta B}{1 + \beta B} K^\alpha \equiv s K^\alpha.$$

Thus the policy is *proportional savings*. Plug the Euler equation from the primal problem,

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} \alpha K_{t+1}^{\alpha-1},$$

together with  $C_t = (1 - s)K_t^\alpha$ ,  $K_{t+1} = sK_t^\alpha$ , to obtain  $s = \alpha\beta$ . Equating the two expressions for  $s$  pins down  $B$ :

$$\frac{\beta B}{1 + \beta B} = \alpha\beta \quad \implies \quad B = \frac{\alpha}{1 - \alpha\beta}.$$

Finally, identify  $A$  by substituting the optimal policy ( $K' = \alpha\beta K^\alpha$ ,  $C = (1 - \alpha\beta)K^\alpha$ ) into the Bellman equation:

$$A = \frac{\log(1 - \alpha\beta) + \beta B \log(\alpha\beta)}{1 - \beta} \quad \text{with} \quad B = \frac{\alpha}{1 - \alpha\beta}.$$

**Closed forms.**

$$\boxed{K_{t+1} = \alpha\beta K_t^\alpha, \quad C_t = (1 - \alpha\beta) K_t^\alpha}$$

$$\boxed{V(K) = A + B \log K, \quad B = \frac{\alpha}{1 - \alpha\beta}, \quad A = \frac{\log(1 - \alpha\beta) + \beta B \log(\alpha\beta)}{1 - \beta}}$$

Feasibility requires only  $\alpha\beta \in (0, 1)$ , which holds given  $\alpha, \beta \in (0, 1)$ .

**Notes.** (i) The proportional-savings rule is the hallmark of log utility plus Cobb–Douglas with full depreciation. (ii) With i.i.d. multiplicative shocks  $Y_t = Z_t K_t^\alpha$  and  $\mathbb{E}[\log Z_t] = 0$ , the same guess delivers  $K_{t+1} = \alpha\beta Z_t K_t^\alpha$  and the value function gains an  $\mathbb{E}[\log Z]$  term in  $A$ ; the *share*  $\alpha\beta$  is unchanged. (iii) These formulas are excellent warm starts for VFI or policy iteration in nearby calibrated models.

**Cookie-Cutter:** Guess and verify under a certain value function form

**Setup.** State  $x$  (e.g. assets  $A$ ), control  $y$  (e.g. next assets  $A'$ ).

Bellman:  $V(x) = \max_y \{ u(x, y) + \beta V(f(x, y)) \}$  with easy-to-eliminate  $y$  via the flow budget.

1) **Guess.** Pick a functional form closed under the operator, e.g.

$$V(x) = \alpha + \kappa g(x)^{1-\rho} \quad \text{or} \quad V(x) = \alpha + \kappa \log g(x).$$

**2) Reduce.** Eliminate consumption using the FBC (e.g.  $C = A - A'/R$ ), so the Bellman is  $V(A) = \max_{A'} \{u(C(A, A')) + \beta V(A')\}$ .

**3) FOC/Policy.** Compute  $\partial/\partial A'$ : solve for  $A' = h(A)$  (often affine). If messy, do a change of variables that linearizes marginal utility (e.g. HARA:  $Z \equiv \frac{a}{\sigma}C + b$ ).

**4) Euler pin-down.** Use envelope  $V'(A) = u'(C)$  to turn FOC into the Euler  $u'(C_t) = \beta R u'(C_{t+1})$ ; this pins the slope/ratio of the policy (e.g.  $Z_{t+1} = \psi Z_t$ ,  $\psi = (\beta R)^{1/\sigma}$ ).

**5) Verify (levels).** Plug  $h(A)$  and the guessed  $V(\cdot)$  back into Bellman:

$$u(C(A, h(A))) + \beta V(h(A)) \stackrel{!}{=} V(A) \quad \forall A.$$

Match coefficients to solve unknown constants  $(\alpha, \kappa, \dots)$ .

**6) Checks.** (i) Domain (e.g.  $Z > 0$ ), (ii) TVC, (iii) corner solutions, (iv) knife-edges ( $\beta R = 1$ ,  $\rho \rightarrow 1$ ), (v) monotonicity/concavity as needed.

## Consumption–savings with $Y = 0$ (log utility)

We revisit the single-agent problem with no exogenous income,  $Y = 0$ . Preferences are  $u(C) = \log C$  and assets evolve with the flow budget constraint

$$A' = A(1 + r) - C.$$

**Idea (interpretation).** If human capital can be traded at the market rate of return, we can subsume labor income into asset income and set  $Y = 0$ . Perfect foresight is not essential for the main insight.

**Bellman equation and guess.**

$$V(A) = \max_{C>0} \{ \log C + \beta V(A') \}, \quad A' = A(1 + r) - C.$$

Guess a linear–affine transformation of felicity:

$$V(A) = E + F \log A.$$

**Verification.** Plugging the guess,

$$E + F \log A = \max_{C>0} \{ \log C + \beta E + \beta F \log((1 + r)A - C) \}.$$

FOC:

$$\frac{1}{C} = \frac{\beta F}{(1 + r)A - C} \implies C = \frac{1 + r}{1 + \beta F} A.$$

Substitute back and compare coefficients on  $\log A$ :

$$F = 1 + \beta F \implies F = \frac{1}{1 - \beta}.$$

Therefore the (policy) consumption rule is

$$\boxed{C = (1 - \beta)(1 + r) A}.$$

*Special case:* if  $\beta = \frac{1}{1+r}$ , then  $C = rA$ . *Note on timing:* under the alternative timing  $A' = (A - C)(1 + r)$  the rule becomes  $C = (1 - \beta)A$ .

**Constants.** The level  $E$  follows from the constant terms in the Bellman equation (uninteresting for policy and comparative statics).

**Generalization: HARA felicitities.** The same logic extends to HARA utility; the value function is a linear–affine transform of felicity. HARA is

$$u(C) = \frac{\sigma}{1-\sigma} \left( \frac{aC}{\sigma} + b \right)^{1-\sigma},$$

with parameters  $(\sigma, a, b)$ . Notable members:

- **Linear:**  $\sigma = 0 \Rightarrow u(C) = aC$  (limit case).
- **Quadratic:**  $\sigma = -1 \Rightarrow u(C) = -\frac{1}{2}(-aC + b)^2 = abC - \frac{1}{2}b^2 - \frac{1}{2}a^2C^2$ .
- **CRRA:**  $a = \sigma, b = 0 \Rightarrow u(C) = \frac{\sigma}{1-\sigma} C^{1-\sigma}$  (the front factor is irrelevant).

In each HARA case (with  $Y = 0$ ),  $V(A) = E + F \cdot \text{felicity}(A)$  with  $F$  pinned down by the same coefficient-matching argument.

**CARA as a limiting case of HARA.** CARA utility can be derived as a limit of the HARA class. Set  $b = 1$  and take  $\sigma \rightarrow \infty$ . Then:

$$\lim_{\sigma \rightarrow \infty} \frac{\sigma}{1-\sigma} \left( \frac{aC}{\sigma} + 1 \right)^{1-\sigma} = \lim_{\sigma \rightarrow \infty} \frac{\sigma}{1-\sigma} \cdot \lim_{\sigma \rightarrow \infty} \left( \frac{aC}{\sigma} + 1 \right)^{1-\sigma}.$$

Clearly,

$$\lim_{\sigma \rightarrow \infty} \frac{\sigma}{1-\sigma} = -1,$$

so we can focus on the second limit.

Inspection shows this is an indeterminate form of type  $1^{-\infty}$ . To apply L'Hôpital's rule, it is convenient to work with logarithms:

$$\lim_{\sigma \rightarrow \infty} (1-\sigma) \log \left( \frac{aC}{\sigma} + 1 \right),$$

which is of the type  $(-\infty) \cdot 0$ . Rewriting gives a  $0/0$  form:

$$\lim_{\sigma \rightarrow \infty} \frac{\log \left( \frac{aC}{\sigma} + 1 \right)}{\frac{1}{1-\sigma}}.$$

Applying L'Hôpital's rule:

$$\lim_{\sigma \rightarrow \infty} \frac{\left( -\frac{1}{\frac{aC}{\sigma} + 1} \right) \frac{aC}{\sigma^2}}{\left( \frac{1}{1-\sigma} \right)^2} = -aC \lim_{\sigma \rightarrow \infty} \frac{1-2\sigma+\sigma^2}{aC\sigma+\sigma^2} = -aC.$$

Since  $-aC$  is the limit of the logarithm, the original expression's limit is its exponential:

$$\lim_{\sigma \rightarrow \infty} \frac{\sigma}{1-\sigma} \left( \frac{aC}{\sigma} + 1 \right)^{1-\sigma} = -e^{-aC}.$$

Hence CARA (constant absolute risk aversion) is obtained as

$$u(C) = -\frac{1}{a} e^{-aC},$$

which is a special case of the HARA family of felicity functions.

### 3 Consumption and Asset Pricing

So far, we have mainly studied labor-income risk, modeled as uncertainty in the household's endowment process. We now turn to *rate-of-return risk*: the return on saving between periods  $t$  and  $t + 1$  may be unknown at time  $t$ .

Throughout this section, let

$$R_{t+1}^i \equiv 1 + r_{t+1}^i$$

denote the gross return on asset  $i$  between periods  $t$  and  $t + 1$ . The return is indexed by  $t + 1$  because it is realized in period  $t + 1$ , although the asset is purchased in period  $t$ .

The central idea of consumption-based asset pricing is that the relevant risk of an asset is not the variance of its return by itself. What matters is how the return covaries with the household's marginal utility of consumption.

#### 3.1 The Euler Equation with a Risky Asset

Consider a household with current assets  $A$ , income  $Y$ , and a stochastic gross return  $1 + r'$  on assets carried into the next period. Using the timing convention

$$C + A' = (1 + r)A + Y,$$

the recursive problem can be written as

$$V(A, r) = \max_{A'} \left\{ u((1 + r)A + Y - A') + \beta \mathbb{E}[V(A', r') \mid r] \right\}.$$

The state variables are  $(A, r)$  and the control is next-period assets  $A'$ .

The first-order condition is

$$u'(C) = \beta \mathbb{E}[V_A(A', r') \mid r].$$

The envelope condition is

$$V_A(A, r) = u'(C)(1 + r).$$

Evaluating the envelope condition one period ahead and substituting it into the first-order condition gives

$$u'(C) = \beta \mathbb{E}[u'(C')(1 + r') \mid r].$$

Returning to time-subscript notation,

$$\boxed{u'(C_t) = \beta \mathbb{E}_t[R_{t+1} u'(C_{t+1})].}$$

Equivalently,

$$\boxed{1 = \mathbb{E}_t \left[ \beta \frac{u'(C_{t+1})}{u'(C_t)} R_{t+1} \right].}$$

The return must remain inside the expectation operator because  $R_{t+1}$  is not generally known at time  $t$ .

**Timing convention.** One can instead write

$$A' = (A + Y - C)(1 + r'),$$

so that the return is explicitly applied after saving takes place. Solving the corresponding Bellman problem delivers the same Euler equation:

$$u'(C_t) = \beta \mathbb{E}_t [R_{t+1} u'(C_{t+1})].$$

Thus, the economics does not depend on which internally consistent timing convention is adopted.

### 3.2 Portfolio Choice

Suppose the household can divide its savings between:

- a safe asset with known gross return  $1 + r$ ;
- a risky asset with stochastic gross return  $1 + z'$ .

Let  $\omega$  be the fraction of saving invested in the safe asset and  $1 - \omega$  the fraction invested in the risky asset. Wealth evolves according to

$$A' = (A + Y - C) [\omega(1 + r) + (1 - \omega)(1 + z')].$$

The Bellman equation is

$$V(A, z) = \max_{C, \omega} \{u(C) + \beta \mathbb{E} [V(A', z') \mid z]\}.$$

Assume an interior portfolio choice,

$$0 < \omega < 1.$$

The first-order condition with respect to consumption is

$$u'(C) = \beta \mathbb{E} [V_A(A', z') (\omega(1 + r) + (1 - \omega)(1 + z')) \mid z].$$

The first-order condition with respect to  $\omega$  is

$$0 = \mathbb{E} [V_A(A', z') (r - z') \mid z].$$

The envelope condition gives

$$V_A(A, z) = u'(C).$$

Therefore, the portfolio first-order condition becomes

$$0 = \mathbb{E} [u'(C') (r - z') \mid z],$$

or

$$\boxed{\mathbb{E} [u'(C') (1 + r) \mid z] = \mathbb{E} [u'(C') (1 + z') \mid z].}$$

In words, the expected returns of the safe and risky assets must be equal after weighting each state by the marginal utility of consumption in that state.

The resulting Euler equations are

$$u'(C) = \beta(1 + r) \mathbb{E} [u'(C') \mid z]$$

for the safe asset and

$$u'(C) = \beta \mathbb{E} [(1 + z') u'(C') \mid z]$$

for the risky asset.

More generally, for every asset  $i$  held in a strictly positive quantity,

$$u'(C_t) = \beta \mathbb{E}_t [R_{t+1}^i u'(C_{t+1})].$$

### 3.3 Interpreting the Risky-Asset Euler Equation

Start from

$$u'(C_t) = \beta \mathbb{E}_t [R_{t+1} u'(C_{t+1})].$$

Using

$$\mathbb{E}_t[XY] = \mathbb{E}_t[X]\mathbb{E}_t[Y] + \text{Cov}_t(X, Y),$$

we obtain

$$u'(C_t) = \beta \{ \mathbb{E}_t[R_{t+1}] \mathbb{E}_t[u'(C_{t+1})] + \text{Cov}_t(R_{t+1}, u'(C_{t+1})) \}.$$

Solving for the expected return,

$$\mathbb{E}_t[R_{t+1}] = \frac{\beta^{-1} u'(C_t) - \text{Cov}_t(R_{t+1}, u'(C_{t+1}))}{\mathbb{E}_t[u'(C_{t+1})]}.$$

This expression shows that the relevant measure of risk is not

$$\text{Var}_t(R_{t+1}),$$

but rather

$$\text{Cov}_t(R_{t+1}, u'(C_{t+1})).$$

Because  $u'(\cdot)$  is decreasing, marginal utility is high when consumption is low. Therefore:

- An asset that pays a high return when consumption is low covaries positively with marginal utility. It provides insurance in bad states and is a good hedging instrument. Agents are willing to hold it even if its average return is relatively low.
- An asset that pays a low return when consumption is low covaries negatively with marginal utility. It performs badly precisely when resources are most valuable and must compensate agents through a higher expected return.

Thus, an asset's risk depends on whether it pays in *good times* or in *bad times*, rather than only on the variability of its payoff.

For example, a worker whose labor income is strongly related to the performance of a particular industry should generally avoid holding too much stock in that same industry. Both labor income and asset income would then fall in the same states. Nevertheless, households often hold stock in their own firms or disproportionately hold domestic assets. These observations are related to the own-company-stock and home-bias puzzles.

### 3.4 The Risk-Free Asset

Let  $R_t^f \equiv 1 + r_t^f$  denote the gross return on a one-period risk-free asset. The return is known at time  $t$ , so it can be taken outside the conditional expectation:

$$u'(C_t) = \beta R_t^f \mathbb{E}_t [u'(C_{t+1})].$$

Therefore,

$$R_t^f = \frac{1}{\beta} \frac{u'(C_t)}{\mathbb{E}_t[u'(C_{t+1})]}.$$

The risk-free rate is determined by the household's willingness to substitute consumption across time and by precautionary motives operating through expected future marginal utility.

### 3.5 The Consumption CAPM

Subtracting the risk-free-return equation from the expected-return equation gives

$$\mathbb{E}_t[r_{t+1}] - r_t^f = -\frac{\text{Cov}_t(R_{t+1}, u'(C_{t+1}))}{\mathbb{E}_t[u'(C_{t+1})]}.$$

This theory of expected excess returns is called the *consumption capital-asset pricing model*, or CCAPM.

Because  $u'(C_t)$  is known at time  $t$ , we can divide the numerator and denominator by it:

$$\mathbb{E}_t[r_{t+1}] - r_t^f = -\frac{\text{Cov}_t\left(R_{t+1}, \frac{u'(C_{t+1})}{u'(C_t)}\right)}{\mathbb{E}_t\left[\frac{u'(C_{t+1})}{u'(C_t)}\right]}.$$

For standard concave utility functions, the marginal-utility ratio is approximately a decreasing function of consumption growth. Therefore, expected excess returns are approximately increasing in the covariance between an asset's return and consumption growth.

An asset with a high return when consumption growth is high is a poor hedge. It pays in states where consumption is already abundant and therefore must offer a high average return.

**Consumption beta.** Let

$$g_{t+1}^C \equiv \frac{C_{t+1} - C_t}{C_t}$$

denote consumption growth. A consumption beta can be defined from the population regression of an asset's excess return on consumption growth:

$$\beta_i^C \equiv \frac{\text{Cov}(r_{t+1}^i - r_t^f, g_{t+1}^C)}{\text{Var}(g_{t+1}^C)}.$$

The CCAPM predicts that average risk premia should be positively related to consumption betas: assets whose returns are especially high when consumption growth is high should offer larger expected excess returns.

### 3.6 The Stochastic Discount Factor

Define the *stochastic discount factor* (SDF) as

$$M_{t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

It combines ordinary time discounting through  $\beta$  with state-dependent valuation through the marginal-utility ratio.

The Euler equation for any asset can then be written compactly as

$$1 = \mathbb{E}_t [M_{t+1} R_{t+1}].$$

Using the covariance decomposition,

$$1 = \mathbb{E}_t [M_{t+1}] \mathbb{E}_t [R_{t+1}] + \text{Cov}_t(M_{t+1}, R_{t+1}),$$

so

$$\mathbb{E}_t [R_{t+1}] = \frac{1 - \text{Cov}_t(M_{t+1}, R_{t+1})}{\mathbb{E}_t [M_{t+1}]}.$$

For the risk-free asset,

$$1 = R_t^f \mathbb{E}_t [M_{t+1}],$$

and hence

$$R_t^f = \frac{1}{\mathbb{E}_t [M_{t+1}]}.$$

Expected excess returns satisfy

$$\mathbb{E}_t [R_{t+1}] - R_t^f = - \frac{\text{Cov}_t(M_{t+1}, R_{t+1})}{\mathbb{E}_t [M_{t+1}]}.$$

Since

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}$$

is high in states with low consumption, an asset is risky when its return is negatively correlated with the SDF.

**State-price interpretation.** The SDF converts units of consumption delivered tomorrow in a particular state into units of consumption today. A unit of consumption delivered in a bad state has a high value because marginal utility is high in that state.

### 3.7 Special Case I: Joint Lognormality

Suppose the gross return and the SDF are jointly conditionally lognormal and homoskedastic. Define

$$x_{t+1} \equiv \log R_{t+1}, \quad m_{t+1} \equiv \log M_{t+1}.$$

Assume that  $(x_{t+1}, m_{t+1})$  is jointly normal conditional on time- $t$  information and that the relevant conditional variances and covariances are constant:

$$\sigma_R^2 \equiv \text{Var}(x_{t+1}),$$

$$\sigma_M^2 \equiv \text{Var}(m_{t+1}),$$

and

$$\sigma_{R,M} \equiv \text{Cov}(x_{t+1}, m_{t+1}).$$

For a normal random variable  $X$ ,

$$\mathbb{E}[e^X] = \exp \left\{ \mathbb{E}[X] + \frac{1}{2} \text{Var}(X) \right\}.$$

Equivalently, if  $Y$  is lognormal,

$$\log \mathbb{E}[Y] = \mathbb{E}[\log Y] + \frac{1}{2} \text{Var}(\log Y).$$

Starting from

$$1 = \mathbb{E}_t [R_{t+1} M_{t+1}] = \mathbb{E}_t [e^{x_{t+1} + m_{t+1}}],$$

joint normality implies

$$0 = \mathbb{E}_t [x_{t+1}] + \mathbb{E}_t [m_{t+1}] + \frac{1}{2} [\sigma_R^2 + \sigma_M^2 + 2\sigma_{R,M}].$$

Thus,

$$0 = \mathbb{E}_t [\log R_{t+1}] + \mathbb{E}_t [\log M_{t+1}] + \frac{1}{2} [\sigma_R^2 + \sigma_M^2 + 2\sigma_{R,M}].$$

For the risk-free asset,  $\log R_t^f$  is known at time  $t$ , so

$$\log R_t^f = -\mathbb{E}_t [\log M_{t+1}] - \frac{1}{2} \sigma_M^2.$$

Subtracting gives the log-return premium:

$$\mathbb{E}_t [\log R_{t+1}] - \log R_t^f + \frac{1}{2} \sigma_R^2 = -\sigma_{R,M}.$$

The term

$$\frac{1}{2} \sigma_R^2$$

is a Jensen-inequality correction generated by the difference between expected gross returns and expected log returns.

### 3.7.1 The SDF Volatility Bound

By the Cauchy–Schwarz inequality,

$$-\sigma_{R,M} \leq \sigma_R \sigma_M.$$

Therefore,

$$\sigma_M \geq \frac{\mathbb{E}_t [\log R_{t+1}] - \log R_t^f + \frac{1}{2} \sigma_R^2}{\sigma_R}.$$

The right-hand side is the asset's logarithmic Sharpe ratio. Thus, the standard deviation of the log SDF must be at least as large as the logarithmic Sharpe ratio of every traded asset, including the asset or portfolio with the largest Sharpe ratio.

Empirically, stock-market returns imply a high lower bound on SDF volatility. The SDF generated by smooth aggregate consumption and conventional preferences is often not volatile enough to satisfy this bound.

**Methodological remark.** The lognormal approach obtains exact expressions by imposing distributional assumptions directly on the endogenous variables  $R_{t+1}$  and  $M_{t+1}$ . An alternative approach is to impose structure on preferences and use a local Taylor approximation. The CRRA case below follows this second route.

### 3.8 Special Case II: CRRA Preferences

Suppose period utility is CRRA:

$$u(C) = \frac{C^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0,$$

with

$$u'(C) = C^{-\sigma}.$$

The parameter  $\sigma$  is the coefficient of relative risk aversion:

$$-\frac{u''(C)C}{u'(C)} = \sigma.$$

Under time-separable CRRA preferences, the intertemporal elasticity of substitution is

$$\text{IES} = \frac{1}{\sigma}.$$

Thus, the same parameter governs both:

- aversion to atemporal consumption risk;
- unwillingness to substitute consumption across time.

The risky-asset Euler equation becomes

$$C_t^{-\sigma} = \beta \mathbb{E}_t [R_{t+1} C_{t+1}^{-\sigma}].$$

Dividing by  $C_t^{-\sigma}$ ,

$$1 = \beta \mathbb{E}_t \left[ R_{t+1} \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \right].$$

Let the subjective discount rate  $\gamma$  satisfy

$$\beta = \frac{1}{1+\gamma}.$$

Then

$$1 + \gamma = \mathbb{E}_t \left[ (1 + r_{t+1}) \left( 1 + \frac{\Delta C_{t+1}}{C_t} \right)^{-\sigma} \right].$$

Define

$$g_{t+1}^C \equiv \frac{\Delta C_{t+1}}{C_t}.$$

A second-order approximation around

$$r_{t+1} = 0, \quad g_{t+1}^C = 0$$

gives

$$(1 + r_{t+1})(1 + g_{t+1}^C)^{-\sigma} \approx 1 + r_{t+1} - \sigma g_{t+1}^C - \sigma r_{t+1} g_{t+1}^C + \frac{1}{2} \sigma(\sigma + 1)(g_{t+1}^C)^2.$$

Taking conditional expectations,

$$\begin{aligned} 1 + \gamma &\approx 1 + \mathbb{E}_t[r_{t+1}] - \sigma \mathbb{E}_t[g_{t+1}^C] \\ &\quad - \sigma \{ \mathbb{E}_t[r_{t+1}] \mathbb{E}_t[g_{t+1}^C] + \text{Cov}_t(r_{t+1}, g_{t+1}^C) \} \\ &\quad + \frac{1}{2} \sigma(\sigma + 1) \{ (\mathbb{E}_t[g_{t+1}^C])^2 + \text{Var}_t(g_{t+1}^C) \}. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{E}_t[r_{t+1}] - \gamma &\approx \sigma \mathbb{E}_t[g_{t+1}^C] \\ &\quad + \sigma \{ \mathbb{E}_t[r_{t+1}] \mathbb{E}_t[g_{t+1}^C] + \text{Cov}_t(r_{t+1}, g_{t+1}^C) \} \\ &\quad - \frac{1}{2} \sigma(\sigma + 1) \{ (\mathbb{E}_t[g_{t+1}^C])^2 + \text{Var}_t(g_{t+1}^C) \}. \end{aligned}$$

Dropping products of conditional means and other small terms gives the approximation emphasized in class:

$$\boxed{\mathbb{E}_t[r_{t+1}] - \gamma \approx \sigma \mathbb{E}_t[g_{t+1}^C] + \sigma \text{Cov}_t(g_{t+1}^C, r_{t+1}) - \frac{1}{2} \sigma(\sigma + 1) \text{Var}_t(g_{t+1}^C).}$$

For two assets  $i$  and  $j$ , the common consumption-growth terms cancel, yielding

$$\boxed{\mathbb{E}_t[r_{t+1}^i] - \mathbb{E}_t[r_{t+1}^j] \approx \sigma \text{Cov}_t(g_{t+1}^C, r_{t+1}^i - r_{t+1}^j).}$$

Asset  $i$  earns a higher expected return than asset  $j$  when its relative return covaries positively with consumption growth. Such an asset performs relatively well in good times and is therefore a relatively poor hedge.

### 3.9 The Equity-Premium Puzzle

Applying the preceding expression to stocks and short-term government bonds gives

$$\mathbb{E}[r_{t+1}^{\text{stock}} - r_{t+1}^{\text{bond}}] \approx \sigma \text{Cov}(g_{t+1}^C, r_{t+1}^{\text{stock}} - r_{t+1}^{\text{bond}}).$$

The numbers discussed in class are approximately

$$\text{Cov}(g_{t+1}^C, r_{t+1}^{\text{stock}} - r_{t+1}^{\text{bond}}) \approx 0.0024$$

and

$$\mathbb{E}[r_{t+1}^{\text{stock}} - r_{t+1}^{\text{bond}}] \approx 0.06.$$

Thus, the implied coefficient of relative risk aversion is

$$\sigma \approx \frac{0.06}{0.0024} = 25.$$

The *equity-premium puzzle* is that explaining the historically large excess return on stocks using the standard consumption-based model requires an implausibly large coefficient of relative risk aversion.

The underlying problem is that aggregate consumption growth is smooth and covaries only weakly with stock returns. Therefore, the model requires a very large value of  $\sigma$  to generate a large equity

premium.

### 3.10 Combining CRRA Preferences and Lognormality

Under CRRA preferences, the SDF is

$$M_{t+1} = \beta \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma}.$$

Taking logs,

$$\boxed{\log M_{t+1} = \log \beta - \sigma \Delta \log C_{t+1}.$$

This expression is exact.

Substitute it into the general lognormal pricing equation:

$$0 = \mathbb{E}_t[\log R_{t+1}] + \mathbb{E}_t[\log M_{t+1}] + \frac{1}{2} [\text{Var}_t(\log R_{t+1}) + \text{Var}_t(\log M_{t+1}) + 2 \text{Cov}_t(\log R_{t+1}, \log M_{t+1})].$$

Using

$$\text{Var}_t(\log M_{t+1}) = \sigma^2 \text{Var}_t(\Delta \log C_{t+1})$$

and

$$\text{Cov}_t(\log R_{t+1}, \log M_{t+1}) = -\sigma \text{Cov}_t(\log R_{t+1}, \Delta \log C_{t+1}),$$

we obtain

$$\begin{aligned} 0 = & \mathbb{E}_t[\log R_{t+1}] + \log \beta - \sigma \mathbb{E}_t[\Delta \log C_{t+1}] \\ & + \frac{1}{2} \left[ \text{Var}_t(\log R_{t+1}) + \sigma^2 \text{Var}_t(\Delta \log C_{t+1}) \right. \\ & \left. - 2\sigma \text{Cov}_t(\log R_{t+1}, \Delta \log C_{t+1}) \right]. \end{aligned}$$

For the risk-free asset,

$$\boxed{\log R_t^f = -\log \beta + \sigma \mathbb{E}_t[\Delta \log C_{t+1}] - \frac{1}{2} \sigma^2 \text{Var}_t(\Delta \log C_{t+1}).}$$

This equation has three central implications:

1. A lower  $\beta$ , meaning greater impatience, raises the risk-free rate. Agents require more compensation to postpone consumption.
2. Higher expected consumption growth raises the risk-free rate. When agents expect consumption to be higher tomorrow, they must be compensated to save rather than consume today.
3. Greater consumption-growth uncertainty lowers the risk-free rate. Precautionary saving increases the demand for safe assets and reduces the return required to induce saving.

Subtracting the risk-free equation from the risky-return equation gives

$$\boxed{\mathbb{E}_t[\log R_{t+1}] - \log R_t^f + \frac{1}{2} \text{Var}_t(\log R_{t+1}) = \sigma \text{Cov}_t(\log R_{t+1}, \Delta \log C_{t+1}).}$$

The economic content is the same as in the second-order CRRA approximation: an asset earns a large premium if it covaries positively with consumption growth and therefore performs poorly as insurance.

### 3.11 The Risk-Free-Rate Puzzle

Taking unconditional expectations of the risk-free-rate equation gives

$$\mathbb{E}[\log R_t^f] = -\log \beta + \sigma g_C - \frac{1}{2}\sigma^2 \text{Var}(\Delta \log C),$$

where

$$g_C \equiv \mathbb{E}[\Delta \log C_{t+1}]$$

is average consumption growth.

The equity-premium puzzle suggests a high value of  $\sigma$ . However:

- average consumption growth is positive;
- the observed risk-free rate is low;
- consumption-growth volatility is small, so the precautionary-saving term is not large enough to offset the effect of  $\sigma g_C$ .

Consequently, reconciling a large  $\sigma$  with the low observed risk-free rate requires  $-\log \beta$  to be extremely small or even negative. Equivalently,  $\beta$  must be close to or possibly greater than one.

This is the *risk-free-rate puzzle*.

The two puzzles are closely connected because under time-separable CRRA preferences the same parameter  $\sigma$  governs both risk aversion and intertemporal substitution:

- a high  $\sigma$  helps generate a large equity premium;
- but a high  $\sigma$  also implies a low intertemporal elasticity of substitution and tends to generate an excessively high risk-free rate when consumption is growing.

### 3.12 Proposed Resolutions of the Asset-Pricing Puzzles

The standard CRRA consumption-based model may fail because of its assumptions about shocks, preferences, or the relevant investor.

**Rare disasters.** Historical samples may understate the probability of extremely large falls in consumption and asset payoffs. Rare-disaster risk can raise the equity premium without requiring ordinary consumption growth to be highly volatile.

**The marginal investor.** Asset prices are determined by the consumption of the marginal holder of an asset, not necessarily by aggregate per-capita consumption. The marginal investor's consumption may be substantially more volatile and more strongly correlated with asset returns.

**Epstein–Zin preferences.** Recursive preferences can separate relative risk aversion from the intertemporal elasticity of substitution. This breaks the tight CRRA link between the equity-premium and risk-free-rate puzzles.

**Habit formation.** If utility depends on consumption relative to a habit stock, small changes in consumption can generate large changes in marginal utility. The resulting SDF is more volatile and effective risk aversion becomes state dependent.

**Loss aversion and behavioral mechanisms.** Investors may evaluate gains and losses asymmetrically or form expectations in ways that depart from the standard rational-expectations benchmark.

## 4 Stocks and Lucas Trees

So far, we have treated  $R_{t+1}$  as a generic risky return. We now impose structure by considering a specific asset: a share of stock.

### 4.1 The Return on a Stock

A share purchased in period  $t$  costs  $P_t$ . In period  $t + 1$ , it pays a dividend  $D_{t+1}$  and can be resold for  $P_{t+1}$ . Its gross return is therefore

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Substituting this return into the consumption Euler equation,

$$1 = \mathbb{E}_t \left[ M_{t+1} \frac{P_{t+1} + D_{t+1}}{P_t} \right].$$

Multiplying by  $P_t$ ,

$$P_t = \mathbb{E}_t [M_{t+1}(P_{t+1} + D_{t+1})].$$

Using the consumption-based SDF,

$$P_t = \beta \mathbb{E}_t \left[ \frac{u'(C_{t+1})}{u'(C_t)} (P_{t+1} + D_{t+1}) \right].$$

The price of the stock is high when its future dividend and resale value are high, especially in states in which marginal utility is high.

### 4.2 Forward Iteration and Fundamental Value

Define the  $k$ -period SDF

$$M_{t,t+k} \equiv \beta^k \frac{u'(C_{t+k})}{u'(C_t)}.$$

Iterating the one-period pricing equation forward  $T$  periods gives

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^T M_{t,t+k} D_{t+k} + M_{t,t+T} P_{t+T} \right].$$

The first term is the discounted value of dividends paid between  $t + 1$  and  $t + T$ . The second term is the discounted resale value of the asset at the terminal date.

Impose the no-bubble or transversality condition

$$\lim_{T \rightarrow \infty} \mathbb{E}_t [M_{t,t+T} P_{t+T}] = 0.$$

Then the stock price equals its fundamental value:

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} M_{t,t+k} D_{t+k} \right].$$

Equivalently,

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} \beta^k D_{t+k} \frac{u'(C_{t+k})}{u'(C_t)} \right].$$

A dividend is valuable when:

- it is large;
- it is paid relatively soon;
- it is likely to occur;
- it is paid in a state with high marginal utility.

The last component is the distinctive implication of consumption-based asset pricing.

### 4.3 Linear Felicity and Rational Bubbles

Suppose felicity is linear:

$$u(C) = aC.$$

Then

$$u'(C) = a$$

is constant, so the household is risk neutral and

$$M_{t+1} = \beta.$$

The stock-pricing equation becomes

$$P_t = \beta \mathbb{E}_t [P_{t+1} + D_{t+1}],$$

or

$$\frac{P_t}{\beta} = \mathbb{E}_t [P_{t+1} + D_{t+1}].$$

A general solution is

$$P_t = \sum_{k=1}^{\infty} \beta^k \mathbb{E}_t [D_{t+k}] + \xi_t \left( \frac{1}{\beta} \right)^t,$$

where

$$\mathbb{E}_t [\xi_{t+1}] = \xi_t.$$

Thus,  $\xi_t$  is a martingale.

The first component,

$$P_t^F \equiv \sum_{k=1}^{\infty} \beta^k \mathbb{E}_t [D_{t+k}],$$

is the fundamental value of the stock.

The second component,

$$B_t \equiv \xi_t \left( \frac{1}{\beta} \right)^t,$$

is a rational bubble.

It satisfies

$$B_t = \beta \mathbb{E}_t[B_{t+1}].$$

The deterministic factor

$$\left(\frac{1}{\beta}\right)^t$$

makes the bubble grow rapidly enough to offset discounting, while the martingale component  $\xi_t$  generates unpredictable changes in its size.

Setting

$$\xi_t \equiv 0$$

selects the bubble-free solution. However, the Euler equation alone does not eliminate rational bubbles; an additional transversality or no-bubble condition is required.

#### 4.4 Shiller's Excess-Volatility Argument

Under linear felicity and no bubbles,

$$P_t = \sum_{k=1}^{\infty} \beta^k \mathbb{E}_t[D_{t+k}].$$

Define the ex post present value of realized future dividends as

$$P_t^* \equiv \sum_{k=1}^{\infty} \beta^k D_{t+k}.$$

Under rational expectations,

$$\boxed{P_t = \mathbb{E}_t[P_t^*].}$$

Therefore,

$$P_t^* = P_t + \varepsilon_{t+1}^P,$$

where the forecast error satisfies

$$\mathbb{E}_t[\varepsilon_{t+1}^P] = 0.$$

By the law of total variance,

$$\text{Var}(P_t^*) = \text{Var}(P_t) + \mathbb{E}[\text{Var}(P_t^* \mid \mathcal{I}_t)],$$

and hence

$$\boxed{\text{Var}(P_t) \leq \text{Var}(P_t^*)}.$$

The observed stock price should not be more volatile than the ex post present value of dividends if prices are rational expectations of discounted dividends under a constant SDF and there are no bubbles.

Shiller's empirical argument is that actual stock prices appear excessively volatile relative to realized dividend fundamentals.

Possible interpretations include:

- expectations may not be fully rational and may fluctuate because of animal spirits or other behavioral forces;
- the SDF may be sufficiently time varying that stock prices move substantially even when expected dividends are smooth;
- rational bubbles may be present.

Thus, the debate can be understood as a disagreement over whether one needs a better theory of expectations, a better theory of the stochastic discount factor, or a theory admitting bubbles.

## 4.5 Lucas Trees

The Lucas-tree model closes the asset-pricing model in general equilibrium.

### 4.5.1 Environment

Assume:

- There is a unit mass of identical, infinitesimal households.
- There is a unit mass of identical trees.
- There is no separate labor-income endowment:

$$Y_t \equiv 0.$$

- Each tree produces  $D_t$  units of fruit in period  $t$ .
- Fruit is perishable and cannot be stored.
- Trees are perfectly durable, never depreciate, and cannot themselves be consumed.
- The current dividend  $D_t$  is known at time  $t$ , while  $D_{t+1}$  may be uncertain.
- The ex-dividend price of a tree is  $P_t$ .

A tree is therefore equivalent to a stock: it pays dividend  $D_{t+1}$  and can be resold for  $P_{t+1}$ . The gross return is

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

The stock-pricing Euler equation is

$$1 = \beta \mathbb{E}_t \left[ \frac{P_{t+1} + D_{t+1}}{P_t} \frac{u'(C_{t+1})}{u'(C_t)} \right].$$

Without bubbles,

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} \beta^k D_{t+k} \frac{u'(C_{t+k})}{u'(C_t)} \right].$$

### 4.5.2 General-Equilibrium Resource Constraint

Because fruit is the only consumption good and it cannot be stored, aggregate feasibility requires

$$C_t = D_t.$$

Substituting the equilibrium resource constraint into the asset-pricing equation gives

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} \beta^k D_{t+k} \frac{u'(D_{t+k})}{u'(D_t)} \right].$$

The dividend process determines aggregate consumption, while the equilibrium asset price adjusts so that households are willing to hold the fixed supply of trees.

### 4.5.3 Log Utility

Suppose

$$u(C) = \log C, \quad u'(C) = \frac{1}{C}.$$

Then

$$\frac{u'(D_{t+k})}{u'(D_t)} = \frac{D_t}{D_{t+k}}.$$

Therefore, each discounted-dividend term simplifies:

$$D_{t+k} \frac{u'(D_{t+k})}{u'(D_t)} = D_{t+k} \frac{D_t}{D_{t+k}} = D_t.$$

Hence

$$P_t = \sum_{k=1}^{\infty} \beta^k D_t = D_t \sum_{k=1}^{\infty} \beta^k.$$

Using

$$\sum_{k=1}^{\infty} \beta^k = \frac{\beta}{1-\beta},$$

we obtain

$$P_t = \frac{\beta}{1-\beta} D_t.$$

Thus, the equilibrium price-dividend ratio is constant:

$$\frac{P_t}{D_t} = \frac{\beta}{1-\beta}.$$

The corresponding gross return is

$$\begin{aligned} R_{t+1} &= \frac{P_{t+1} + D_{t+1}}{P_t} \\ &= \frac{\frac{\beta}{1-\beta} D_{t+1} + D_{t+1}}{\frac{\beta}{1-\beta} D_t} \\ &= \frac{1}{\beta} \frac{D_{t+1}}{D_t}. \end{aligned}$$

Therefore,

$$R_{t+1} = \frac{1}{\beta} \frac{D_{t+1}}{D_t}.$$

The SDF is

$$M_{t+1} = \beta \frac{u'(D_{t+1})}{u'(D_t)} = \beta \frac{D_t}{D_{t+1}}.$$

Consequently,

$$M_{t+1} R_{t+1} = 1$$

state by state, verifying the Euler equation.

**Economic interpretation.** A high future dividend has two opposing effects:

1. It directly raises the payoff of the tree.

2. In equilibrium, it also raises aggregate consumption, reducing the marginal utility with which that dividend is valued.

With logarithmic felicity, these two effects exactly cancel, generating a constant price-dividend ratio.

#### 4.5.4 Partial Equilibrium versus General Equilibrium

The Lucas-tree model is the polar opposite of the earlier consumption-saving problem with a given rate of return.

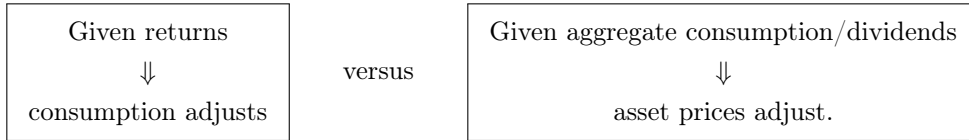
**Consumption-saving problem.** Asset prices and returns are taken as given. The household adjusts its consumption and saving decisions to satisfy the Euler equation.

**Lucas-tree equilibrium.** The dividend process fixes aggregate consumption through

$$C_t = D_t.$$

The supply of trees is fixed, so asset prices and returns must adjust to make households willing to hold the available trees.

Thus:



In reality, asset supply is typically neither perfectly elastic, as in a small-open-economy consumption-saving problem, nor perfectly inelastic, as in the simplest Lucas-tree model.

## 4.6 Core Asset-Pricing Logic

The main results can be summarized by the following chain:

$$\begin{array}{c}
 \boxed{u'(C_t) = \beta \mathbb{E}_t [R_{t+1} u'(C_{t+1})]} \\
 \Downarrow \\
 \boxed{M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}} \\
 \Downarrow \\
 \boxed{1 = \mathbb{E}_t [M_{t+1} R_{t+1}]} \\
 \Downarrow \\
 \boxed{R_t^f = \frac{1}{\mathbb{E}_t [M_{t+1}]}}
 \end{array}$$

and

$$\boxed{\mathbb{E}_t [R_{t+1}] - R_t^f = - \frac{\text{Cov}_t(M_{t+1}, R_{t+1})}{\mathbb{E}_t [M_{t+1}]}.}$$

The central economic lesson is:

An asset commands a high expected return when it performs poorly in states in which consumption is low and marginal utility is high. Risk is determined by covariance with marginal utility, not merely by return volatility.

## 5 Additional Time-Series Tools

This section collects a number of tools that were originally introduced during the empirical and time-series part of the course. They are presented here as an appendix because they are also useful for formulating and solving stochastic dynamic economic models.

The central connection is the following:

|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| Continuous stochastic process $\xrightarrow{\text{discretization}}$ finite-state Markov chain |
|-----------------------------------------------------------------------------------------------|

and finite-state Markov chains can then be incorporated directly into dynamic-programming problems.

### 5.1 Higher Moments and Time-Varying Volatility

A standard linear time-series model often assumes that innovations have constant unconditional and conditional variances. In macroeconomic and financial data, however, periods of large fluctuations frequently cluster together.

This motivates studying:

- third moments, such as skewness;
- fourth moments, such as kurtosis;
- changes in volatility across subsamples or moving windows;
- structural breaks in volatility, such as the Great Moderation;
- direct measures of risk and uncertainty;
- models of conditional heteroskedasticity;
- models of stochastic volatility.

The key distinction throughout this section is between the *unconditional variance* of an innovation and its *conditional variance* given current information.

#### 5.1.1 Conditional versus Unconditional Variance

Consider a demeaned AR(1):

$$y_t = \phi y_{t-1} + \varepsilon_t,$$

where  $\varepsilon_t$  is white noise:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] = \sigma^2,$$

and

$$\mathbb{E}[\varepsilon_{t_1} \varepsilon_{t_2}] = 0 \quad \text{for } t_1 \neq t_2.$$

The assumption

$$\mathbb{E}[\varepsilon_t^2] = \sigma^2$$

states that the *unconditional* variance is constant over time.

It does not necessarily imply that

$$\text{Var}_{t-1}(\varepsilon_t)$$

is constant. Thus, a process can be unconditionally homoskedastic but conditionally heteroskedastic.

A white-noise process need not be i.i.d. In particular, zero serial correlation in  $\varepsilon_t$  does not rule out serial dependence in  $\varepsilon_t^2$ .

## 5.2 ARCH Models

ARCH stands for *autoregressive conditional heteroskedasticity*.

The object being modeled is the innovation  $\varepsilon_t$ , or more specifically its conditional variance, rather than the original series  $y_t$  itself.

### 5.2.1 Weak ARCH Representation

Suppose squared innovations follow

$$\varepsilon_t^2 = (1 - \alpha)\sigma^2 + \alpha\varepsilon_{t-1}^2 + w_t,$$

where  $w_t$  is white noise:

$$\mathbb{E}[w_t] = 0, \quad \mathbb{E}[w_t^2] = \eta^2, \quad \mathbb{E}[w_{t_1}w_{t_2}] = 0 \quad \text{for } t_1 \neq t_2.$$

The best linear predictor of squared innovations is then

$$\boxed{\widehat{\mathbb{E}}_{t-1}[\varepsilon_t^2] = (1 - \alpha)\sigma^2 + \alpha\varepsilon_{t-1}^2.}$$

This is called a *weak ARCH(1)* process when the equation is interpreted as a linear-projection relationship rather than as the exact conditional variance.

Under this representation, conditional heteroskedasticity can be detected by estimating a regression of squared residuals on their lags and testing whether

$$\alpha = 0.$$

If

$$\alpha > 0,$$

a large squared innovation in period  $t - 1$  predicts a large squared innovation in period  $t$ : volatility is persistent.

### 5.2.2 Strong ARCH Representation

A more explicit ARCH(1) specification is

$$\boxed{\varepsilon_t = \sqrt{h_t} v_t,}$$

where

$$v_t \stackrel{\text{i.i.d.}}{\sim} (0, 1)$$

and

$$\boxed{h_t = (1 - \alpha)\sigma^2 + \alpha\varepsilon_{t-1}^2.}$$

More precisely,

$$\mathbb{E}[v_t] = 0, \quad \mathbb{E}[v_t^2] = 1,$$

and  $v_t$  is independent of past information.

Because  $h_t$  is known at time  $t - 1$ ,

$$\mathbb{E}_{t-1}[\varepsilon_t] = \sqrt{h_t} \mathbb{E}_{t-1}[v_t] = 0.$$

Moreover,

$$\begin{aligned} \text{Var}_{t-1}(\varepsilon_t) &= \mathbb{E}_{t-1}[\varepsilon_t^2] \\ &= h_t \mathbb{E}_{t-1}[v_t^2] \\ &= h_t. \end{aligned}$$

Therefore,

$$\boxed{\text{Var}_{t-1}(\varepsilon_t) = h_t = (1 - \alpha)\sigma^2 + \alpha\varepsilon_{t-1}^2.}$$

The parameterization of the intercept ensures that, under stationarity, the unconditional variance is  $\sigma^2$ :

$$\begin{aligned} \mathbb{E}[h_t] &= (1 - \alpha)\sigma^2 + \alpha\mathbb{E}[\varepsilon_{t-1}^2] \\ &= (1 - \alpha)\sigma^2 + \alpha\sigma^2 \\ &= \sigma^2. \end{aligned}$$

Usual sufficient restrictions are

$$0 \leq \alpha < 1.$$

### 5.3 GARCH Models

GARCH stands for *generalized autoregressive conditional heteroskedasticity*.

A GARCH(1,1) process generalizes ARCH by allowing the conditional variance to depend both on past squared innovations and on its own past value:

$$\boxed{h_t = (1 - \alpha - \delta)\sigma^2 + \alpha\varepsilon_{t-1}^2 + \delta h_{t-1}.}$$

Here:

- $\alpha$  measures the immediate response of volatility to a new squared innovation;
- $\delta$  measures the persistence of the conditional variance;
- $\alpha + \delta$  governs the overall persistence of volatility.

The usual sufficient restrictions are

$$\alpha \geq 0, \quad \delta \geq 0, \quad \alpha + \delta < 1.$$

Under these conditions,

$$\mathbb{E}[h_t] = \sigma^2.$$

Indeed,

$$\begin{aligned}\mathbb{E}[h_t] &= (1 - \alpha - \delta)\sigma^2 + \alpha\mathbb{E}[\varepsilon_{t-1}^2] + \delta\mathbb{E}[h_{t-1}] \\ &= (1 - \alpha - \delta)\sigma^2 + (\alpha + \delta)\mathbb{E}[h_t],\end{aligned}$$

so

$$\mathbb{E}[h_t] = \sigma^2.$$

ARCH can be estimated by first estimating the main time-series equation and then regressing squared residuals on their lags. ARCH and GARCH models can also be estimated by maximum likelihood after imposing a distribution on the standardized innovation  $v_t$ .

### 5.3.1 Limitation of ARCH and GARCH in Macroeconomics

In ARCH and GARCH models,  $h_t$  is determined by variables already observed in the past:

$$h_t = h(\varepsilon_{t-1}^2, h_{t-1}, \dots).$$

Therefore, at time  $t - 1$ ,  $h_t$  is known. The conditional variance varies over time, but there is no independent innovation to uncertainty itself.

A high  $h_t$  occurs because a large innovation to the main process occurred in the past. Thus, ARCH and GARCH partly mix two distinct concepts:

|                        |     |                                       |
|------------------------|-----|---------------------------------------|
| a large realized shock | and | high uncertainty about future shocks. |
|------------------------|-----|---------------------------------------|

This motivates stochastic-volatility models, which explicitly introduce a separate shock to volatility.

## 5.4 Stochastic Volatility

Consider again

$$y_t = \phi y_{t-1} + \varepsilon_t,$$

but now let

|                                     |
|-------------------------------------|
| $\varepsilon_t = e^{\sigma_t} v_t,$ |
|-------------------------------------|

where

$$v_t \stackrel{\text{i.i.d.}}{\sim} (0, 1).$$

Thus,

$$\mathbb{E}[v_t] = 0, \quad \mathbb{E}[v_t^2] = 1.$$

The variable  $\sigma_t$  is the *log-volatility*. Modeling the logarithm of volatility ensures that the standard deviation

$$e^{\sigma_t}$$

is always positive.

The log-volatility process can itself be modeled as an AR(1):

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| $\sigma_t = (1 - \phi_\sigma)\bar{\sigma} + \phi_\sigma\sigma_{t-1} + \sqrt{1 - \phi_\sigma^2}\eta w_t,$ |
|----------------------------------------------------------------------------------------------------------|

where

$$\mathbb{E}[w_t] = 0, \quad \mathbb{E}[w_t^2] = 1,$$

and

$$w_t \perp v_t.$$

The two innovations have distinct meanings:

$$v_t = \text{innovation to the level of } y_t,$$

whereas

$$w_t = \text{innovation to the volatility of } y_t.$$

The parameter  $\phi_\sigma$  determines the persistence of volatility. Under stationarity, the chosen normalization implies

$$\mathbb{E}[\sigma_t] = \bar{\sigma}$$

and

$$\text{Var}(\sigma_t) = \eta^2.$$

#### 5.4.1 Conditional Variance

The relevant information set must be specified carefully because the uncertainty shock  $w_t$  is a second, independent innovation.

Suppose  $w_t$  has been observed, so that  $\sigma_t$  is known, but  $v_t$  has not yet been realized. Conditional on that information,

$$\mathbb{E}[\varepsilon_t \mid \sigma_t] = e^{\sigma_t} \mathbb{E}[v_t] = 0.$$

Moreover,

$$\begin{aligned} \text{Var}(\varepsilon_t \mid \sigma_t) &= \mathbb{E}[\varepsilon_t^2 \mid \sigma_t] \\ &= e^{2\sigma_t} \mathbb{E}[v_t^2] \\ &= e^{2\sigma_t}. \end{aligned}$$

Hence,

$$\text{SD}(\varepsilon_t \mid \sigma_t) = e^{\sigma_t}.$$

Before the uncertainty shock  $w_t$  is observed, however,

$$\text{Var}_{t-1}(\varepsilon_t) = \mathbb{E}_{t-1} [e^{2\sigma_t}],$$

rather than simply  $e^{2\sigma_t}$ .

Therefore, the timing of information matters:

$$\boxed{\text{When is } w_t \text{ observed relative to } v_t?}$$

In structural models with uncertainty shocks, the modeler must state explicitly whether the volatility shock is revealed before, simultaneously with, or after the innovation to the main economic variable.

### 5.4.2 ARCH/GARCH versus Stochastic Volatility

| ARCH/GARCH                              | Stochastic volatility                                |
|-----------------------------------------|------------------------------------------------------|
| $h_t$ is a function of past observables | $\sigma_t$ receives a separate innovation $w_t$      |
| $h_t$ is known at $t - 1$               | $\sigma_t$ may not be known before $w_t$ is revealed |
| one primitive innovation                | two independent primitive innovations                |
| large past shocks imply high volatility | volatility can rise without a large level shock      |

The main conceptual advantage of stochastic volatility in macroeconomics is that it provides a direct representation of an *uncertainty shock*.

## 6 Discretization of Continuous-State Processes

Discretization approximates a continuous-state stochastic process by a finite-state stochastic process.

Every discretization method produces two objects:

1. a finite grid of possible realizations;
2. a transition matrix assigning probabilities to movements between grid points.

Thus,

Discretization produces a finite-state Markov chain.

Although the class discussion uses an AR(1), the same logic can be extended to VAR(1) processes. Since AR( $p$ ) and VAR( $p$ ) processes can be written in first-order companion form, the methods extend more generally.

### 6.1 The Continuous AR(1)

Consider

$$y_t = \phi y_{t-1} + \varepsilon_t,$$

where

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad |\phi| < 1.$$

The conditional mean is

$$\mathbb{E}_{t-1}[y_t] = \phi y_{t-1},$$

and the conditional variance is

$$\text{Var}_{t-1}(y_t) = \sigma^2.$$

The unconditional variance satisfies

$$\text{Var}(y_t) = \phi^2 \text{Var}(y_{t-1}) + \sigma^2.$$

Under stationarity,

$$\text{Var}(y_t) = \text{Var}(y_{t-1}),$$

so

$$\text{Var}(y_t) = \frac{\sigma^2}{1 - \phi^2}.$$

Define the unconditional standard deviation

$$\sigma_y \equiv \sqrt{\text{Var}(y_t)} = \frac{\sigma}{\sqrt{1 - \phi^2}}.$$

Do not confuse:

$$\underbrace{\sigma^2}_{\text{conditional innovation variance}} \neq \underbrace{\frac{\sigma^2}{1 - \phi^2}}_{\text{unconditional variance of } y_t}.$$

## 6.2 Tauchen's Method

Tauchen's method uses the distribution of the innovation  $\varepsilon_t$  to allocate probability mass from the continuous AR(1) process to a finite grid.

Assume

$$\Pr(\varepsilon_t \leq \bar{\varepsilon}) = F\left(\frac{\bar{\varepsilon}}{\sigma}\right),$$

where  $F$  is a specified cumulative distribution function. A common choice is the standard normal CDF, although normality is not essential.

### 6.2.1 Step 1: Choose the Grid

Let

$$\bar{y}_1 < \bar{y}_2 < \cdots < \bar{y}_N$$

denote the  $N$  possible realizations of the discretized process.

A conventional choice is an equally spaced and symmetric grid:

$$\bar{y}_1 = -m\sigma_y, \quad \bar{y}_N = m\sigma_y.$$

The distance between adjacent points is

$$w = \bar{y}_j - \bar{y}_{j-1} = \frac{2m\sigma_y}{N-1}.$$

Tauchen recommends approximately

$$m = 3,$$

but in practice the grid should be evaluated by checking whether simulations from the discrete process reproduce the important properties of the original AR(1).

### 6.2.2 Step 2: Construct Transition Probabilities

Define

$$P_{ij} \equiv \Pr(y_{t+1} = \bar{y}_j \mid y_t = \bar{y}_i).$$

For an interior grid point  $j = 2, \dots, N-1$ , assign to  $\bar{y}_j$  all continuous realizations falling between

the midpoints of its neighboring grid cells:

$$\begin{aligned} P_{ij} &= \Pr\left(\bar{y}_j - \frac{w}{2} \leq \phi \bar{y}_i + \varepsilon_{t+1} \leq \bar{y}_j + \frac{w}{2}\right) \\ &= \left[ F\left(\frac{\bar{y}_j - \phi \bar{y}_i + w/2}{\sigma}\right) - F\left(\frac{\bar{y}_j - \phi \bar{y}_i - w/2}{\sigma}\right) \right]. \end{aligned}$$

At the lower endpoint, all probability mass below the first cutoff is assigned to  $\bar{y}_1$ :

$$P_{i1} = F\left(\frac{\bar{y}_1 - \phi \bar{y}_i + w/2}{\sigma}\right).$$

At the upper endpoint, all probability mass above the last cutoff is assigned to  $\bar{y}_N$ :

$$P_{iN} = 1 - F\left(\frac{\bar{y}_N - \phi \bar{y}_i - w/2}{\sigma}\right).$$

By construction,

$$P_{ij} \geq 0$$

and

$$\sum_{j=1}^N P_{ij} = 1 \quad \text{for every } i.$$

**Core intuition.** Tauchen's method collects the probability mass of the continuous conditional distribution lying around each grid point and assigns all of that mass to the grid point itself.

### 6.3 Rouwenhorst's Method

Rouwenhorst's method provides an alternative finite-state approximation, particularly useful for highly persistent AR(1) processes.

Let

$$p = q = \frac{1 + \phi}{2}.$$

For  $N = 2$ , define

$$\Theta_2 = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix}.$$

For  $N \geq 3$ , construct the transition matrix recursively:

$$\begin{aligned} \Theta_N &= p \begin{pmatrix} \Theta_{N-1} & 0 \\ 0 & 0 \end{pmatrix} + (1-p) \begin{pmatrix} 0 & \Theta_{N-1} \\ 0 & 0 \end{pmatrix} \\ &\quad + (1-q) \begin{pmatrix} 0 & 0 \\ \Theta_{N-1} & 0 \end{pmatrix} + q \begin{pmatrix} 0 & 0 \\ 0 & \Theta_{N-1} \end{pmatrix}. \end{aligned}$$

After constructing the matrix, divide all rows except the first and last rows by two so that every row sums to one.

The grid is symmetric and equally spaced, with endpoints

$$\bar{y}_1 = -\sqrt{N-1} \sigma_y, \quad \bar{y}_N = \sqrt{N-1} \sigma_y.$$

The resulting discrete process reproduces the original AR(1)'s:

- unconditional mean;
- conditional mean;
- unconditional variance;
- conditional variance;
- first-order autocorrelation.

## 6.4 Tauchen versus Rouwenhorst

Both methods produce

$$\{\bar{y}_1, \dots, \bar{y}_N\} \quad \text{and} \quad P.$$

They differ in what they are designed to preserve.

**Tauchen.** Tauchen uses the assumed CDF  $F$  of the innovation and therefore explicitly captures the distributional shape of  $\varepsilon_t$ .

**Rouwenhorst.** Rouwenhorst does not use higher-order distributional information about  $\varepsilon_t$ . It instead reproduces the first and second conditional and unconditional moments and the first-order autocorrelation.

Rouwenhorst is particularly useful when

$$0.95 \leq \phi < 1,$$

because Tauchen's method can perform poorly for highly persistent processes.

However, if skewness, kurtosis, or other higher moments of the innovation distribution are central to the application, Tauchen may be preferable.

| Tauchen                            | Rouwenhorst                       |
|------------------------------------|-----------------------------------|
| Uses the innovation CDF            | Does not use the innovation CDF   |
| Captures distributional shape      | Matches first and second moments  |
| Can struggle when $\phi \approx 1$ | Works well under high persistence |

## 7 Finite-State Markov Chains

### 7.1 Definition and Markov Property

A Markov chain is a discrete-time, discrete-state stochastic process satisfying the Markov property.

Let the process take values in the finite state space

$$\mathcal{S} = \{\bar{y}_1, \bar{y}_2, \dots, \bar{y}_N\}.$$

The Markov property is

$$\Pr(y_{t+1} \mid y_t, y_{t-1}, \dots, y_{t-k}) = \Pr(y_{t+1} \mid y_t)$$

for every  $k \geq 1$ .

In words, once the current state  $y_t$  is known, additional information about the past does not improve the conditional distribution of  $y_{t+1}$ . The current state summarizes all relevant history.

## 7.2 Elements of a Finite-State Markov Chain

A finite-state Markov chain has three components.

### 1. State grid.

$$\mathcal{S} = \{\bar{y}_1, \dots, \bar{y}_N\}.$$

### 2. Transition matrix.

The  $N \times N$  matrix  $P$  has elements

$$P_{ij} = \Pr(y_{t+1} = \bar{y}_j \mid y_t = \bar{y}_i).$$

The current state indexes the row and the future state indexes the column.

If the transition probabilities do not depend on  $t$ , the chain is *time invariant* or *homogeneous*.

### 3. Initial distribution.

The vector  $\pi_0$  describes probabilities over the initial state:

$$\pi_{0i} = \Pr(y_0 = \bar{y}_i),$$

with

$$\pi_{0i} \geq 0$$

and

$$\sum_{i=1}^N \pi_{0i} = 1.$$

## 7.3 Stochastic Matrices

A transition matrix must satisfy

$$P_{ij} \geq 0 \quad \text{for every } i, j$$

and

$$\sum_{j=1}^N P_{ij} = 1 \quad \text{for every row } i.$$

A nonnegative matrix whose rows sum to one is called a *stochastic matrix*.

Be careful: it is the rows, not the columns, that sum to one under the notation

$$P_{ij} = \Pr(y_{t+1} = \bar{y}_j \mid y_t = \bar{y}_i).$$

## 7.4 Evolution of Probability Distributions

Treat probability distributions as column vectors but write their transposes as row vectors when multiplying by  $P$ .

The one-period-ahead unconditional distribution is

$$\pi'_{t+1} = \pi'_t P.$$

Iterating,

$$\pi'_{t+k} = \pi'_t P^k.$$

In particular,

$$\pi'_k = \pi'_0 P^k.$$

The  $(i, j)$  element of  $P^k$  satisfies

$$(P^k)_{ij} = \Pr(y_{t+k} = \bar{y}_j \mid y_t = \bar{y}_i).$$

Thus, powers of the transition matrix characterize multi-period transition probabilities. Conditional expectations can also be computed directly. For any function  $g$ ,

$$\mathbb{E}[g(y_{t+1}) \mid y_t = \bar{y}_i] = \sum_{j=1}^N P_{ij} g(\bar{y}_j).$$

More generally,

$$\mathbb{E}[g(y_{t+k}) \mid y_t = \bar{y}_i] = \sum_{j=1}^N (P^k)_{ij} g(\bar{y}_j).$$

## 7.5 Stationary Distributions

A probability distribution  $\pi$  is *stationary* or *invariant* if applying the transition matrix leaves it unchanged:

$$\pi' = \pi' P.$$

Equivalently,

$$\pi'(I - P) = 0,$$

or, after transposing,

$$P' \pi = \pi.$$

Therefore,  $\pi$  is a right eigenvector of  $P'$  associated with the unit eigenvalue:

$$P' \pi = 1 \cdot \pi.$$

Equivalently,  $\pi'$  is a left eigenvector of  $P$  associated with the unit eigenvalue.

The eigenvector must be normalized to satisfy

$$\pi_i \geq 0$$

and

$$\sum_{i=1}^N \pi_i = 1.$$

## 7.6 Why a Unit Eigenvalue Always Exists

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Because every row of  $P$  sums to one,

$$P \mathbf{1} = \mathbf{1}.$$

Thus, one is an eigenvalue of  $P$ . Since  $P$  and  $P'$  have the same eigenvalues, one is also an eigenvalue of  $P'$ .

A finite stochastic matrix therefore has at least one stationary distribution.

**Important distinction.** The existence of a unit eigenvalue does not imply:

- that the unit eigenvalue is unique;
- that the stationary distribution is unique;
- that the chain converges to a stationary distribution from every initial condition.

These are separate questions.

## 7.7 Multiplicity and the Number of Stationary Distributions

Suppose  $\pi^a$  and  $\pi^b$  are two different stationary distributions:

$$(\pi^a)'P = (\pi^a)'$$

and

$$(\pi^b)'P = (\pi^b)'.$$

Then every convex combination

$$\pi^\lambda = \lambda\pi^a + (1 - \lambda)\pi^b, \quad \lambda \in [0, 1],$$

is also stationary:

$$\begin{aligned} (\pi^\lambda)'P &= \lambda(\pi^a)'P + (1 - \lambda)(\pi^b)'P \\ &= \lambda(\pi^a)' + (1 - \lambda)(\pi^b)' \\ &= (\pi^\lambda)'. \end{aligned}$$

Therefore,

Two distinct stationary distributions imply a continuum of stationary distributions.

The number of unit eigenvalues and the number of stationary distributions should therefore not be reported as the same object. For example, two linearly independent unit eigenvectors can generate infinitely many normalized stationary distributions through convex combinations.

## 7.8 Uniqueness versus Asymptotic Stationarity

A Markov chain has a *unique stationary distribution* if exactly one probability vector  $\pi$  satisfies

$$\pi' = \pi'P.$$

The chain is *asymptotically stationary with a unique invariant distribution* if, in addition,

$$\lim_{k \rightarrow \infty} \pi_0' P^k = \pi'$$

for every possible initial distribution  $\pi_0$ .

These properties are distinct:

Unique stationary distribution  $\not\Rightarrow$  convergence from every initial distribution.

A chain can possess a unique stationary distribution and nevertheless cycle forever because of periodicity.

## 7.9 A Sufficient Positivity Condition

A sufficient condition for uniqueness and convergence is

$$P_{ij} > 0 \quad \text{for every } i, j.$$

If every transition has strictly positive probability, then:

- the unit eigenvalue is unique;
- the stationary distribution is unique;
- all other eigenvalues have modulus strictly smaller than one;
- the chain is asymptotically stationary;
- the limiting distribution is independent of the initial distribution.

Thus,

$$P_{ij} > 0 \quad \forall i, j \quad \implies \quad \pi'_0 P^k \longrightarrow \pi' \quad \text{for every } \pi_0.$$

Strict positivity is a sufficient condition, not a necessary one.

## 7.10 Exam-Style Examples

### 7.10.1 Example 1: Multiple Absorbing States

Consider

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}.$$

States 1 and 2 are absorbing:

$$P_{11} = 1, \quad P_{22} = 1.$$

Therefore,

$$\pi^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

and

$$\pi^2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

are stationary distributions.

There are two unit eigenvalues corresponding to the two absorbing classes.

Every convex combination

$$\pi(\lambda) = \begin{pmatrix} \lambda \\ 1 - \lambda \\ 0 \end{pmatrix}, \quad \lambda \in [0, 1],$$

is also stationary.

Hence:

two unit eigenvalues, but a continuum of stationary distributions.

The initial condition matters in the long run because it determines how much probability ultimately reaches each absorbing state.

### 7.10.2 Example 2: A Periodic Chain

Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The stationary-distribution condition

$$\pi' = \pi' P$$

implies

$$\pi_1 = \pi_2.$$

Together with

$$\pi_1 + \pi_2 = 1,$$

this gives

$$\pi = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

The stationary distribution is unique.

However, starting from state 1,

$$\begin{pmatrix} 1 & 0 \end{pmatrix} P = \begin{pmatrix} 0 & 1 \end{pmatrix},$$

and

$$\begin{pmatrix} 1 & 0 \end{pmatrix} P^2 = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

The distribution alternates forever:

$$(1, 0) \longrightarrow (0, 1) \longrightarrow (1, 0) \longrightarrow \cdots.$$

Therefore,

the chain has a unique stationary distribution but does not converge to it from every initial condition.

This example demonstrates why uniqueness and asymptotic stationarity are not equivalent.

### 7.10.3 Example 3: Strictly Positive Transition Matrix

Consider

$$P = \begin{pmatrix} 0.9 & 0.05 & 0.05 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}.$$

Every element is strictly positive:

$$P_{ij} > 0 \quad \forall i, j.$$

Therefore, without computing any eigenvalues or eigenvectors, we can conclude:

there is one unit eigenvalue,  
there is one stationary distribution,  
the chain is asymptotically stationary.

The limiting distribution is independent of the initial distribution.

## 7.11 Inspection Rules for Markov-Chain Questions

For short exam questions, the following inspection rules are useful.

1. A row with a one on the diagonal and zeros elsewhere identifies an absorbing state.
2. Each separate absorbing class can support its own stationary distribution.
3. If there are two distinct stationary distributions, all their convex combinations are also stationary; hence there is a continuum.
4. A strictly positive transition matrix has a unique stationary distribution and converges to it from every initial distribution.
5. A unique stationary distribution does not by itself guarantee convergence. Check for deterministic or periodic cycling.
6. Always distinguish:

existence,      uniqueness,      asymptotic convergence.

## 7.12 Markov Chains in Dynamic Programming

Suppose an economic agent has:

- an endogenous state  $x$ ;
- an exogenous Markov state

$$y \in \{\bar{y}_1, \dots, \bar{y}_N\};$$

- a transition matrix  $P$  for the exogenous state.

If the current exogenous state is  $\bar{y}_i$ , then

$$\Pr(y' = \bar{y}_j \mid y = \bar{y}_i) = P_{ij}.$$

A stochastic Bellman equation can therefore be written as

$$V(x, \bar{y}_i) = \max_{x' \in \Gamma(x, \bar{y}_i)} \left\{ R(x, x', \bar{y}_i) + \beta \sum_{j=1}^N P_{ij} V(x', \bar{y}_j) \right\}.$$

The expectation over future states becomes a finite weighted sum:

$$\mathbb{E}[V(x', y') \mid y = \bar{y}_i] = \sum_{j=1}^N P_{ij} V(x', \bar{y}_j).$$

This is the main reason discretization is useful in quantitative macroeconomics. It replaces an integral over a continuous shock with a finite sum over Markov states:

$$\int V(x', y') dF(y' \mid y) \longrightarrow \sum_{j=1}^N P_{ij} V(x', \bar{y}_j).$$

The accuracy of the numerical economic solution then depends partly on how well the chosen grid and transition matrix approximate the original continuous-state stochastic process.

### 7.13 Core Summary

$$\begin{aligned} & y_t = \phi y_{t-1} + \varepsilon_t \\ & \Downarrow \text{Tauchen or Rouwenhorst} \\ & \{\bar{y}_1, \dots, \bar{y}_N\}, P \\ & \Downarrow \\ & \pi'_{t+k} = \pi'_t P^k \\ & \Downarrow \\ & \pi' = \pi' P \\ & \Downarrow \\ & V(x, \bar{y}_i) = \max_{x'} \left\{ R(x, x', \bar{y}_i) + \beta \sum_j P_{ij} V(x', \bar{y}_j) \right\}. \end{aligned}$$

The essential conceptual lessons are:

- ARCH and GARCH produce time-varying conditional variance from past realized shocks.
- Stochastic volatility introduces a separate innovation to uncertainty.
- Discretization transforms a continuous-state stochastic process into a finite grid and transition matrix.
- A Markov chain summarizes relevant history through the current state.
- A stationary distribution is a normalized unit-eigenvalue eigenvector of  $P'$ .
- Existence, uniqueness, and convergence of stationary distributions are distinct properties.

- A strictly positive transition matrix is sufficient for a unique invariant distribution and asymptotic stationarity.

## 8 The Neoclassical Growth Model and Dynamic Programming

### 8.1 The neoclassical environment

#### 8.1.1 Technology

Aggregate output is produced according to

$$Y_t = F(K_t, A_t L_t),$$

where  $K_t$  is capital,  $L_t$  is labor, and  $A_t$  is labor-augmenting technology. A neoclassical production function is constant returns to scale in  $(K, AL)$ ,

$$F(\lambda K, \lambda AL) = \lambda F(K, AL), \quad \lambda > 0,$$

and satisfies

$$F_K > 0, \quad F_L > 0, \quad F_{KK} < 0, \quad F_{LL} < 0,$$

together with essentiality of both inputs and the usual Inada conditions. Complementarity of capital and labor is represented by  $F_{KL} > 0$ .

**Example** (Cobb–Douglas production). For

$$F(K, AL) = K^\alpha (AL)^{1-\alpha}, \quad 0 < \alpha < 1,$$

the parameter  $\alpha$  is the capital share and  $1 - \alpha$  is the labor share.

Labor and technology grow exogenously according to

$$L_t = (1 + n)L_{t-1}, \quad A_t = (1 + \gamma)A_{t-1}.$$

Capital is predetermined and evolves according to the one-period time-to-build equation

$$K_{t+1} = (1 - \delta)K_t + I_t, \quad 0 < \delta < 1.$$

In a closed economy,

$$Y_t = C_t + I_t,$$

so aggregate saving equals aggregate investment.

#### 8.1.2 Intensive form

Define variables per efficiency unit of labor:

$$k_t \equiv \frac{K_t}{A_t L_t}, \quad y_t \equiv \frac{Y_t}{A_t L_t}, \quad c_t \equiv \frac{C_t}{A_t L_t}, \quad i_t \equiv \frac{I_t}{A_t L_t}.$$

Constant returns to scale imply

$$y_t = F\left(\frac{K_t}{A_t L_t}, 1\right) \equiv f(k_t).$$

The intensive production function inherits the standard neoclassical properties:

$$f'(k) > 0, \quad f''(k) < 0, \quad f(0) = 0, \quad \lim_{k \rightarrow 0^+} f'(k) = \infty, \quad \lim_{k \rightarrow \infty} f'(k) = 0.$$

The remaining aggregate conditions become

$$(1 + \gamma)(1 + n)k_{t+1} = (1 - \delta)k_t + i_t, \\ y_t = c_t + i_t.$$

Equivalently,

$$k_{t+1} = \frac{(1 - \delta)k_t + i_t}{(1 + \gamma)(1 + n)}.$$

When  $n = \gamma = 0$ , the key feasibility condition simplifies to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (8.1)$$

For Cobb–Douglas production,  $f(k) = k^\alpha$ .

### 8.1.3 Preferences

The representative household has lifetime utility

$$\sum_{t=0}^T \beta^t u(c_t) \quad \text{or} \quad \sum_{t=0}^{\infty} \beta^t u(c_t), \quad 0 < \beta < 1,$$

depending on whether the horizon is finite or infinite. Period utility satisfies

$$u'(c) > 0, \quad u''(c) < 0, \quad \lim_{c \rightarrow 0^+} u'(c) = \infty, \quad \lim_{c \rightarrow \infty} u'(c) = 0.$$

**Example** (CRRA utility).

$$u(c) = \begin{cases} \frac{c^{1-\sigma} - 1}{1 - \sigma}, & \sigma \neq 1, \\ \log c, & \sigma = 1, \end{cases} \quad \sigma > 0.$$

Then

$$u'(c) = c^{-\sigma}, \quad -\frac{cu''(c)}{u'(c)} = \sigma > 0, \quad \text{IES} = \frac{1}{\sigma}.$$

Thus the Arrow–Pratt coefficients include a minus sign and are positive for a risk-averse household.

## 8.2 Finite-horizon optimization

### 8.2.1 The planner's problem

Given  $k_0 > 0$ , the social planner solves

$$\max_{\{c_t, i_t, k_{t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t u(c_t)$$

subject to production, capital accumulation, feasibility, and nonnegativity. Using Equation (8.1) to eliminate  $c_t$ , the problem can be written with future capital as the only choice:

$$\max_{\{k_{t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t u(f(k_t) + (1 - \delta)k_t - k_{t+1}), \quad (8.2)$$

subject to

$$0 \leq k_{t+1} \leq f(k_t) + (1 - \delta)k_t.$$

### 8.2.2 Generic finite-horizon result

Consider

$$\max_{\{x_{t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t R(x_t, x_{t+1}), \quad x_{t+1} \in \Gamma(x_t) = [\underline{x}, \bar{\gamma}(x_t)],$$

with  $x_0$  given. If  $R$  is differentiable and jointly concave, and the solution is interior through period  $T - 1$ , optimality is characterized by

$$R_2(x_t, x_{t+1}) + \beta R_1(x_{t+1}, x_{t+2}) = 0, \quad t = 0, \dots, T - 1, \quad (8.3)$$

$$x_{T+1} = \underline{x}. \quad (8.4)$$

The Euler equation equates the marginal cost today of increasing  $x_{t+1}$  with its discounted marginal benefit tomorrow. The terminal condition follows because the state has no value beyond the last period.

### 8.2.3 Euler equation and terminal condition in the NGM

In Equation (8.2),

$$R(k_t, k_{t+1}) = u(f(k_t) + (1 - \delta)k_t - k_{t+1}).$$

Applying Equations (8.3) and (8.4) gives

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)], \quad t = 0, \dots, T - 1, \quad (8.5)$$

$$k_{T+1} = 0. \quad (8.6)$$

The left-hand side of Equation (8.5) is the marginal utility cost of sacrificing one unit of current consumption. The right-hand side is the discounted marginal utility benefit of the extra output and undepreciated capital available next period. The terminal condition says that all remaining resources are consumed in period  $T$ :

$$c_T = f(k_T) + (1 - \delta)k_T.$$

With CRRA utility, the Euler equation can also be written

$$\frac{c_{t+1}}{c_t} = \{\beta [f'(k_{t+1}) + (1 - \delta)]\}^{1/\sigma}.$$

#### Deriving a discrete-time Euler equation

1. Use all static constraints to express current consumption or current profit as a function of the current state and next period's state.
2. Differentiate the lifetime objective with respect to the chosen next-period state. It appears negatively in today's payoff and positively in tomorrow's payoff.

3. Cancel common discount factors and multipliers.

4. Write the result as

$$\text{marginal cost today} = \text{discounted expected marginal benefit tomorrow.}$$

5. Add the appropriate terminal condition in finite horizon or transversality condition in infinite horizon.

### 8.3 Infinite-horizon optimization

#### 8.3.1 Sequential formulation

The infinite-horizon planner solves

$$\max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(f(k_t) + (1 - \delta)k_t - k_{t+1}), \quad (8.7)$$

subject to

$$0 \leq k_{t+1} \leq f(k_t) + (1 - \delta)k_t \quad \forall t.$$

Besides feasibility, one must require that lifetime utility be well defined. Relative to finite horizon, the substantive change is that there is no last period and therefore no terminal value  $k_{T+1} = 0$ .

#### 8.3.2 Euler equation, interiority, and the TVC

For the generic infinite-horizon problem

$$\max_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t R(x_t, x_{t+1}), \quad x_{t+1} \in \Gamma(x_t),$$

the relevant conditions are

$$R_2(x_t, x_{t+1}) + \beta R_1(x_{t+1}, x_{t+2}) = 0, \quad t \geq 0, \quad (8.8)$$

$$x_{t+1} \in \text{int } \Gamma(x_t), \quad t \geq 0, \quad (8.9)$$

$$\lim_{t \rightarrow \infty} \beta^t R_1(x_t, x_{t+1}) x_t = 0. \quad (8.10)$$

The transversality condition rules out paths that satisfy the local Euler equations but overaccumulate the state asymptotically.

For the NGM, these conditions become

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + (1 - \delta)], \quad t \geq 0, \quad (8.11)$$

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) [f'(k_t) + (1 - \delta)] k_t = 0. \quad (8.12)$$

The Euler equation alone does not select the optimal infinite path; the TVC eliminates paths with excessive asymptotic saving.

### 8.3.3 KKT derivation and the role of Inada conditions

Write

$$\begin{aligned}\mathcal{L} = & \sum_{t=0}^{\infty} \beta^t u(f(k_t) + (1-\delta)k_t - k_{t+1}) \\ & + \sum_{t=0}^{\infty} \lambda_t (f(k_t) + (1-\delta)k_t - k_{t+1}) + \sum_{t=0}^{\infty} \mu_t k_{t+1}.\end{aligned}$$

The complementary-slackness conditions are

$$\lambda_t c_t = 0, \quad \mu_t k_{t+1} = 0.$$

The Inada condition  $\lim_{c \rightarrow 0^+} u'(c) = \infty$  implies  $c_t > 0$ , while  $\lim_{k \rightarrow 0^+} f'(k) = \infty$  implies  $k_{t+1} > 0$ . Hence

$$\lambda_t = \mu_t = 0,$$

and the first-order condition with respect to  $k_{t+1}$  reduces to Equation (8.11). Thus the Inada conditions provide the economic justification for focusing on an interior solution.

### 8.3.4 Graphical interpretation

Fix  $k_t$  and  $k_{t+2}$ . As a function of  $k_{t+1}$ , the marginal cost of saving is

$$\underbrace{u'(f(k_t) + (1-\delta)k_t - k_{t+1})}_{\text{LHS}},$$

which is increasing because larger saving lowers  $c_t$  and  $u'' < 0$ . The marginal benefit is

$$\underbrace{\beta u'(f(k_{t+1}) + (1-\delta)k_{t+1} - k_{t+2}) [f'(k_{t+1}) + (1-\delta)]}_{\text{RHS}},$$

which is decreasing because both marginal utility tomorrow and the marginal product of capital fall with  $k_{t+1}$ . Their intersection determines optimal saving.

### 8.3.5 Steady state and transition dynamics

In a steady state,

$$k_t = k_{t+1} = \bar{k}, \quad c_t = \bar{c} = f(\bar{k}) - \delta \bar{k}.$$

The Euler equation reduces to

$$1 = \beta [f'(\bar{k}) + (1-\delta)] \iff f'(\bar{k}) = \frac{1}{\beta} - 1 + \delta. \quad (8.13)$$

Strict concavity and the Inada conditions imply that  $\bar{k}$  exists and is unique. With  $f(k) = k^\alpha$ ,

$$\bar{k} = \left( \frac{\alpha}{\beta^{-1} - 1 + \delta} \right)^{\frac{1}{1-\alpha}}.$$

The model is globally stable and exhibits monotone convergence:

$$k_0 < \bar{k} \implies k_t \uparrow \bar{k}, \quad k_0 > \bar{k} \implies k_t \downarrow \bar{k}.$$

When capital is scarce, its marginal product is high and the planner accumulates it. When capital is abundant, its marginal product is low and the planner decumulates it.

**Remark.** Capital accumulation generates transitional growth but not sustained growth in income per person. Diminishing returns eventually force capital per efficiency unit toward  $\bar{k}$ .

### 8.3.6 Decentralization

A competitive firm solves

$$\max_{K_t, L_t} F(K_t, A_t L_t) - R_t^k K_t - W_t L_t,$$

so

$$R_t^k = F_K(K_t, A_t L_t), \quad W_t = F_L(K_t, A_t L_t).$$

Euler's theorem and constant returns imply

$$F(K_t, A_t L_t) = R_t^k K_t + W_t L_t,$$

so profits are zero. The household solves

$$\max_{\{C_t, K_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(C_t)$$

subject to

$$C_t + K_{t+1} = W_t L_t + [R_t^k + (1 - \delta)] K_t.$$

The gross return to saving in capital is therefore

$$1 + r_t = R_t^k + (1 - \delta).$$

Substituting the firm's first-order conditions and using Euler's theorem turns the household budget constraint into the planner's resource constraint. Hence the competitive equilibrium and planner allocation coincide: the decentralized equilibrium is efficient.

### 8.3.7 Positive productivity and population growth

When  $\gamma, n > 0$ , the balanced-growth conclusion is

$$k_t \rightarrow \bar{k}, \quad y_t \rightarrow \bar{y} = f(\bar{k}),$$

while

$$\frac{Y_t}{L_t} = A_t y_t \longrightarrow \bar{y} A_0 (1 + \gamma)^t.$$

Thus long-run income per person grows at the technology growth rate  $\gamma$ . Population growth raises aggregate output but does not by itself raise output per person. Along the balanced growth path, aggregate  $Y_t, K_t, C_t$ , and  $I_t$  grow at the common gross rate  $(1 + \gamma)(1 + n)$ .

## 8.4 Dynamic programming

### 8.4.1 Sequential and recursive formulations

Consider the stationary infinite-horizon problem

$$\max_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t R(x_t, x_{t+1}), \quad x_{t+1} \in \Gamma(x_t),$$

with  $x_0$  given. Its recursive representation is the Bellman equation

$$V(x) = \max_{x' \in \Gamma(x)} \{R(x, x') + \beta V(x')\}. \quad (8.14)$$

The value function  $V(x)$  is optimized lifetime utility starting from state  $x$ . The policy function is

$$g(x) = \operatorname{argmax}_{x' \in \Gamma(x)} \{R(x, x') + \beta V(x')\}.$$

Bellman's principle of optimality states that the recursive policy generates the same optimal sequence as the sequential problem:

$$x_{t+1}^* = g(x_t^*), \quad x_0^* = x_0.$$

The NGM Bellman equation is

$$V(k) = \max_{0 \leq k' \leq f(k) + (1-\delta)k} \{u(f(k) + (1-\delta)k - k') + \beta V(k')\}. \quad (8.15)$$

Here  $k$  is the state,  $k'$  is the control as well as next period's state, and consumption is implied by feasibility.

### 8.4.2 Properties of the value function

Under the standard regularity conditions:

- if  $R_1 > 0$  and  $\Gamma(x)$  expands with  $x$ , then  $V$  is increasing;
- if  $R$  is concave and  $\Gamma$  is convex, then  $V$  is concave;
- with suitable differentiability and interiority,  $V'(x)$  exists.

Accordingly, the NGM value function is increasing, concave, and smooth in capital.

### 8.4.3 The Bellman operator and value function iteration

Define the Bellman operator

$$(T\tilde{V})(x) \equiv \max_{x' \in \Gamma(x)} \{R(x, x') + \beta \tilde{V}(x')\}.$$

Starting from a guess  $V_0$ , value function iteration sets

$$V_{n+1} = TV_n.$$

The operator is a contraction in the sup norm:

$$\|TV - TW\|_{\infty} \leq \beta \|V - W\|_{\infty}.$$

Therefore, the fixed point is unique and

$$\|V_n - V\|_\infty \leq \beta^n \|V_0 - V\|_\infty \longrightarrow 0.$$

Computationally, one chooses a grid for the state, guesses  $V_0$ , solves the one-period maximization at every grid point, stores both  $V_{n+1}$  and the maximizing policy, and repeats until the sup-norm difference is small.

#### 8.4.4 FOC, envelope condition, and functional Euler equation

For an interior optimum in Equation (8.14), the first-order condition is

$$R_2(x, g(x)) + \beta V'(g(x)) = 0. \quad (8.16)$$

Differentiating the Bellman equation and using Equation (8.16) eliminates the indirect effect operating through  $g(x)$ :

$$V'(x) = R_1(x, g(x)). \quad (8.17)$$

This is the envelope or Benveniste–Scheinkman condition. Combining the FOC and envelope condition gives the functional Euler equation

$$R_2(x, g(x)) + \beta R_1(g(x), g(g(x))) = 0. \quad (8.18)$$

Writing  $x' = g(x)$  and  $x'' = g(x')$  reduces this to the familiar expression

$$R_2(x, x') + \beta R_1(x', x'') = 0.$$

For the NGM, define

$$c = f(k) + (1 - \delta)k - k', \quad c' = f(k') + (1 - \delta)k' - k''.$$

The Bellman FOC and envelope condition are

$$\begin{aligned} u'(c) &= \beta V'(k'), \\ V'(k) &= u'(c) [f'(k) + (1 - \delta)]. \end{aligned}$$

Evaluating the envelope condition at  $k'$  and substituting into the FOC yields

$$u'(c) = \beta u'(c') [f'(k') + (1 - \delta)],$$

which is exactly the sequential Euler equation. Recursive and sequential methods therefore encode the same optimal allocation.

#### Bellman derivation under exam conditions

1. Identify predetermined endogenous states and current exogenous states.
2. Use feasibility to eliminate as many controls as possible.
3. Write  $V(\text{state}) = \max\{\text{current payoff} + \beta E[V(\text{next state})]\}$ .
4. Take one FOC for each control.

5. Take one envelope condition for each endogenous state.
6. Evaluate the envelope condition at next period's state and substitute it into the FOC.
7. Interpret the final Euler equation as marginal cost equals discounted expected marginal benefit.

#### 8.4.5 Useful extension: several capital stocks

Suppose output is  $f(k, q) = k^\alpha q^\nu$  and the two capital stocks depreciate at rates  $\delta_k$  and  $\delta_q$ . The recursive problem is

$$V(k, q) = \max_{k', q'} \{u(k^\alpha q^\nu + (1 - \delta_k)k + (1 - \delta_q)q - k' - q') + \beta V(k', q')\}.$$

The two Euler equations are

$$\begin{aligned} u'(c) &= \beta u'(c') [\alpha(k')^{\alpha-1}(q')^\nu + (1 - \delta_k)] , \\ u'(c) &= \beta u'(c') [\nu(k')^\alpha(q')^{\nu-1} + (1 - \delta_q)] . \end{aligned}$$

The logic is unchanged: each asset has the same marginal utility cost today but its own marginal product and undepreciated payoff tomorrow.

**Summary 3.** *The core chain of results is*

$$\begin{aligned} \text{feasibility} &\implies \text{planner problem} \implies \text{Euler equation} + \text{boundary condition} \\ &\implies \text{steady state and transition dynamics.} \end{aligned}$$

*Sequential and recursive methods are equivalent. In finite horizon, the boundary condition is  $k_{T+1} = 0$ ; in infinite horizon it is replaced by the TVC. Diminishing returns imply convergence of capital per efficiency unit, so only productivity can generate sustained growth in income per person.*

## 9 Endogenous Growth

### 9.1 Why the neoclassical model does not explain long-run growth

In the neoclassical growth model, capital per efficiency unit of labor converges to a steady state:

$$k_t \equiv \frac{K_t}{A_t L_t} \longrightarrow \bar{k}.$$

By continuity of the intensive production function,

$$y_t \equiv \frac{Y_t}{A_t L_t} = f(k_t) \longrightarrow \bar{y} = f(\bar{k}),$$

and therefore

$$\frac{Y_t}{L_t} = A_t y_t \longrightarrow A_t \bar{y}.$$

Capital accumulation can generate transitional growth, but diminishing returns prevent it from generating permanent growth in income per worker. On the balanced growth path, long-run growth in living standards is entirely driven by productivity  $A_t$ . Since the neoclassical model takes the path of  $A_t$  as exogenous, it does not provide a fundamental explanation for long-run growth.

**Intuition.** Endogenous growth theory keeps the neoclassical lesson that ideas are the ultimate source of sustained growth, but makes the production of ideas an economic activity. Resources allocated to research sacrifice current output in exchange for a larger future stock of knowledge.

## 9.2 Growth mechanics in continuous time

For a differentiable variable  $X(t)$ , define

$$\dot{X}(t) \equiv \frac{dX(t)}{dt}, \quad g_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{d \log X(t)}{dt}.$$

If  $X$  grows at a constant instantaneous rate  $g$ , the following representations are equivalent:

$$\frac{\dot{X}(t)}{X(t)} = g, \tag{9.1}$$

$$X(t) = X(0)e^{gt}, \tag{9.2}$$

$$\log X(t) = \log X(0) + gt. \tag{9.3}$$

Useful growth-accounting rules are

$$X(t) = Y(t)Z(t) \implies g_X = g_Y + g_Z,$$

$$X(t) = \frac{Y(t)}{Z(t)} \implies g_X = g_Y - g_Z,$$

$$X(t) = Y(t)^\alpha Z(t)^\nu \implies g_X = \alpha g_Y + \nu g_Z.$$

**Remark.** A dot always denotes a derivative with respect to time:

$$\dot{X}(t) = \frac{dX(t)}{dt}.$$

It is the ratio  $\dot{X}(t)/X(t)$ , not  $\dot{X}(t)$  itself, that equals  $d \log X(t)/dt$ .

## 9.3 A baseline idea-driven growth model

### 9.3.1 Technology and labor allocation

Consider

$$Y(t) = A(t)L(t), \tag{9.4}$$

$$\dot{A}(t) = A(t)S(t), \tag{9.5}$$

$$L(t) + S(t) = N(t). \tag{9.6}$$

Here  $A(t)$  is the stock of ideas,  $L(t)$  is labor producing final output, and  $S(t)$  is scientific labor producing new ideas. Equation (9.5) contains two forces:

- more scientific labor raises current innovation;
- a larger stock of existing ideas makes future discoveries easier.

The second force is an intertemporal knowledge spillover, or *standing on the shoulders of giants*.

### 9.3.2 Exogenous research intensity

Suppose

$$N(t) = \bar{N}, \quad S(t) = \bar{s}\bar{N}, \quad L(t) = (1 - \bar{s})\bar{N}, \quad 0 < \bar{s} < 1.$$

Then

$$\frac{\dot{A}(t)}{A(t)} = S(t) = \bar{s}\bar{N} \equiv g_A,$$

so

$$A(t) = A(0)e^{\bar{s}\bar{N}t}.$$

Output per person is

$$\begin{aligned} \frac{Y(t)}{N(t)} &= (1 - \bar{s})A(t) \\ &= (1 - \bar{s})A(0)e^{\bar{s}\bar{N}t}, \end{aligned} \tag{9.7}$$

and hence

$$g_{Y/N} = g_A = \bar{s}\bar{N}.$$

**Intuition** (Level effect versus growth effect). An increase in  $\bar{s}$  has two opposing effects:

- *negative level effect*: fewer workers produce current output, so the factor  $1 - \bar{s}$  falls;
- *positive growth effect*: more workers create ideas, so  $g_A = \bar{s}\bar{N}$  rises.

Research is analogous to saving: it is costly today but expands future productive capacity.

## 9.4 Two interpretations of ideas

### 9.4.1 Expanding varieties

In the Romer expanding-variety interpretation, the stock  $A(t)$  is the measure of available intermediate varieties  $j \in [0, A(t)]$ . Final output is produced with the CES aggregator

$$Y(t) = \left( \int_0^{A(t)} y_j(t)^\rho dj \right)^{1/\rho}, \quad 0 < \rho < 1, \tag{9.8}$$

where

$$\sigma \equiv \frac{1}{1 - \rho} > 1$$

is the elasticity of substitution across varieties. Each intermediate is produced linearly with labor:

$$y_j(t) = l_j(t), \quad \int_0^{A(t)} l_j(t) dj = L(t).$$

Taking  $A(t)$  and  $L(t)$  as given, the efficient allocation of production labor solves

$$\max_{\{l_j(t)\}} \left( \int_0^{A(t)} l_j(t)^\rho dj \right)^{1/\rho} \quad \text{subject to} \quad \int_0^{A(t)} l_j(t) dj \leq L(t).$$

The first-order condition is identical for every  $j$ , so symmetry implies

$$l_j(t) = y_j(t) = \frac{L(t)}{A(t)} \quad \forall j. \tag{9.9}$$

Substitution into Equation (9.8) gives

$$\begin{aligned} Y(t) &= \left[ A(t) \left( \frac{L(t)}{A(t)} \right)^\rho \right]^{1/\rho} \\ &= A(t)^{\frac{1-\rho}{\rho}} L(t). \end{aligned} \quad (9.10)$$

Because  $(1 - \rho)/\rho > 0$ , the technology exhibits gains from variety.

With the exogenous research allocation  $S(t) = \bar{s}\bar{N}$  and  $L(t) = (1 - \bar{s})\bar{N}$ ,

$$g_A = \bar{s}\bar{N}, \quad g_{Y/N} = \frac{1-\rho}{\rho} g_A = \frac{1-\rho}{\rho} \bar{s}\bar{N}.$$

The later Romer analysis imposes

$$\rho = \frac{1}{2},$$

in which case Equation (9.10) simplifies to

$$Y(t) = A(t)L(t). \quad (9.11)$$

#### 9.4.2 Quality ladders and creative destruction

In the Schumpeterian interpretation, the set of product lines is fixed,  $j \in [0, 1]$ , but the quality  $A_j(t)$  within each line improves. Let

$$\log Y(t) = \int_0^1 \log y_j(t) dj, \quad (9.12)$$

$$y_j(t) = A_j(t)l_j(t). \quad (9.13)$$

Efficient allocation of a fixed amount  $L(t)$  of production labor implies

$$l_j(t) = L(t) \quad \forall j,$$

because the continuum of product lines has unit mass. Define aggregate productivity as the geometric mean of line-specific qualities:

$$A(t) \equiv \exp \left\{ \int_0^1 \log A_j(t) dj \right\}.$$

Then Equations (9.12) and (9.13) aggregate to

$$Y(t) = A(t)L(t).$$

Suppose a fraction  $\bar{s}$  of labor is allocated to research. During a small interval  $\Delta t$ , each product line innovates with probability  $\bar{s}\Delta t$ . An innovation replaces the previous technology with one that is  $1 + \lambda$  times as productive, where  $\lambda > 0$ . By the law of large numbers,

$$\log A(t + \Delta t) - \log A(t) \approx \bar{s}\Delta t \log(1 + \lambda).$$

Taking  $\Delta t \rightarrow 0$ ,

$$\frac{d \log A(t)}{dt} = \bar{s} \log(1 + \lambda) \equiv g_A.$$

Therefore,

$$A(t) = A(0)e^{\bar{s} \log(1 + \lambda)t},$$

and, with  $L(t) = (1 - \bar{s})\bar{N}$ ,

$$\frac{Y(t)}{N(t)} = (1 - \bar{s})A(0)e^{\bar{s}\log(1+\lambda)t}, \quad g_{Y/N} = g_A = \bar{s}\log(1 + \lambda).$$

Growth now comes from *creative destruction*: new high-quality versions displace old low-quality versions.

**Remark.** Expanding-variety and quality-ladder models give different microeconomic meanings to  $A(t)$ , but share the same central mechanism:

$$\text{resources devoted to research} \implies \text{innovation} \implies \text{sustained productivity growth.}$$

## 9.5 Continuous-time dynamic optimization

### 9.5.1 The current-value Hamiltonian

Consider the optimal-control problem

$$\max_{\{u(t)\}_{t \geq 0}} \int_0^\infty e^{-\delta t} h(x(t), u(t)) dt \quad \text{subject to} \quad \dot{x}(t) = g(x(t), u(t)), \quad (9.14)$$

with  $x(0)$  given and  $\delta > 0$ . The variable  $x(t)$  is a state and  $u(t)$  is a control. The current-value Hamiltonian is

$$H(x, u, \lambda) = h(x, u) + \lambda g(x, u), \quad (9.15)$$

where  $\lambda(t)$  is the current-value shadow value of the state.

For an interior solution, the canonical conditions are

$$H_u(x(t), u(t), \lambda(t)) = 0, \quad (9.16)$$

$$\dot{\lambda}(t) = \delta \lambda(t) - H_x(x(t), u(t), \lambda(t)), \quad (9.17)$$

$$\dot{x}(t) = H_\lambda(x(t), u(t), \lambda(t)) = g(x(t), u(t)), \quad (9.18)$$

$$\lim_{t \rightarrow \infty} e^{-\delta t} \lambda(t) x(t) = 0. \quad (9.19)$$

Under the usual differentiability and concavity conditions, these conditions are also sufficient.

**Remark** (Present-value versus current-value Hamiltonians). The present-value Hamiltonian is

$$H^{PV}(x, u, \mu, t) = e^{-\delta t} h(x, u) + \mu g(x, u).$$

Its costate equation is

$$\dot{\mu}(t) = -H_x^{PV},$$

with no  $\delta \mu(t)$  term. The two formulations are equivalent because

$$\mu(t) = e^{-\delta t} \lambda(t).$$

#### Solving a continuous-time control problem

1. Identify each predetermined state, each control, and every law of motion.
2. Use static constraints to eliminate redundant controls whenever possible.
3. Write the *current-value* Hamiltonian  $H = \text{flow payoff} + \sum_j \lambda_j \dot{x}_j$ .

4. Take one first-order condition for each interior control.
5. For each state, write  $\dot{\lambda}_j = \delta\lambda_j - H_{x_j}$ .
6. Append the state equations and one transversality condition for each state.
7. Log-differentiate the control FOC when it links the costate to marginal utility; then combine it with the costate equation.
8. On a balanced growth path, impose the growth-rate restrictions implied by resource constraints and bounded factor supplies.

## 9.6 The Romer social planner problem

### 9.6.1 Reduction to one state and one control

Return to the expanding-variety economy and impose

$$N(t) = \bar{N}, \quad \rho = \frac{1}{2}, \quad \gamma > 0,$$

and CRRA utility

$$u(C) = \begin{cases} \frac{C^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \log C, & \gamma = 1. \end{cases}$$

There is no physical capital, so

$$C(t) = Y(t) = A(t)L(t).$$

Using  $L(t) + S(t) = \bar{N}$ ,

$$S(t) = \bar{N} - L(t) = \bar{N} - \frac{C(t)}{A(t)}.$$

The idea-production technology becomes

$$\begin{aligned} \dot{A}(t) &= A(t)S(t) \\ &= A(t)\bar{N} - C(t). \end{aligned} \tag{9.20}$$

Hence the planner's problem reduces to

$$\max_{\{C(t)\}_{t \geq 0}} \int_0^\infty e^{-\delta t} u(C(t)) dt \quad \text{subject to} \quad \dot{A}(t) = \bar{N}A(t) - C(t), \tag{9.21}$$

with  $A(0)$  given. Consumption is the control and the stock of ideas is the state.

### 9.6.2 Optimality conditions and balanced growth

The current-value Hamiltonian is

$$H = u(C(t)) + \lambda(t) [\bar{N}A(t) - C(t)].$$

The control and costate equations are

$$u'(C(t)) = \lambda(t), \quad (9.22)$$

$$\dot{\lambda}(t) = \delta\lambda(t) - \bar{N}\lambda(t). \quad (9.23)$$

Under CRRA utility,

$$\lambda(t) = C(t)^{-\gamma},$$

so log differentiation gives

$$\frac{\dot{\lambda}(t)}{\lambda(t)} = -\gamma \frac{\dot{C}(t)}{C(t)}.$$

Combining this result with Equation (9.23),

$$-\gamma \frac{\dot{C}(t)}{C(t)} = \delta - \bar{N},$$

and therefore

$$g_{SP} \equiv \frac{\dot{C}(t)}{C(t)} = \frac{\bar{N} - \delta}{\gamma}. \quad (9.24)$$

Since  $C(t) = Y(t)$ ,  $g_C = g_Y$ . Moreover,

$$Y(t) = A(t)L(t), \quad 0 \leq L(t) \leq \bar{N}.$$

On a balanced growth path, labor cannot grow without bound, so  $L(t)$  is constant and

$$g_C = g_Y = g_A.$$

The idea-production function implies

$$g_A = \frac{\dot{A}(t)}{A(t)} = S(t).$$

Thus the planner's complete balanced-growth allocation satisfies

$$g_C = g_Y = g_{Y/N} = g_A = g_{SP} = \frac{\bar{N} - \delta}{\gamma}, \quad (9.25)$$

$$S^{SP} = g_{SP}, \quad (9.26)$$

$$L^{SP} = \bar{N} - g_{SP}, \quad (9.27)$$

$$A(t) = A(0)e^{g_{SP}t}, \quad (9.28)$$

$$C(t) = Y(t) = A(t)L^{SP}. \quad (9.29)$$

An interior positive-growth solution requires

$$0 < g_{SP} < \bar{N}.$$

In particular,  $\bar{N} > \delta$  is necessary for positive growth.

### 9.6.3 Comparative statics

The expression

$$g_{SP} = \frac{\bar{N} - \delta}{\gamma}$$

has three central implications:

- A larger discount rate  $\delta$  makes the household more impatient, shifts labor toward current production, and lowers growth.
- Since  $1/\gamma$  is the intertemporal elasticity of substitution, a larger  $\gamma$  means stronger consumption-smoothing motives and lowers the willingness to sacrifice current consumption for future growth.<sup>4</sup>
- A larger population  $\bar{N}$  provides more potential researchers and raises the long-run growth rate. This is the model's strong scale effect.

## 9.7 The increasing-returns problem

Ideas are a special input:

- they are *nonrival*: the same idea can be used by several producers at the same time;
- they may be *partially excludable*: patents or other institutions can restrict their use.

Suppose  $F(K, AL)$  is constant returns to scale in the rival inputs  $K$  and  $L$ , taking the idea stock  $A$  as given:

$$F(\eta K, A\eta L) = \eta F(K, AL).$$

If  $A$  is also treated as an input and all three inputs are scaled by  $\eta > 1$ , then

$$F(\eta K, \eta A \eta L) = \eta F(K, \eta AL) > \eta F(K, AL).$$

Production therefore exhibits increasing returns in  $(K, L, A)$  jointly.

Under perfect competition, capital and labor are paid their marginal products:

$$R^K = F_K, \quad W = F_L.$$

Euler's theorem then implies

$$F(K, AL) = R^K K + WL.$$

All output is exhausted paying rival factors; no residual remains to cover the fixed cost of inventing or acquiring an idea. Consequently, a model with costly innovation generally cannot sustain innovation under perfect competition in every market.

**Intuition** (Static versus dynamic efficiency). Market power creates markups and a static deadweight loss, but the profits generated by those markups give firms an incentive to incur the fixed cost of innovation. Patent protection therefore trades off inefficient use of existing ideas against incentives to create new ideas.

## 9.8 The decentralized Romer equilibrium

### 9.8.1 Agents and their problems

Maintain

$$Y(t) = \left( \int_0^{A(t)} y_j(t)^\rho dj \right)^{1/\rho}, \quad y_j(t) = l_j(t), \quad \dot{A}(t) = A(t)S(t),$$

---

<sup>4</sup>An increase in  $\gamma$  means a lower IES =  $1/\gamma$ . The agent is less willing to exchange present consumption for future consumption. Look that in the Euler Eq. or in Eq. (9.24) when  $\gamma$  increases then  $C(t)$  increases, ceteris paribus.

together with

$$L(t) + S(t) = \bar{N}, \quad C(t) = Y(t).$$

**Representative household.** Taking  $r(t)$ ,  $W(t)$ , and aggregate profits  $\Pi(t)$  as given, the household solves

$$\max_{\{C(t)\}_{t \geq 0}} \int_0^\infty e^{-\delta t} u(C(t)) dt$$

subject to

$$\dot{B}(t) = r(t)B(t) - C(t) + W(t)\bar{N} + \Pi(t), \quad (9.30)$$

with financial assets in zero net supply in equilibrium:

$$B(t) = 0.$$

The household's optimality conditions imply

$$r(t) = \delta + \gamma \frac{\dot{C}(t)}{C(t)}. \quad (9.31)$$

**Competitive final-goods firm.** Taking intermediate prices  $\{p_j(t)\}$  as given, the firm solves

$$\max_{\{y_j(t)\}} \left\{ Y(t) - \int_0^{A(t)} p_j(t) y_j(t) dj \right\}.$$

The demand condition for each variety is

$$p_j(t) = Y(t)^{1-\rho} y_j(t)^{\rho-1}, \quad (9.32)$$

$$y_j(t) = Y(t) p_j(t)^{-\sigma}, \quad \sigma = \frac{1}{1-\rho}. \quad (9.33)$$

**Monopolistic intermediate-goods firms.** The patent owner for variety  $j$  chooses its price:

$$\max_{p_j(t)} \pi_j(t) = [p_j(t) - W(t)] y_j(p_j(t)).$$

Since demand has constant elasticity  $\sigma$ , optimal pricing satisfies

$$p_j(t) = \frac{\sigma}{\sigma-1} W(t) = \frac{1}{\rho} W(t). \quad (9.34)$$

Aggregate operating profits are

$$\Pi(t) = \int_0^{A(t)} \pi_j(t) dj.$$

Free entry into ownership of a newly invented variety makes its patent price equal to the present discounted value of monopoly profits:

$$P_A(t) = \int_t^\infty \exp \left\{ - \int_t^\tau r(s) ds \right\} \pi_j(\tau) d\tau. \quad (9.35)$$

Equivalently, the patent satisfies the asset-pricing.<sup>5</sup>

$$r(t)P_A(t) = \pi_j(t) + \dot{P}_A(t).$$

**Competitive research firm.** Taking  $A(t)$ ,  $P_A(t)$ , and  $W(t)$  as given, the research firm solves

$$\max_{S(t)} \left\{ P_A(t)\dot{A}(t) - W(t)S(t) \right\} = \max_{S(t)} \left\{ P_A(t)A(t)S(t) - W(t)S(t) \right\}.$$

For an interior research allocation,

$$P_A(t)A(t) = W(t). \quad (9.36)$$

The private marginal value of scientific labor is the value of the new patents it creates.

**Definition** (Romer equilibrium). An equilibrium is a collection of price paths

$$\{r(t), W(t), P_A(t), \{p_j(t)\}_{j \in [0, A(t)]}\}_{t \geq 0}$$

and quantity paths

$$\{C(t), B(t), Y(t), A(t), S(t), L(t), \{y_j(t)\}_{j \in [0, A(t)]}, \Pi(t)\}_{t \geq 0}$$

such that the household, final-goods firm, intermediate firms, and research firm optimize; the final-good, intermediate-good, labor, and asset markets clear; and all production, innovation, aggregation, and resource constraints hold. A balanced-growth-path equilibrium additionally requires all variables to grow at mutually consistent constant rates.

### 9.8.2 Derivation of the equilibrium growth rate

Set  $\rho = 1/2$ , so

$$\sigma = 2, \quad Y(t) = A(t)L(t), \quad y_j(t) = \frac{L(t)}{A(t)}.$$

The markup rule becomes  $p_j(t) = 2W(t)$ , and profits per variety and aggregate profits satisfy

$$\pi_j(t) = W(t)y_j(t) = \frac{W(t)L(t)}{A(t)}, \quad (9.37)$$

$$\Pi(t) = A(t)\pi_j(t) = W(t)L(t). \quad (9.38)$$

With  $B(t) = \dot{B}(t) = 0$ , the household budget constraint gives

$$C(t) = W(t)\bar{N} + \Pi(t) = W(t) [\bar{N} + L(t)].$$

---

<sup>5</sup>By no arbitrage, the return over the patent,  $r(t)P_A(t)$  needs to be equal to the dividends/flow profits,  $\pi_j$ , plus a capital gain,  $\dot{P}_A(t)$ . To derive this expression, define

$$D(t, \tau) \equiv \exp \left\{ - \int_t^\tau r(s) ds \right\}$$

as the discount factor and differentiate with respect to  $t$  using Leibniz's rule, keeping track of the minus sign inside the exponent. This gives

$$\dot{P}_A(t) = -\pi_j(t) + r(t) \int_t^\infty D(t, \tau) \pi_j(\tau) d\tau.$$

Since the integral is exactly  $P_A(t)$ , we obtain

$$r(t)P_A(t) = \pi_j(t) + \dot{P}_A(t).$$

On a balanced growth path,  $g_A$  is constant. Since  $g_A = \dot{A}(t)/A(t) = S(t)$ , scientific labor is constant; labor-market clearing then makes  $L(t) = \bar{N} - S(t)$  constant as well. Consequently,

$$g_C = g_W.$$

The resource constraint and aggregate production function imply

$$g_C = g_Y = g_A,$$

and the idea-production function implies

$$g_A = S(t).$$

Let their common growth rate be  $g$ . Then

$$S(t) = g, \quad L(t) = \bar{N} - g. \quad (9.39)$$

Since  $g_W = g_A$  and  $L(t)$  is constant, Equation (9.37) implies that  $\pi_j(t) = \bar{\pi}$  is constant. Equation (9.35) therefore reduces to

$$P_A = \int_t^\infty e^{-r(\tau-t)} \bar{\pi} d\tau = \frac{\bar{\pi}}{r}.$$

Combining this with the research firm's FOC,

$$\frac{W(t)}{A(t)} = P_A = \frac{\bar{\pi}}{r} = \frac{1}{r} \frac{W(t)L(t)}{A(t)},$$

gives

$$r = L(t) = \bar{N} - g. \quad (9.40)$$

Finally, the household Euler equation gives

$$r = \delta + \gamma g.$$

Combining it with Equation (9.40),

$$\delta + \gamma g = \bar{N} - g,$$

so the decentralized equilibrium growth rate is

$$g_{EQ} = \frac{\bar{N} - \delta}{\gamma + 1}. \quad (9.41)$$

The main BGP quantities satisfy

$$S^{EQ} = g_{EQ}, \quad L^{EQ} = \bar{N} - g_{EQ}, \quad (9.42)$$

$$r^{EQ} = L^{EQ} = \frac{\gamma \bar{N} + \delta}{\gamma + 1}, \quad A(t) = A(0)e^{g_{EQ}t}. \quad (9.43)$$

The associated BGP growth rates are

$$g_C = g_Y = g_{Y/N} = g_A = g_W = g_{p_j} = g_\Pi = g_{EQ},$$

while

$$g_S = g_L = g_r = g_{P_A} = g_{\pi_j} = 0, \quad g_{y_j} = -g_{EQ}.$$

#### Romer equilibrium growth under exam conditions

1. Derive the household Euler equation:  $r = \delta + \gamma g_C$ .
2. Derive the research condition:  $P_A A = W$ .
3. Use the CES demand curve to obtain the constant markup  $p = \sigma W / (\sigma - 1)$ .
4. Under  $\rho = 1/2$  and symmetry, use  $Y = AL$ ,  $y_j = L/A$ , and  $\pi_j = WL/A$ .
5. Use balanced-growth restrictions to establish  $g_C = g_Y = g_A = g_W \equiv g$ , so  $S = g$  and  $L = \bar{N} - g$ .
6. Since  $\pi_j$  and  $r$  are constant, use free entry to write  $P_A = \pi_j / r$ .
7. Combine  $P_A A = W$ ,  $P_A = \pi_j / r$ , and  $\pi_j = WL/A$  to obtain  $r = L = \bar{N} - g$ .
8. Equate this return to the household Euler equation and solve:

$$\delta + \gamma g = \bar{N} - g \implies g = \frac{\bar{N} - \delta}{\gamma + 1}.$$

### 9.8.3 Equilibrium versus the social optimum

For  $\bar{N} > \delta$ ,

$$g_{SP} = \frac{\bar{N} - \delta}{\gamma} > \frac{\bar{N} - \delta}{\gamma + 1} = g_{EQ}. \quad (9.44)$$

The equilibrium underinvests in research for two principal reasons:

1. The research firm takes  $A(t)$  as given and does not internalize the intertemporal knowledge spillover: a new idea raises the stock of knowledge that makes future ideas easier to discover.
2. A patent is valued using the producer surplus generated by the new variety. It does not include the consumer surplus created by that variety, so the private value of innovation is below its social value.

These distortions provide a rationale for a welfare-improving research subsidy. At the same time, the monopoly power that generates patent profits produces a static markup distortion. The policy problem therefore combines the positive dynamic incentive to innovate with the negative static effect of market power.

## 9.9 Scale effects and semiendogenous growth

### 9.9.1 Strong scale effects

In the Romer technology

$$\dot{A}(t) = A(t)S(t),$$

the growth rate of ideas is

$$g_A = \frac{\dot{A}(t)}{A(t)} = S(t).$$

It depends on the *level* of research labor. A larger population can support more scientists and therefore permanently raises the growth rate:

$$\frac{\partial g_{SP}}{\partial \bar{N}} > 0, \quad \frac{\partial g_{EQ}}{\partial \bar{N}} > 0.$$

This is a strong scale effect, and models with this property are called fully endogenous growth models.

The empirical difficulty is that research employment and expenditure have risen substantially over time without a corresponding sustained rise in productivity growth. The fully endogenous model predicts that a permanent increase in the level of research effort should permanently increase  $g_A$ .

### 9.9.2 Ideas become harder to find

A semiendogenous alternative is

$$\dot{A}(t) = A(t)^{1-\beta} S(t), \quad \beta > 0, \quad (9.45)$$

where  $\beta$  is a technology parameter unrelated to the discrete-time discount factor. Now

$$g_A = \frac{\dot{A}(t)}{A(t)} = \frac{S(t)}{A(t)^\beta}.$$

On a balanced growth path,  $g_A$  is constant. Log differentiating the last expression gives

$$0 = g_S - \beta g_A,$$

and hence

$$g_A = \frac{g_S}{\beta}. \quad (9.46)$$

If a constant fraction of a population growing at rate  $n$  performs research, then  $g_S = n$  and

$$g_A = \frac{n}{\beta}.$$

Long-run productivity growth now depends on the *growth rate* of research labor, not its level. This is a weak scale effect.

**Remark** (Policy implication). In a semiendogenous model, a permanent increase in the research share can raise productivity during the transition and permanently raise its *level*, but it does not permanently raise the asymptotic growth rate unless it also changes  $g_S$ . With a constant population and therefore  $g_S = 0$ , long-run idea growth converges to zero.

**Summary 4.** *The essential sequence is*

$$\begin{aligned} \text{research labor} &\implies \text{new varieties or higher quality} \\ &\implies \text{productivity growth} \implies \text{growth in living standards.} \end{aligned}$$

*In the simplified Romer model,*

$$g_{SP} = \frac{\bar{N} - \delta}{\gamma}, \quad g_{EQ} = \frac{\bar{N} - \delta}{\gamma + 1}, \quad g_{SP} > g_{EQ}.$$

*The planner internalizes the social value of ideas; the decentralized economy does not. Fully endogenous models make growth depend on the level of research labor and exhibit strong scale effects. Semiendogenous*

models make long-run growth depend on the growth rate of research labor and therefore exhibit only weak scale effects.

## 10 Risk Sharing and Complete Markets

### 10.1 Uncertainty, events, and histories

Time is discrete,  $t = 0, 1, 2, \dots$ . At each date, a stochastic event  $s_t \in \mathcal{S}$  is realized. The event history through date  $t$  is

$$s^t \equiv (s_0, s_1, \dots, s_t) \in \mathcal{S}^t \equiv \prod_{\tau=0}^t \mathcal{S}.$$

Thus,  $s_t$  denotes the event realized at date  $t$ , whereas  $s^t$  contains all information available at that date. A continuation of  $s^t$  is denoted  $s^\tau \succeq s^t$  for  $\tau \geq t$ .

Let  $\pi(s^t)$  be the date-0 probability of history  $s^t$ . Then

$$\pi(s^t) \in [0, 1], \quad \sum_{s^t \in \mathcal{S}^t} \pi(s^t) = 1.$$

For any continuation  $s^\tau \succeq s^t$  with  $\pi(s^t) > 0$ ,

$$\pi(s^\tau \mid s^t) = \frac{\pi(s^\tau)}{\pi(s^t)}.$$

In particular,

$$\begin{aligned} \pi(s^t) &= \pi(s_t \mid s^{t-1})\pi(s^{t-1}) \\ &= \pi(s_t \mid s^{t-1})\pi(s_{t-1} \mid s^{t-2}) \cdots \pi(s_1 \mid s_0)\pi(s_0). \end{aligned} \tag{10.1}$$

The initial event  $s_0$  is usually known, so  $\pi(s_0) = 1$ .

**Remark** (Two useful special cases). If events are i.i.d., then

$$\pi(s_t \mid s^{t-1}) = \pi(s_t).$$

If the process is first-order Markov, then

$$\pi(s_t \mid s^{t-1}) = \pi(s_t \mid s_{t-1}).$$

The general history notation accommodates both cases.

### 10.2 A risky endowment economy

#### 10.2.1 Endowments and feasibility

There are  $I$  consumers indexed by  $i = 1, \dots, I$ . Consumer  $i$  receives a state-contingent endowment

$$y_t^i(s^t) \geq 0$$

of a homogeneous, nonstorable consumption good. The aggregate endowment is

$$Y_t(s^t) \equiv \sum_{i=1}^I y_t^i(s^t). \quad (10.2)$$

An endowment can be interpreted as the fruit produced by a *Lucas tree*: the tree is the asset and  $y_t^i(s^t)$  is its history-dependent fruit or dividend.

An allocation is a collection of state-contingent consumption plans

$$c^i = \{c_t^i(s^t)\}_{t,s^t}, \quad i = 1, \dots, I.$$

**Definition** (Feasible allocation). An allocation  $\{c^i\}_{i=1}^I$  is feasible if

$$\sum_{i=1}^I c_t^i(s^t) \leq \sum_{i=1}^I y_t^i(s^t) = Y_t(s^t) \quad \forall t, s^t. \quad (10.3)$$

Because the good is nonstorable, feasibility must hold separately after every history.

It is useful to distinguish two types of risk:

- *Idiosyncratic risk*:  $y_t^i(s^t)$  varies across agents or histories while  $Y_t(s^t)$  is unchanged.
- *Aggregate risk*: the total endowment  $Y_t(s^t)$  itself varies across histories.

Risk sharing can reallocate idiosyncratic endowment risk, but it cannot remove aggregate resource risk.

### 10.2.2 Expected utility

All agents have time-separable expected utility

$$U(c^i) = \sum_{t=0}^{\infty} \sum_{s^t \in \mathcal{S}^t} \beta^t \pi(s^t) u(c_t^i(s^t)) = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i), \quad (10.4)$$

where  $\beta \in (0, 1)$  and

$$u'(c) > 0, \quad u''(c) < 0, \quad \lim_{c \downarrow 0} u'(c) = \infty.$$

The last condition is an Inada condition that helps guarantee strictly positive consumption.

The CRRA specification is

$$u(c) = \begin{cases} \frac{c^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1, \end{cases} \quad \gamma > 0. \quad (10.5)$$

For  $\gamma \neq 1$ ,

$$u'(c) = c^{-\gamma}.$$

The Arrow–Pratt coefficients include a minus sign:

$$A(c) \equiv -\frac{u''(c)}{u'(c)} = \frac{\gamma}{c} > 0, \quad R(c) \equiv -c \frac{u''(c)}{u'(c)} = \gamma > 0.$$

Thus  $\gamma$  is the coefficient of relative risk aversion.

### 10.3 The planner's risk-sharing problem

**Definition** (Pareto efficiency). A feasible allocation  $\{c^i\}_{i=1}^I$  is Pareto efficient if there is no other feasible allocation  $\{\tilde{c}^i\}_{i=1}^I$  such that

$$U(\tilde{c}^i) \geq U(c^i) \quad \forall i, \quad U(\tilde{c}^j) > U(c^j) \quad \text{for at least one } j.$$

Fix Pareto weights  $\lambda_i > 0$ , normalized without loss of generality so that  $\sum_i \lambda_i = 1$ . The planner solves

$$\max_{\{c_t^i(s^t)\}} \sum_{i=1}^I \lambda_i \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(c_t^i(s^t)) \quad (10.6)$$

subject to Equation (10.3) after every  $t, s^t$ .

Let  $\theta_t(s^t)$  denote the multiplier on the history-specific resource constraint. The first-order condition for agent  $i$  is

$$\lambda_i \beta^t \pi(s^t) u'(c_t^i(s^t)) = \theta_t(s^t). \quad (10.7)$$

Since utility is increasing, the resource constraint binds. Taking ratios of Equation (10.7) for agents  $i$  and  $j$  gives

$$\frac{u'(c_t^i(s^t))}{u'(c_t^j(s^t))} = \frac{\lambda_j}{\lambda_i} \quad \forall t, s^t. \quad (10.8)$$

Equivalently, using agent 1 as a reference,

$$c_t^i(s^t) = (u')^{-1} \left[ \frac{\lambda_1}{\lambda_i} u'(c_t^1(s^t)) \right]. \quad (10.9)$$

Substituting Equation (10.9) into feasibility yields

$$\sum_{i=1}^I (u')^{-1} \left[ \frac{\lambda_1}{\lambda_i} u'(c_t^1(s^t)) \right] = Y_t(s^t). \quad (10.10)$$

This equation determines  $c_t^1(s^t)$  solely as a function of the aggregate endowment  $Y_t(s^t)$ ; Equation (10.9) then determines every other agent's consumption.

**Proposition** (Perfect risk sharing). *At a Pareto-efficient allocation with common beliefs and common period utility:*

1. each  $c_t^i(s^t)$  depends on the history only through aggregate resources  $Y_t(s^t)$ ;
2. marginal utilities are proportional across agents, as in Equation (10.8);
3. idiosyncratic endowment realizations do not affect an agent's consumption after conditioning on aggregate resources.

**Intuition.** The planner transfers resources toward agents with relatively high marginal utility until weighted marginal utilities are equalized. Idiosyncratic shocks are therefore insured. Aggregate shocks cannot be insured away because they change the economy's total resource constraint.

### 10.3.1 CRRA allocations

Under CRRA utility, Equation (10.8) becomes

$$\frac{(c_t^i(s^t))^{-\gamma}}{(c_t^j(s^t))^{-\gamma}} = \frac{\lambda_j}{\lambda_i},$$

so

$$\frac{c_t^i(s^t)}{c_t^j(s^t)} = \left( \frac{\lambda_i}{\lambda_j} \right)^{1/\gamma}.$$

Combining this condition with feasibility gives

$$c_t^i(s^t) = \frac{\lambda_i^{1/\gamma}}{\sum_{j=1}^I \lambda_j^{1/\gamma}} Y_t(s^t) \equiv \omega_i Y_t(s^t), \quad \sum_i \omega_i = 1. \quad (10.11)$$

Every agent consumes a constant share of the aggregate endowment. Hence

$$\Delta \log c_t^i = \Delta \log Y_t \quad \forall i. \quad (10.12)$$

**Remark** (Empirical implication). A stylized test of full insurance is

$$\Delta \log c_t^i = \alpha_i + \eta \Delta \log Y_t + \xi \Delta \log y_t^i + \varepsilon_t^i.$$

The exact CRRA full-insurance benchmark implies

$$\eta = 1, \quad \xi = 0.$$

Once aggregate resources are controlled for, idiosyncratic income growth should not predict individual consumption growth. Empirically, full insurance is best viewed as a benchmark: actual institutions generally provide only partial insurance.

#### Solving a risk-sharing planner problem

1. Write one feasibility constraint for every date and history.
2. Attach Pareto weights  $\lambda_i$  to agents and a multiplier  $\theta_t(s^t)$  to each resource constraint.
3. Derive

$$\lambda_i u'(c_t^i) = \lambda_j u'(c_t^j).$$

Common factors  $\beta^t \pi(s^t)$  cancel across agents.

4. Express every  $c_t^i$  relative to one reference agent.
5. Use feasibility to determine levels.
6. Under CRRA, jump directly to constant aggregate consumption shares:

$$c_t^i = \frac{\lambda_i^{1/\gamma}}{\sum_j \lambda_j^{1/\gamma}} Y_t.$$

## 10.4 Markov chains

### 10.4.1 Transition probabilities

Let the finite state space be

$$\mathcal{S} = \{\bar{s}_1, \dots, \bar{s}_n\}.$$

The stochastic process  $\{s_t\}_{t \geq 0}$  is a first-order Markov chain if

$$\mathbb{P}(s_{t+1} \mid s_t, s_{t-1}, \dots, s_0) = \mathbb{P}(s_{t+1} \mid s_t). \quad (10.13)$$

Its transition matrix  $P$  has elements

$$P_{ij} \equiv \mathbb{P}(s_{t+1} = \bar{s}_j \mid s_t = \bar{s}_i), \quad P_{ij} \geq 0, \quad \sum_{j=1}^n P_{ij} = 1. \quad (10.14)$$

Thus  $P$  is row-stochastic. The  $k$ -step transition probabilities are

$$\mathbb{P}(s_{t+k} = \bar{s}_j \mid s_t = \bar{s}_i) = (P^k)_{ij}. \quad (10.15)$$

For any function  $g$ ,

$$\mathbb{E}[g(s_{t+1}) \mid s_t = \bar{s}_i] = \sum_{j=1}^n P_{ij} g(\bar{s}_j), \quad (10.16)$$

$$\mathbb{E}[g(s_{t+k}) \mid s_t = \bar{s}_i] = \sum_{j=1}^n (P^k)_{ij} g(\bar{s}_j). \quad (10.17)$$

### 10.4.2 Unconditional and stationary distributions

Let  $\pi_t$  be the column vector whose  $i$ -th element is  $\mathbb{P}(s_t = \bar{s}_i)$ . Then

$$\pi_{t+1}^\top = \pi_t^\top P, \quad \pi_{t+k}^\top = \pi_t^\top P^k. \quad (10.18)$$

**Definition** (Stationary distribution). A probability vector  $\pi$  is stationary or invariant if

$$\pi^\top = \pi^\top P, \quad \pi \geq 0, \quad \mathbf{1}^\top \pi = 1. \quad (10.19)$$

Equivalently,

$$P^\top \pi = \pi,$$

so  $\pi$  is the normalized eigenvector of  $P^\top$  associated with eigenvalue one.

**Theorem 5** (A useful sufficient condition for convergence). *If  $P_{ij} > 0$  for all  $i, j$ , or more generally if every element of  $P^k$  is strictly positive for some finite  $k$ , then the Markov chain has a unique stationary distribution and*

$$\pi_{t+k} \longrightarrow \pi \quad \text{as } k \rightarrow \infty$$

*from any initial distribution.*

### 10.4.3 Dynamic programming under Markov risk

Consider

$$\max_{\{x_{t+1}\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t R(x_t, x_{t+1}, s_t) \quad \text{subject to} \quad x_{t+1} \in \Gamma(x_t, s_t). \quad (10.20)$$

Its recursive representation is

$$V(x, \bar{s}_i) = \max_{x' \in \Gamma(x, \bar{s}_i)} \left\{ R(x, x', \bar{s}_i) + \beta \sum_{j=1}^n P_{ij} V(x', \bar{s}_j) \right\}. \quad (10.21)$$

For an interior choice,

$$0 = R_2(x, x', \bar{s}_i) + \beta \sum_{j=1}^n P_{ij} V_1(x', \bar{s}_j), \quad (10.22)$$

$$V_1(x, \bar{s}_i) = R_1(x, x', \bar{s}_i). \quad (10.23)$$

Combining the FOC and envelope condition yields the stochastic Euler equation

$$0 = R_2(x, x', s) + \beta \mathbb{E} [R_1(x', x'', s') \mid s]. \quad (10.24)$$

If  $x' = g(x, s)$  is the optimal policy, then

$$0 = R_2(x, g(x, s), s) + \beta \mathbb{E} [R_1(g(x, s), g(g(x, s), s'), s') \mid s]. \quad (10.25)$$

This is a functional equation that must hold for every endogenous state  $x$  and exogenous state  $s$ .

**Remark** (Continuous state space). If  $s'$  is continuous with conditional density  $\pi(s' \mid s)$ , sums become integrals:

$$\mathbb{E}[g(s') \mid s] = \int g(s') \pi(s' \mid s) ds'.$$

The Bellman equation and Euler equation retain the same structure.

## 10.5 Complete markets with time-zero trading

### 10.5.1 Arrow–Debreu claims

At date 0, households trade a complete set of contingent claims. A claim for history  $s^t$  delivers one unit of consumption at date  $t$  if and only if history  $s^t$  occurs. Its date-0 price is  $q_t^0(s^t)$ .

Consumer  $i$  faces one lifetime budget constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) c_t^i(s^t) \leq \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) y_t^i(s^t). \quad (10.26)$$

Prices are defined only up to a positive normalization, so the date-0 good can be chosen as numeraire:

$$q_0^0(s_0) = 1.$$

**Definition** (Arrow–Debreu equilibrium). An Arrow–Debreu competitive equilibrium consists of a price system  $\{q_t^0(s^t)\}_{t, s^t}$  and a feasible allocation  $\{c_t^i(s^t)\}_{i, t, s^t}$  such that:

1. taking state prices as given, each household maximizes Equation (10.4) subject to Equation (10.26);

2. goods markets clear after every history:

$$\sum_i c_t^i(s^t) = \sum_i y_t^i(s^t).$$

### 10.5.2 Household optimality and risk sharing

Let  $\mu_i > 0$  be the multiplier on household  $i$ 's date-0 budget constraint. The FOC for consumption at  $t, s^t$  is

$$\beta^t \pi(s^t) u'(c_t^i(s^t)) = \mu_i q_t^0(s^t). \quad (10.27)$$

Taking ratios across agents,

$$\frac{u'(c_t^i(s^t))}{u'(c_t^j(s^t))} = \frac{\mu_i}{\mu_j}. \quad (10.28)$$

This is identical to the planner's condition Equation (10.8) when

$$\lambda_i \propto \frac{1}{\mu_i}. \quad (10.29)$$

Therefore, an Arrow–Debreu equilibrium is Pareto efficient and features perfect risk sharing.

| Planner's problem                   | Arrow–Debreu equilibrium               |
|-------------------------------------|----------------------------------------|
| Pareto weight $\lambda_i$           | inverse budget multiplier $\mu_i^{-1}$ |
| resource multiplier $\theta_t(s^t)$ | state price $q_t^0(s^t)$               |

Under CRRA utility, Equation (10.28) and feasibility imply

$$c_t^i(s^t) = \frac{\mu_i^{-1/\gamma}}{\sum_{j=1}^I \mu_j^{-1/\gamma}} Y_t(s^t). \quad (10.30)$$

**Example** (Constant aggregate endowment). Suppose  $Y_t(s^t) = \bar{Y}$  for all  $t, s^t$ . Perfect risk sharing implies

$$c_t^i(s^t) = \bar{c}_i \quad \forall t, s^t.$$

Using the household budget constraint and the FOC-implied state prices,

$$\bar{c}_i = (1 - \beta) \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) y_t^i(s^t). \quad (10.31)$$

Households consume the annuity value of their expected discounted endowment. All idiosyncratic risk is insured and consumption is constant. If aggregate endowment is risky instead, idiosyncratic risk is still insured, but every household remains exposed to aggregate risk through  $Y_t(s^t)$ .

## 10.6 Sequential trading in Arrow securities

### 10.6.1 Budget constraints and natural debt limits

Markets now reopen after every history. In history  $s^t$ , an Arrow security for next-period event  $s_{t+1}$  costs

$$Q_t(s_{t+1} \mid s^t)$$

units of current consumption and pays one unit next period if  $s_{t+1}$  occurs. Let  $a_{t+1}^i(s^{t+1})$  denote household  $i$ 's holdings of that claim. The sequential budget constraint is

$$c_t^i(s^t) + \sum_{s_{t+1} \in \mathcal{S}} Q_t(s_{t+1} | s^t) a_{t+1}^i(s^{t+1}) \leq y_t^i(s^t) + a_t^i(s^t). \quad (10.32)$$

Current financial wealth  $a_t^i(s^t)$  is the payoff from claims purchased in the preceding period.

Define the price at  $s^t$  of one unit delivered in continuation history  $s^\tau \succeq s^t$  by

$$q_\tau^t(s^\tau) = \prod_{k=t}^{\tau-1} Q_k(s_{k+1} | s^k), \quad q_t^t(s^t) = 1. \quad (10.33)$$

The natural debt limit is the value of household  $i$ 's entire continuation endowment:

$$A_t^i(s^t) = \sum_{\tau=t}^{\infty} \sum_{s^\tau \succeq s^t} q_\tau^t(s^\tau) y_\tau^i(s^\tau). \quad (10.34)$$

The borrowing restriction is

$$a_t^i(s^t) \geq -A_t^i(s^t). \quad (10.35)$$

It is the least restrictive debt limit that rules out Ponzi schemes. It obeys the recursion

$$A_t^i(s^t) = y_t^i(s^t) + \sum_{s_{t+1}} Q_t(s_{t+1} | s^t) A_{t+1}^i(s^{t+1}). \quad (10.36)$$

**Definition** (Sequential complete-markets equilibrium). A sequential complete-markets equilibrium consists of pricing kernels  $\{Q_t(s_{t+1} | s^t)\}$ , consumption allocations  $\{c_t^i(s^t)\}$ , and Arrow-security positions  $\{a_{t+1}^i(s^{t+1})\}$  such that:

1. households optimize subject to Equations (10.32) and (10.35);
2. goods markets clear:

$$\sum_i c_t^i(s^t) = \sum_i y_t^i(s^t);$$

3. Arrow-security markets clear after every history:

$$\sum_i a_{t+1}^i(s^{t+1}) = 0 \quad \forall s^{t+1}.$$

### 10.6.2 The state-price kernel

Let  $\Lambda_t^i(s^t)$  be the multiplier on household  $i$ 's sequential budget constraint. The FOC for consumption is

$$\Lambda_t^i(s^t) = \beta^t \pi(s^t) u'(c_t^i(s^t)). \quad (10.37)$$

Because reaching the natural debt limit would require zero consumption in all continuation histories, the Inada condition implies that the limit does not bind at the optimum. The FOC for an Arrow security is therefore

$$\Lambda_t^i(s^t) Q_t(s_{t+1} | s^t) = \Lambda_{t+1}^i(s^{t+1}). \quad (10.38)$$

Combining Equations (10.37) and (10.38),

$$Q_t(s_{t+1} | s^t) = \beta \pi(s_{t+1} | s^t) \frac{u'(c_{t+1}^i(s^{t+1}))}{u'(c_t^i(s^t))}. \quad (10.39)$$

An Arrow claim is more expensive when its payoff state is more likely and when consumption in that state has high marginal utility.

### 10.6.3 Equivalence with date-zero trading

The date-0 and sequential state prices are related by

$$Q_t(s_{t+1} | s^t) = \frac{q_{t+1}^0(s^{t+1})}{q_t^0(s^t)}. \quad (10.40)$$

Repeated substitution of Equation (10.40) replicates any date-0 contingent claim using a sequence of one-period Arrow securities. Conversely, the date-0 system prices every sequence of one-period trades. Hence both market structures support the same consumption allocation.

**Proposition** (Equivalence). *Under the maintained assumptions, the Arrow–Debreu equilibrium with date-0 trading and the sequential equilibrium with a complete set of one-period Arrow securities have:*

- *the same state-contingent consumption allocations;*
- *consistent relative state prices, as in Equation (10.40);*
- *Pareto efficiency and perfect risk sharing.*

**Remark** (A common distinction). The Arrow-security price includes the conditional probability:

$$Q_t(s_{t+1} | s^t) = \pi(s_{t+1} | s^t) m_{t+1}(s^{t+1}),$$

where the stochastic discount factor

$$m_{t+1} \equiv \beta \frac{u'(c_{t+1})}{u'(c_t)}$$

does *not* include that probability. Expectations incorporate the probabilities separately.

## 10.7 Asset-pricing implications

### 10.7.1 Pricing by replication

A security is a claim to a state-contingent dividend stream  $\{d_t(s^t)\}_{t,s^t}$ . Complete markets allow its dividends to be replicated with Arrow–Debreu claims. By no arbitrage, the security and the replicating portfolio must have the same price.

The date-0 price of the full dividend stream is

$$p_0^0(s_0) = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) d_t(s^t). \quad (10.41)$$

More generally, the price in history  $s^t$ , in units of current consumption, of the tail dividend stream beginning at  $t$  is

$$p_t^t(s^t) = \sum_{\tau=t}^{\infty} \sum_{s^\tau \succeq s^t} q_\tau^t(s^\tau) d_\tau(s^\tau), \quad q_\tau^t(s^\tau) = \frac{q_\tau^0(s^\tau)}{q_t^0(s^t)}. \quad (10.42)$$

Using household optimality,

$$q_\tau^t(s^\tau) = \beta^{\tau-t} \pi(s^\tau | s^t) \frac{u'(c_\tau^i(s^\tau))}{u'(c_t^i(s^t))}. \quad (10.43)$$

Assets are especially valuable when they pay large dividends in likely, nearby histories in which marginal utility is high.

### 10.7.2 One-period returns and the stochastic discount factor

Let an asset purchased at  $t$  cost  $p_t$  and pay  $\omega_{t+1}(s^{t+1})$  next period. Define its gross return by

$$R_{t+1} \equiv \frac{\omega_{t+1}}{p_t}.$$

Its price satisfies

$$\begin{aligned} p_t &= \sum_{s_{t+1}} Q_t(s_{t+1} | s^t) \omega_{t+1}(s^{t+1}) \\ &= \mathbb{E}_t[m_{t+1} \omega_{t+1}], \quad m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}. \end{aligned} \quad (10.44)$$

Dividing by  $p_t$  gives the fundamental return equation

$$1 = \mathbb{E}_t[m_{t+1} R_{t+1}]. \quad (10.45)$$

For CRRA preferences,

$$m_{t+1} = \beta \left( \frac{c_{t+1}}{c_t} \right)^{-\gamma}. \quad (10.46)$$

A one-period risk-free asset has a return  $R_{t+1}^f$  known at date  $t$ . Therefore,

$$R_{t+1}^f = \frac{1}{\mathbb{E}_t[m_{t+1}]} = \left\{ \mathbb{E}_t \left[ \beta \frac{u'(c_{t+1})}{u'(c_t)} \right] \right\}^{-1}. \quad (10.47)$$

**Intuition** (Why marginal utility prices risk). One unit of consumption is especially valuable in a state in which the household would otherwise consume little. An asset that pays in those states provides insurance and commands a high price. The stochastic discount factor  $m_{t+1}$  is therefore high precisely in states in which marginal utility is high.

## 10.8 Recursive complete-markets equilibrium

Assume from now on that  $s_t$  is Markov and that endowments depend only on the current event:

$$y_t^i(s^t) = y^i(s_t).$$

Then equilibrium Arrow prices and natural debt limits can be written as stationary functions

$$Q_t(s_{t+1} | s^t) = Q(s' | s), \quad A_t^i(s^t) = A^i(s).$$

For household  $i$ , the endogenous state is current financial wealth  $a$  and the exogenous state is  $s$ . Its Bellman equation is

$$v^i(a, s) = \max_{c, \{a'(s')\}_{s'}} \left\{ u(c) + \beta \sum_{s'} \pi(s' | s) v^i(a'(s'), s') \right\} \quad (10.48)$$

subject to

$$c + \sum_{s'} Q(s' | s) a'(s') \leq y^i(s) + a, \quad (10.49)$$

$$a'(s') \geq -A^i(s') \quad \forall s'. \quad (10.50)$$

When the natural debt constraint is slack, the FOC for each  $a'(s')$  and the envelope condition are

$$u'(c)Q(s' | s) = \beta\pi(s' | s)v_a^i(a'(s'), s'), \quad (10.51)$$

$$v_a^i(a, s) = u'(c). \quad (10.52)$$

Therefore,

$$Q(s' | s) = \beta\pi(s' | s) \frac{u'(c^i(a'(s'), s'))}{u'(c^i(a, s))}, \quad (10.53)$$

which is the recursive version of Equation (10.39).

**Definition** (Recursive competitive equilibrium). A recursive competitive equilibrium consists of:

$$Q(s' | s), \quad \{A^i(s)\}_i, \quad \{v^i(a, s), c^i(a, s), a^i(a, s, s')\}_i,$$

together with an initial wealth distribution, such that:

1. each household's value and policy functions solve Equation (10.48) subject to Equations (10.49) and (10.50);

2. natural debt limits satisfy

$$A^i(s) = y^i(s) + \sum_{s'} Q(s' | s)A^i(s');$$

3. the goods market clears in every state;

4. every Arrow-security market clears in every state.

#### Recursive equilibrium under Markov risk

1. Use the state  $(a, s)$ : current wealth is endogenous and the Markov event is exogenous.
2. Choose one continuation asset  $a'(s')$  for every possible next state.
3. Write expected continuation value using  $\sum_{s'} \pi(s' | s)v(a'(s'), s')$ .
4. Derive one FOC for each  $a'(s')$ .
5. Combine those FOCs with the envelope condition to obtain the state price kernel.
6. Impose goods-market and state-by-state asset-market clearing.

## 10.9 The representative-agent representation

Complete markets allow individual endowment heterogeneity to be aggregated. Let  $\mu_i$  be the household budget multipliers in an Arrow–Debreu equilibrium and choose Pareto weights

$$\lambda_i = \frac{1}{\mu_i},$$

up to an irrelevant common normalization. Define the representative period utility aggregator

$$u^R(C) \equiv \max_{\{c_i\}_{i=1}^I} \left\{ \sum_{i=1}^I \lambda_i u(c_i) : \sum_{i=1}^I c_i \leq C \right\}. \quad (10.54)$$

The intratemporal FOCs and the envelope theorem imply

$$(u^R)'(C) = \lambda_i u'(c_i) \quad \forall i. \quad (10.55)$$

Using  $\lambda_i = 1/\mu_i$  in the individual Arrow–Debreu FOC,

$$\beta^t \pi(s^t) (u^R)'(C_t(s^t)) = q_t^0(s^t),$$

after a common price normalization. This is precisely the consumption FOC of a single household endowed with aggregate resources

$$C_t(s^t) = Y_t(s^t).$$

**Proposition** (Representative-agent equivalence). *The heterogeneous-agent complete-markets equilibrium has the same aggregate allocation and the same relative state prices as an economy populated by a single representative agent with period utility  $u^R$  and endowment  $Y_t(s^t)$ .*

**Remark.** The representative agent does not mean that households are literally identical. Individual endowments affect the budget multipliers  $\mu_i$ , hence the Pareto weights embedded in  $u^R$  and the distribution of consumption. Complete markets make those distributional details unnecessary for determining aggregate consumption and state prices.

### 10.9.1 CRRA aggregation

If every household has CRRA utility with the same  $\gamma$ , the solution to Equation (10.54) assigns

$$c_i = \frac{\lambda_i^{1/\gamma}}{\sum_j \lambda_j^{1/\gamma}} C.$$

Substitution into the aggregator shows that  $u^R(C)$  is, up to a positive affine transformation, CRRA with the same coefficient  $\gamma$ . Thus one may normalize the representative household's period utility to

$$u^R(C) = \frac{C^{1-\gamma} - 1}{1-\gamma} \quad \text{or} \quad u^R(C) = \log C \quad \text{when } \gamma = 1. \quad (10.56)$$

**Remark** (Identical-household shortcut). An alternative justification uses a unit mass of ex ante identical households. Symmetry implies  $c_t^i(s^t) = C_t(s^t)$  and  $y_t^i(s^t) = Y_t(s^t)$  for every  $i$ , so household indices can be suppressed from the outset. This shortcut is distinct from the aggregation result above, which allows heterogeneous endowments but relies on complete markets.

#### Core complete-markets derivations

1. *Planner:*

$$\lambda_i u'(c_t^i) = \lambda_j u'(c_t^j).$$

2. *Date-0 competitive equilibrium:*

$$\beta^t \pi(s^t) u'(c_t^i) = \mu_i q_t^0(s^t), \quad \lambda_i \propto \mu_i^{-1}.$$

3. *Sequential Arrow prices:*

$$Q_t(s_{t+1} | s^t) = \beta \pi(s_{t+1} | s^t) \frac{u'(c_{t+1}^i)}{u'(c_t^i)}.$$

4. *Equivalence of market structures:*

$$Q_t(s_{t+1} \mid s^t) = \frac{q_{t+1}^0(s^{t+1})}{q_t^0(s^t)}.$$

5. *Asset-return Euler equation:*

$$1 = \mathbb{E}_t[m_{t+1}R_{t+1}], \quad m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

6. *Recursive formulation:* use  $(a, s)$  as the state and choose one  $a'(s')$  for every continuation state.

**Summary 6.** *The central logic of the block is*

*complete contingent claims  $\implies$  perfect insurance of idiosyncratic risk*  
 *$\implies$  proportional marginal utilities*  
 *$\implies$  Pareto-efficient allocations*  
 *$\implies$  representative-agent pricing of aggregate risk.*

*Under CRRA utility,*

$$c_t^i(s^t) = \omega_i Y_t(s^t), \quad m_{t+1} = \beta \left( \frac{C_{t+1}}{C_t} \right)^{-\gamma}, \quad 1 = \mathbb{E}_t[m_{t+1}R_{t+1}].$$

*Complete markets eliminate exposure to individual endowment risk, not to aggregate resource risk. The Arrow–Debreu, sequential, and recursive formulations are alternative representations of the same allocation and price system.*

## 11 Real Business Cycle Models

### 11.1 Lecture 10: Real Business Cycle Models

#### 11.1.1 Motivation

The real business cycle (RBC), stochastic growth, or neoclassical business cycle model adds uncertainty and short-run fluctuations to the neoclassical growth model. Its historical motivation has two central components:

- The deterministic neoclassical growth model provides a theory of capital accumulation and long-run convergence, but not a theory of fluctuations around trend.
- The Lucas critique argues that reduced-form empirical relationships need not remain invariant after a policy change. Macroeconomic models should therefore build aggregate outcomes from micro-founded optimization problems whose decisions may themselves change with policy.

The baseline RBC model treats short-run movements in total factor productivity (TFP), or the Solow residual, as the primitive source of business cycle fluctuations. It also features competitive markets and fully flexible prices. Because its competitive equilibrium coincides with the planner’s allocation, baseline RBC fluctuations are efficient: there is no welfare-improving role for business cycle stabilization within the undistorted model. Later business cycle models retain the RBC core while adding nominal rigidities, demand effects, financial frictions, and policy.

### 11.1.2 Uncertainty, preferences, and technology

Let

$$s^t = (s_0, \dots, s_t)$$

denote the history through date  $t$ , with probability  $\pi(s^t)$ . Capital chosen at  $t-1$ ,  $K_t(s^{t-1})$ , is predetermined when  $s_t$  is realized.

**Representative household.** The household has one unit of time,

$$L_t(s^t) + N_t(s^t) = 1, \quad (11.1)$$

where  $L_t$  is leisure and  $N_t$  is labor. Expected lifetime utility is

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(C_t(s^t), L_t(s^t)), \quad 0 < \beta < 1. \quad (11.2)$$

The period utility function satisfies

$$u_C > 0, \quad u_L > 0, \quad u_{CC} < 0, \quad u_{LL} < 0,$$

together with Inada conditions as consumption or leisure approaches zero. Two examples used in the notes are

$$u(C, L) = \log C + \psi \log L, \quad \psi > 0, \quad (11.3)$$

$$u(C, L) = \frac{(C^\theta L^{1-\theta})^{1-\gamma} - 1}{1-\gamma}, \quad 0 < \theta < 1, \quad \gamma > 0. \quad (11.4)$$

**Production.** An aggregate competitive firm produces

$$Y_t(s^t) = A_t(s^t) F(K_t(s^{t-1}), N_t(s^t)), \quad (11.5)$$

where  $A_t$  is exogenous TFP. The production function is constant returns to scale in  $(K, N)$ , strictly increasing, and concave:

$$F_K > 0, \quad F_N > 0, \quad F_{KK} < 0, \quad F_{NN} < 0.$$

The main example is Cobb–Douglas production,

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}, \quad 0 < \alpha < 1. \quad (11.6)$$

**Capital and resources.** Capital takes one period to build:

$$K_{t+1}(s^t) = (1 - \delta)K_t(s^{t-1}) + I_t(s^t), \quad 0 < \delta < 1. \quad (11.7)$$

The goods-market resource constraint is

$$Y_t(s^t) = C_t(s^t) + I_t(s^t). \quad (11.8)$$

Combining (11.5)–(11.8),

$$C_t(s^t) + K_{t+1}(s^t) = A_t(s^t) F(K_t(s^{t-1}), N_t(s^t)) + (1 - \delta)K_t(s^{t-1}). \quad (11.9)$$

The output good is the numeraire, so all variables and factor prices are real. Negative investment is permitted: the economy can consume part of the undepreciated capital stock.

### 11.1.3 The planner's problem

Given  $K_0$ , the social planner solves

$$\begin{aligned} \max_{\{C_t(s^t), K_{t+1}(s^t), N_t(s^t)\}} \quad & \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(C_t(s^t), 1 - N_t(s^t)) \\ \text{subject to} \quad & C_t(s^t) + K_{t+1}(s^t) = A_t(s^t) F(K_t(s^{t-1}), N_t(s^t)) + (1 - \delta) K_t(s^{t-1}) \end{aligned} \quad (11.10)$$

for every  $t, s^t$ .

Let  $\lambda_t(s^t)$  be the multiplier on the history-specific resource constraint. The first-order conditions are

$$\lambda_t(s^t) = \beta^t \pi(s^t) u_C(C_t(s^t), 1 - N_t(s^t)), \quad (11.11)$$

$$\lambda_t(s^t) = \sum_{s_{t+1}|s^t} \lambda_{t+1}(s^{t+1}) [A_{t+1}(s^{t+1}) F_K(K_{t+1}(s^t), N_{t+1}(s^{t+1})) + (1 - \delta)], \quad (11.12)$$

$$\beta^t \pi(s^t) u_L(C_t(s^t), 1 - N_t(s^t)) = \lambda_t(s^t) A_t(s^t) F_N(K_t(s^{t-1}), N_t(s^t)). \quad (11.13)$$

Combining (11.11) and (11.13) gives the intratemporal labor condition

$$\frac{u_L(C_t, L_t)}{u_C(C_t, L_t)} = A_t F_N(K_t, N_t). \quad (11.14)$$

The left-hand side is the household's marginal rate of substitution between consumption and leisure. An additional unit of labor costs the marginal utility of foregone leisure; its benefit is the marginal utility of the additional output produced by the marginal product of labor.

Combining (11.11) and (11.12) gives the stochastic intertemporal Euler equation

$$u_C(C_t, L_t) = \mathbb{E}_t \{ \beta u_C(C_{t+1}, L_{t+1}) [A_{t+1} F_K(K_{t+1}, N_{t+1}) + (1 - \delta)] \}. \quad (11.15)$$

The marginal cost of one more unit of investment is the foregone marginal utility of consumption today. The marginal benefit is tomorrow's discounted and probability-weighted marginal utility times the marginal product of the new capital plus its undepreciated value.

Using the notation

$$MPN_t = A_t F_N(K_t, N_t), \quad MPK_{t+1} = A_{t+1} F_K(K_{t+1}, N_{t+1}),$$

the two core optimality conditions are simply

$$\frac{u_L(C_t, L_t)}{u_C(C_t, L_t)} = MPN_t, \quad (11.16)$$

$$u_C(C_t, L_t) = \mathbb{E}_t [\beta u_C(C_{t+1}, L_{t+1}) (MPK_{t+1} + 1 - \delta)]. \quad (11.17)$$

### 11.1.4 Competitive equilibrium

In the decentralized economy, the household owns the capital stock and rents it to the firm at the real rental rate  $R_t(s^t)$ . It supplies labor at the real wage  $W_t(s^t)$ . The representative firm makes a static profit-maximizing decision in each date and history.

**Household problem.** Taking  $\{W_t(s^t), R_t(s^t)\}$  as given, the household maximizes (11.2) subject to

$$C_t(s^t) + K_{t+1}(s^t) = R_t(s^t)K_t(s^{t-1}) + W_t(s^t)N_t(s^t) + (1 - \delta)K_t(s^{t-1}). \quad (11.18)$$

Its labor-supply and saving conditions are

$$\frac{u_L(C_t, L_t)}{u_C(C_t, L_t)} = W_t, \quad (11.19)$$

$$u_C(C_t, L_t) = \mathbb{E}_t [\beta u_C(C_{t+1}, L_{t+1})(R_{t+1} + 1 - \delta)]. \quad (11.20)$$

**Firm problem.** At every  $t, s^t$ , the firm solves

$$\max_{K_t^d, N_t^d} \{A_t F(K_t^d, N_t^d) - W_t N_t^d - R_t K_t^d\}. \quad (11.21)$$

Its factor-demand conditions are

$$W_t = A_t F_N(K_t^d, N_t^d), \quad (11.22)$$

$$R_t = A_t F_K(K_t^d, N_t^d). \quad (11.23)$$

**Market clearing.** For every date and history,

$$N_t^d = N_t, \quad K_t^d = K_t, \quad L_t + N_t = 1, \quad (11.24)$$

$$C_t + K_{t+1} = A_t F(K_t, N_t) + (1 - \delta)K_t. \quad (11.25)$$

**Planner-equilibrium equivalence.** Impose factor-market clearing and substitute (11.22)–(11.23) into the household conditions (11.19)–(11.20). The resulting conditions are exactly (11.16)–(11.17), and the resource constraint is also identical. Hence the competitive equilibrium allocation coincides with the social planner's allocation. This is the relevant special case of the First Welfare Theorem.

### 11.1.5 The equilibrium as nine equations in nine unknowns

Given  $K_0$  and  $A_0$ , the equilibrium determines

$$\{C_t, K_{t+1}, I_t, N_t, L_t, Y_t, W_t, R_t, A_t\}.$$

The nine equations or stochastic processes are

1. investment Euler equation:

$$u_C(C_t, L_t) = \mathbb{E}_t [\beta u_C(C_{t+1}, L_{t+1})(A_{t+1} F_K(K_{t+1}, N_{t+1}) + 1 - \delta)];$$

2. labor optimality:

$$u_L(C_t, L_t) = u_C(C_t, L_t) A_t F_N(K_t, N_t);$$

3. goods-market clearing:  $Y_t = C_t + I_t$ ;

4. production:  $Y_t = A_t F(K_t, N_t)$ ;

5. capital accumulation:  $K_{t+1} = I_t + (1 - \delta)K_t$ ;

6. time:  $L_t + N_t = 1$ ;

7. labor demand:  $W_t = A_t F_N(K_t, N_t)$ ;
8. capital demand:  $R_t = A_t F_K(K_t, N_t)$ ;
9. the exogenous stochastic process for  $A_t$ .

The gross one-period return from capital is

$$1 + r_{t+1} = R_{t+1} + 1 - \delta,$$

so the net real interest rate is

$$r_{t+1} = R_{t+1} - \delta. \quad (11.26)$$

## 11.2 Lecture 11: Dynamics, Labor Supply, Calibration, and Solution

### 11.2.1 A positive TFP shock and comovement

Specialize TFP to the stationary log AR(1)

$$\log A_t = \rho \log A_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma^2), \quad 0 < \rho < 1. \quad (11.27)$$

A positive innovation raises current TFP. Its effect decays geometrically with  $\rho$ , which gives the familiar hump-free declining impulse response of TFP in the notes.

**Consumption and investment.** Use

$$C_t = A_t F(K_t, N_t) + (1 - \delta)K_t - K_{t+1}$$

in the investment Euler equation. Treating both sides as functions of  $K_{t+1}$ :

- A higher  $K_{t+1}$  lowers  $C_t$ , so  $u_C(C_t, L_t)$  rises. The marginal cost of investment is increasing.
- A higher  $K_{t+1}$  raises future resources and lowers the future marginal product of capital. Consequently, the discounted marginal benefit on the right-hand side is decreasing.

A positive  $A_t$  shock raises resources and therefore raises  $C_t$  for any fixed  $K_{t+1}$ . Since marginal utility falls, the left-hand side of the Euler equation shifts down, inducing higher  $K_{t+1}$  and investment. Thus both  $C_t$  and  $I_t$  rise. The economic mechanism is consumption smoothing: the household consumes part of the extra resources immediately and carries another part into the future through capital, the economy's storage technology.

**Output.** Since  $Y_t = A_t F(K_t, N_t)$ , output rises directly after a positive productivity shock, holding inputs fixed. Hence  $Y_t$ ,  $C_t$ , and  $I_t$  comove.

**Labor and wages.** Labor-market equilibrium is

$$\underbrace{\frac{u_L(C_t, 1 - N_t)}{u_C(C_t, 1 - N_t)}}_{\text{household MRS}} = W_t = \underbrace{A_t F_N(K_t, N_t)}_{\text{firm MPN}}. \quad (11.28)$$

A higher wage generates a substitution effect toward work. But the associated increase in wealth and consumption can generate an income effect toward leisure. If the substitution effect dominates, labor supply slopes upward. With  $K_t$  predetermined, firm labor demand slopes downward because  $F_{NN} < 0$ .

A positive  $A_t$  shock shifts labor demand upward, raising the wage and, absent a dominant wealth effect, labor. Strong income effects can make labor rise only slightly, remain unchanged, or even fall.

**Comovement result.** In the standard case, a positive TFP shock produces

$$A_t \uparrow \implies Y_t \uparrow, \quad C_t \uparrow, \quad I_t \uparrow, \quad N_t \uparrow.$$

The ability to generate positive comovement among the central aggregates is a major success of the baseline RBC model. The arguments above isolate direct effects; in general equilibrium, future productivity, future consumption, labor, and leisure all move jointly.

### 11.2.2 The Frisch elasticity and preference classes

**Definition.** Let  $\mu_t$  be the marginal utility of wealth. The labor condition can be written as

$$u_L(C_t, 1 - N_t) = W_t \mu_t. \quad (11.29)$$

The Frisch elasticity of labor supply is the elasticity of labor with respect to the wage holding  $\mu_t$  fixed:

$$\mathcal{E}_{N,W}^F \equiv \left. \frac{\partial \log N_t}{\partial \log W_t} \right|_{\mu_t}. \quad (11.30)$$

Holding the marginal utility of wealth fixed removes the income effect and isolates the substitution response. This elasticity is quantitatively important for the response of labor to TFP shocks.

**Log-log preferences.** If

$$u(C, L) = \log C + \psi \log L,$$

then

$$\frac{\psi}{1 - N_t} = W_t \mu_t$$

and

$$\mathcal{E}_{N,W}^F = \frac{1 - N_t}{N_t}. \quad (11.31)$$

The elasticity therefore depends on equilibrium hours.

**Separable CRRA preferences.** For

$$u(C, L) = \frac{C^{1-\gamma} - 1}{1-\gamma} - \psi \frac{(1-L)^{1+\epsilon}}{1+\epsilon}, \quad \gamma, \epsilon > 0, \quad (11.32)$$

the labor condition is

$$\psi N_t^\epsilon = W_t \mu_t,$$

so

$$\mathcal{E}_{N,W}^F = \frac{1}{\epsilon}. \quad (11.33)$$

Here the Frisch elasticity is directly controlled by a parameter. The notes emphasize that microeconomic estimates have often found relatively low values, roughly between 0 and 0.5, although estimates depend on whether the intensive or extensive margin is considered.

**GHH preferences: no wealth effect on labor.** Greenwood–Hercowitz–Huffman (GHH) preferences take the form

$$u(C, L) = \tilde{u}(C - G(1 - L)) = \tilde{u}(C - G(N)), \quad (11.34)$$

where

$$\tilde{u}' > 0, \quad \tilde{u}'' < 0, \quad G'(N) > 0, \quad G''(N) > 0.$$

Their key implication is

$$\frac{u_L(C_t, L_t)}{u_C(C_t, L_t)} = G'(N_t) = W_t. \quad (11.35)$$

The MRS depends on labor alone, so consumption and wealth do not shift labor supply. Since  $G'' > 0$ , labor supply is strictly upward sloping. A natural special case is

$$u(C, L) = \log \left[ C - \psi \frac{(1 - L)^{1+\epsilon}}{1 + \epsilon} \right] = \log \left[ C - \psi \frac{N^{1+\epsilon}}{1 + \epsilon} \right]. \quad (11.36)$$

**KPR or balanced-growth preferences: wealth effects remain.** If consumption and wages grow along a balanced growth path while hours per person remain stationary, the preference class highlighted in the notes is

$$u(C, L) = \frac{[C v(L)]^{1-\gamma} - 1}{1 - \gamma} = \frac{[C v(1 - N)]^{1-\gamma} - 1}{1 - \gamma}, \quad (11.37)$$

with  $\gamma > 0$ ,  $v'(L) > 0$ , and  $v''(L) < 0$ . When  $\gamma = 1$ ,

$$u(C, L) = \log C + \log v(L). \quad (11.38)$$

Log-log preferences are the special case  $v(L) = L^\psi$ . KPR preferences retain a wealth effect. Along a balanced growth path, their income and substitution effects exactly cancel, permitting constant hours even as consumption and wages grow.

### 11.2.3 The nonstochastic steady state

Specialize to

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}, \quad u(C_t, L_t) = \log C_t + \psi \log L_t,$$

and the TFP process in (11.27). In the nonstochastic steady state every endogenous aggregate is constant and the shock is zero. Since

$$\log A^* = \rho \log A^*,$$

stationarity implies

$$A^* = 1. \quad (11.39)$$

**Step 1: capital-labor ratio.** The investment Euler equation gives

$$1 = \beta \left[ \alpha \left( \frac{K^*}{N^*} \right)^{\alpha-1} + (1 - \delta) \right], \quad (11.40)$$

$$\frac{K^*}{N^*} = \left( \frac{\alpha}{\beta^{-1} - 1 + \delta} \right)^{\frac{1}{1-\alpha}}. \quad (11.41)$$

**Step 2: consumption-capital ratio.** Because steady-state investment replaces depreciation,  $I^* = \delta K^*$ . The resource constraint implies

$$\frac{C^*}{K^*} = \left( \frac{K^*}{N^*} \right)^{\alpha-1} - \delta. \quad (11.42)$$

**Step 3: levels and hours.** The labor condition is

$$\frac{\psi C^*}{1 - N^*} = (1 - \alpha) \left( \frac{K^*}{N^*} \right)^\alpha. \quad (11.43)$$

Given the two ratios in (11.41) and (11.42), this condition pins down  $N^*$ , and hence  $K^*$  and  $C^*$ . Equivalently, if

$$x \equiv \frac{K^*}{N^*}, \quad c_K \equiv \frac{C^*}{K^*},$$

then

$$\frac{N^*}{1 - N^*} = \frac{(1 - \alpha)x^{\alpha-1}}{\psi c_K}. \quad (11.44)$$

**Step 4: remaining variables and prices.** Once  $K^*$  and  $N^*$  are known,

$$L^* = 1 - N^*, \quad Y^* = (K^*)^\alpha (N^*)^{1-\alpha}, \quad I^* = \delta K^*, \quad (11.45)$$

$$W^* = (1 - \alpha) \left( \frac{K^*}{N^*} \right)^\alpha, \quad R^* = \alpha \left( \frac{K^*}{N^*} \right)^{\alpha-1}, \quad r^* = R^* - \delta. \quad (11.46)$$

This maps  $(\beta, \alpha, \delta, \psi)$  into all steady-state quantities and factor prices.

#### 11.2.4 Calibration

Calibration chooses parameters so that selected model moments match their empirical counterparts. In a first-order solution, the nonstochastic steady state is also the mean around which simulated variables fluctuate. The notes describe calibration as an informal, exactly identified moment-matching exercise, similar in spirit to exactly identified GMM.

| Parameter | Target or identifying relation                          | Choice in Lecture 11                        |
|-----------|---------------------------------------------------------|---------------------------------------------|
| $\beta$   | $1 = \beta(1 + r^*)$                                    | data suggest about 0.995; conventional 0.99 |
| $\alpha$  | $1 - \alpha = W^* N^* / Y^*$                            | data suggest about 0.37; conventional 1/3   |
| $\delta$  | $I^* / Y^* = \alpha \delta / (\beta^{-1} - 1 + \delta)$ | 0.015 quarterly                             |
| $\psi$    | labor condition with $N^* \simeq 1/3$                   | approximately 1.67                          |
| $\rho$    | autocorrelation of detrended Solow residual             | 0.979                                       |
| $\sigma$  | innovation volatility of detrended Solow residual       | 0.0072                                      |

The details behind these targets are:

- A quarterly real interest rate of roughly 0.41% gives  $\beta \simeq 0.995$ , although  $\beta = 0.99$  is retained for comparability with the early quarterly RBC literature.
- A measured labor share near 0.63 implies  $\alpha \simeq 0.37$ . The conventional older value  $1 - \alpha = 2/3$  gives  $\alpha = 1/3$ .

- With  $I^*/Y^* \simeq 0.143$ , the formula above gives a quarterly depreciation rate around 0.76%. Raising the target investment share to 0.20 to account for omitted intangible investment gives the conventional quarterly value  $\delta \simeq 1.5\%$ , or about 6% annually.
- Targeting one third of available time in work gives  $\psi \simeq 1.67$ .
- Under Cobb–Douglas production, the Solow residual is

$$\log A_t = \log Y_t - \alpha \log K_t - (1 - \alpha) \log N_t. \quad (11.47)$$

For the AR(1), the autocorrelation is  $\rho$  and

$$\text{Var}(\log A_t) = \frac{\sigma^2}{1 - \rho^2}.$$

The Lecture 11 values use the moments reported by King and Rebelo for their detrended TFP series.

### 11.2.5 Numerical solution

Let

$$X_t = (C_t, K_{t+1}, I_t, N_t, L_t, Y_t, W_t, R_t, A_t)'. \quad (11.48)$$

The nonlinear equilibrium system can be written as

$$\mathbb{E}_t f(X_t, X_{t+1}, \varepsilon_{t+1}) = 0, \quad f(X^*, X^*, 0) = 0. \quad (11.49)$$

The desired solution is a law of motion

$$X_{t+1} = g(X_t, \varepsilon_{t+1}). \quad (11.50)$$

Except in special cases,  $g$  has no analytical closed form.

**Projection or global methods.** These methods approximate  $g$  and require the equilibrium conditions to hold over a grid of points. Dynamic programming and interpolation methods belong to this class. Their advantage is that they can preserve important nonlinearities and perform well over a broad region of the state space; their disadvantage is computational cost.

**Perturbation or local methods.** These methods take a Taylor approximation around the nonstochastic steady state. At first order,

$$\tilde{f}(X_t, X_{t+1}, \varepsilon_{t+1}) = F_1 \hat{X}_{t+1} + F_2 \hat{X}_t + F_3 \varepsilon_{t+1}, \quad \hat{X}_t = X_t - X^*, \quad (11.51)$$

and the corresponding linear rational expectations solution has the form

$$\hat{X}_{t+1} = A_1 \hat{X}_t + A_2 \varepsilon_{t+1}. \quad (11.52)$$

The approximation is accurate near the steady state and for small shocks. Software such as DYNARE forms and solves these systems while handling the distinction between predetermined variables, such as capital, and jump variables, such as consumption.

If the system is written in logs, then

$$100\hat{X}_{it} = 100 \log \left( \frac{X_{it}}{X_i^*} \right) \simeq 100 \left( \frac{X_{it}}{X_i^*} - 1 \right), \quad (11.53)$$

so the solution has the convenient interpretation of percentage deviations from steady state.

#### Limits of first-order perturbation.

- First-order coefficients do not change with the variance of the shock, so agents do not respond to changes in risk at this order.
- Asset-pricing applications are sensitive to precisely these risk effects; the notes therefore warn against using a first-order linearized model for asset pricing.
- Local approximations can deteriorate for very large shocks and should not be used to study disasters.

## 11.3 Lecture 12: Quantitative RBC, Extensions, Policy, and Recursion

### 11.3.1 Measuring business cycles

The raw US series for output, investment, consumption, hours, and TFP all contain strong trends. Since the RBC model studied here describes fluctuations around a trend, the data must be detrended. The notes mention three common approaches:

1. a linear trend;
2. the Hodrick–Prescott (HP) filter;
3. a band-pass filter.

The quantitative exercise uses the HP filter with the conventional quarterly smoothing parameter 1600, and reports variables in 100 times logs, so cyclical components are approximately percentage deviations from trend.

For quarterly US data over 1972Q2–2017Q2, the notes report:

| Series $X$  | $\sigma(X)$ | $\sigma(X)/\sigma(Y)$ | $\text{Corr}(X, Y)$ |
|-------------|-------------|-----------------------|---------------------|
| Output      | 1.4800      | 1.0000                | 1.0000              |
| Investment  | 6.8757      | 4.6457                | 0.9155              |
| Consumption | 1.2140      | 0.8203                | 0.8682              |
| Hours       | 1.9130      | 1.2926                | 0.8743              |
| TFP         | 1.2122      | 0.8190                | 0.7677              |

The two central sets of stylized facts are:

- *Comovement*: investment, consumption, hours, and TFP are all positively correlated with output.
- *Relative volatility*: investment is much more volatile than output; hours are about as volatile as, or somewhat more volatile than, output; consumption and TFP are less volatile than output.

### 11.3.2 Quantitative performance of the baseline model

The simulated baseline model uses

$$u(C, L) = \log C + \psi \log L, \quad Y = AK^\alpha N^{1-\alpha},$$

and the calibration

$$\alpha = 0.33, \quad \beta = 0.99, \quad \delta = 0.015, \quad \psi = 1.67, \quad \rho = 0.8254, \quad \sigma = 0.006843. \quad (11.54)$$

The values of  $\rho$  and  $\sigma$  differ from those in Lecture 11 because Lecture 12 re-estimates the TFP process over a different sample and after applying a different detrending procedure. This is a change in empirical implementation, not a change in the AR(1) model.

Using a first-order log-linear solution in DYNARE, the notes report:

| Series $X$  | US data     |                       |                     | Baseline RBC |                       |                     |
|-------------|-------------|-----------------------|---------------------|--------------|-----------------------|---------------------|
|             | $\sigma(X)$ | $\sigma(X)/\sigma(Y)$ | $\text{Corr}(X, Y)$ | $\sigma(X)$  | $\sigma(X)/\sigma(Y)$ | $\text{Corr}(X, Y)$ |
| Output      | 1.4800      | 1.0000                | 1.0000              | 1.6278       | 1.0000                | 1.0000              |
| Investment  | 6.8757      | 4.6457                | 0.9155              | 7.2455       | 4.4510                | 0.9775              |
| Consumption | 1.2140      | 0.8203                | 0.8682              | 0.4732       | 0.2907                | 0.6085              |
| Hours       | 1.9130      | 1.2926                | 0.8743              | 0.9278       | 0.5700                | 0.9629              |
| TFP         | 1.2122      | 0.8190                | 0.7677              | 0.9632       | 0.5917                | 0.9977              |

The evaluation in the notes is:

- *Not bad overall:* the model reproduces positive comovement, high investment volatility, and low consumption volatility.
- *Hours are too smooth:* the wealth effect in log-log preferences dampens the labor response.
- *Correlations are too strong:* a single shock drives essentially all fluctuations, so model variables are more tightly correlated with output than in the data.

### 11.3.3 Indivisible labor and the extensive margin

Most observed variation in total hours comes from the extensive margin (employment), not from large movements in hours per employed worker. To capture this, suppose the representative household contains a unit mass of individuals. A fraction  $\phi_t$  works exactly  $\bar{n} > 0$  hours and a fraction  $1 - \phi_t$  does not work. The household chooses a lottery over employment:

$$\begin{aligned} \max_{\{C_t, K_{t+1}, \phi_t\}} \quad & \sum_t \sum_{s^t} \beta^t \pi(s^t) [\log C_t + (1 - \phi_t) \psi \log(1) + \phi_t \psi \log(1 - \bar{n})] \\ \text{subject to} \quad & C_t + K_{t+1} = R_t K_t + W_t \phi_t \bar{n} + (1 - \delta) K_t. \end{aligned} \quad (11.55)$$

Aggregate hours are  $N_t = \phi_t \bar{n}$ . Since  $\log(1) = 0$ , the period utility can be rewritten as

$$\log C_t - A N_t + B, \quad A = \frac{\psi}{\bar{n}} \log \left( \frac{1}{1 - \bar{n}} \right) > 0, \quad (11.56)$$

where the constant  $B$  is irrelevant.

Utility is linear in aggregate labor. Relative to

$$\log C - \frac{\psi}{1 + \epsilon} N^{1+\epsilon},$$

the indivisible-labor case has  $\epsilon = 0$ , and therefore an infinite Frisch elasticity. Aggregate hours can move by changing the fraction employed; no individual worker must move up a convex intensive-margin labor-supply curve. The only changed equilibrium condition is

$$A = \frac{W_t}{C_t}. \quad (11.57)$$

The corresponding simulated moments are:

| Series $X$  | US data     |                       |                     | Indivisible-labor RBC |                       |                     |
|-------------|-------------|-----------------------|---------------------|-----------------------|-----------------------|---------------------|
|             | $\sigma(X)$ | $\sigma(X)/\sigma(Y)$ | $\text{Corr}(X, Y)$ | $\sigma(X)$           | $\sigma(X)/\sigma(Y)$ | $\text{Corr}(X, Y)$ |
| Output      | 1.4800      | 1.0000                | 1.0000              | 2.4348                | 1.0000                | 1.0000              |
| Investment  | 6.8757      | 4.6457                | 0.9155              | 11.1010               | 4.5593                | 0.9827              |
| Consumption | 1.2140      | 0.8203                | 0.8682              | 0.6162                | 0.2531                | 0.5731              |
| Hours       | 1.9130      | 1.2926                | 0.8743              | 2.1420                | 0.8798                | 0.9718              |
| TFP         | 1.2122      | 0.8190                | 0.7677              | 0.9632                | 0.3956                | 0.9999              |

Hours become substantially more volatile and much closer to output volatility, which is the targeted improvement of the extension.

#### 11.3.4 Other RBC extensions

**Investment shocks.** An investment-specific shock  $\mu_t$  can enter the capital law of motion:

$$K_{t+1} = \mu_t I_t + (1 - \delta_t) K_t. \quad (11.58)$$

The shock changes the efficiency with which investment creates future capital.

**Variable capacity utilization.** Let  $u_t$  denote the intensity with which installed capital is used:

$$Y_t = A_t (u_t K_t)^\alpha N_t^{1-\alpha}, \quad \delta_t = \delta(u_t), \quad \delta'(u_t) > 0. \quad (11.59)$$

Higher utilization produces more output today but depreciates capital faster. After a positive TFP shock, utilization rises to exploit the temporarily high marginal product of capital, amplifying output fluctuations.

**News shocks.** Agents may learn in advance about future productivity:

$$\log A_{t+1} = \rho \log A_t + \varepsilon_{t+1} + \varepsilon_t^{\text{news}}, \quad (11.60)$$

where  $\varepsilon_t^{\text{news}}$  is observed at  $t$ . News can generate strong current consumption and investment responses. Labor responses are more delicate because wealth effects can work in the opposite direction.

The purpose of these extensions is targeted: indivisible labor raises hours volatility, utilization amplifies output, and news shocks introduce anticipated movements in fundamentals. Their success is judged using the same comovement and volatility moments used for the baseline model.

#### 11.3.5 How should TFP shocks be interpreted?

**Slower technological progress, not literal regress.** The model describes deviations from a trend. A negative cyclical TFP shock usually means technology grows more slowly than its trend, not that society literally forgets productive knowledge.

**A measure of ignorance.** The Solow residual is the component of output not explained by observed inputs. Treating all of it as an exogenous technological shock makes TFP a “measure of our ignorance.” This motivates models that generate large endogenous fluctuations from smaller and more interpretable primitive shocks.

**A placeholder for other shocks.** The notes give two examples:

- With search frictions between consumers and producers, demand shocks can alter output conditional on measured inputs and therefore appear as TFP shocks.
- Higher uncertainty can reduce activity and measured productivity, generating an endogenous efficiency decline during recessions.

**Business cycle accounting.** The RBC model can also be used as an accounting device with wedges:

- an *efficiency wedge*, represented by measured TFP;
- a *labor wedge*, which distorts equality between the household MRS and the firm’s marginal product of labor;
- an *investment wedge*, which distorts the intertemporal Euler equation.

A proposed model can then be evaluated by asking whether it generates the empirical behavior of these wedges. Financial frictions typically appear in the investment wedge; uncertainty can appear in the efficiency wedge; and the labor wedge is quantitatively important in the data discussed by the notes.

### 11.3.6 Policy basics: do not distort an efficient RBC equilibrium

Suppose the government introduces a marginal labor-income tax  $\tau_{Nt}(s^t)$  and rebates its revenue through a lump-sum transfer  $T_t(s^t)$ . The household budget becomes

$$C_t + K_{t+1} = R_t K_t + (1 - \tau_{Nt}) W_t N_t + (1 - \delta) K_t + T_t. \quad (11.61)$$

In equilibrium the government budget is

$$T_t = \tau_{Nt} W_t N_t, \quad (11.62)$$

but an individual household takes the transfer as a lump sum. Its optimality conditions are

$$u_C(C_t, L_t) = \mathbb{E}_t [\beta u_C(C_{t+1}, L_{t+1})(R_{t+1} + 1 - \delta)], \quad (11.63)$$

$$\frac{u_L(C_t, 1 - N_t)}{u_C(C_t, 1 - N_t)} = (1 - \tau_{Nt}) W_t. \quad (11.64)$$

The tax drives a wedge between the MRS and the marginal product of labor. Whenever it changes the allocation relative to the no-tax equilibrium, the new allocation is feasible but was not selected by the planner. Since the original competitive equilibrium already solved the planner’s problem, the distortionary allocation lowers welfare. The government may stabilize some quantities, but within the RBC model the distortion costs more than that stabilization is worth.

The notes emphasize three qualifications:

1. the government may need to raise exogenous revenue and may lack nondistortionary instruments;
2. public spending financed by the tax may itself create benefits;

3. the assumptions supporting RBC efficiency, especially fully flexible prices, may be empirically inappropriate.

Subject to the baseline RBC assumptions, however, any fiscal policy that distorts the efficient competitive equilibrium produces a welfare loss.

### 11.3.7 Recursive formulation

With stationary TFP, the predetermined state is the pair  $(A, K)$ . Dropping time subscripts and using primes for next-period variables, the planner's Bellman equation is

$$V(A, K) = \max_{K', N} \left\{ u(AF(K, N) - K' + (1 - \delta)K, 1 - N) + \beta \mathbb{E}[V(A', K') \mid A] \right\}, \quad (11.65)$$

$$\log A' = \rho \log A + \varepsilon'.$$

Consumption is

$$C = AF(K, N) - K' + (1 - \delta)K. \quad (11.66)$$

The first-order conditions are

$$u_C(C, 1 - N) = \beta \mathbb{E}[V_K(A', K') \mid A], \quad (11.67)$$

$$u_L(C, 1 - N) = u_C(C, 1 - N)AF_N(K, N), \quad (11.68)$$

and the envelope condition is

$$V_K(A, K) = u_C(C, 1 - N)[AF_K(K, N) + (1 - \delta)]. \quad (11.69)$$

Substituting the future envelope condition into (11.67) yields

$$u_C(C, 1 - N) = \beta \mathbb{E} \{ u_C(C', 1 - N') [A' F_K(K', N') + (1 - \delta)] \mid A \}, \quad (11.70)$$

which is exactly the sequential investment Euler equation. Equation (11.68) is exactly the sequential intratemporal labor condition. The recursive and sequential formulations therefore characterize the same planner allocation.

### 11.3.8 RBC block: central logic

The core chain of the three lectures is

$$\begin{aligned} \text{TFP shocks} &\implies \text{joint household and firm responses} \implies \text{business cycle comovement,} \\ \text{competitive factor prices} &= \text{marginal products} \implies \text{competitive allocation} = \text{planner allocation,} \\ \text{efficient baseline equilibrium} &\implies \text{distortionary stabilization lowers welfare within the model.} \end{aligned}$$

Quantitatively, baseline RBC performs well on broad comovement and investment volatility but produces hours that are too smooth and correlations that are too high. Indivisible labor improves hours volatility; other extensions target amplification or alternative shocks. The model can be solved globally, locally around the steady state, or recursively with state  $(A, K)$ .

## 12 Supplemental Non-Tested Lecture: Asset Pricing

*This section corresponds to the lecture explicitly labeled supplemental and non-tested. It is included for completeness but should remain separate from the tested RBC material.*

### 12.1 The stochastic discount factor

**Definition.** A stochastic discount factor (SDF) is a stochastic process  $\{M_{t,t+1}\}_t$  such that the time- $t$  price of every asset with time- $t+1$  payoff  $X_{t+1}$  satisfies

$$P_t = \mathbb{E}_t[M_{t,t+1}X_{t+1}]. \quad (12.1)$$

If

$$R_{t+1} \equiv \frac{X_{t+1}}{P_t},$$

then every return obeys the fundamental moment condition

$$1 = \mathbb{E}_t[M_{t,t+1}R_{t+1}]. \quad (12.2)$$

#### 12.1.1 Law of one price and no arbitrage

Suppose assets  $k = 1, \dots, N$  have prices  $P_t^k$  and payoffs  $X_{t+1}^k$ . A portfolio with holdings  $\theta = (\theta_1, \dots, \theta_N)$  has payoff

$$X_{t+1}^\theta = \sum_{k=1}^N \theta_k X_{t+1}^k.$$

Linearity of conditional expectations implies

$$P_t^\theta = \mathbb{E}_t \left[ M_{t,t+1} \sum_{k=1}^N \theta_k X_{t+1}^k \right] = \sum_{k=1}^N \theta_k P_t^k. \quad (12.3)$$

Thus portfolios with identical payoffs must have identical prices. If instead  $P_t^\theta > \sum_k \theta_k P_t^k$ , one could short the expensive composite asset and buy the underlying portfolio. The date- $t$  cost would be negative and the date- $t+1$  net payoff would be zero: an arbitrage. The notes emphasize the deeper result that existence of an SDF and absence of arbitrage are equivalent.

#### 12.1.2 Risk-free assets, equity, and excess returns

**Risk-free asset.** If the payoff is one unit for sure, then

$$R_{t+1}^f = \frac{1}{P_t^f}$$

is known at  $t$ . Equation (12.2) gives

$$R_{t+1}^f = \frac{1}{\mathbb{E}_t[M_{t,t+1}]}. \quad (12.4)$$

**Equity.** A share of equity pays its dividend plus its resale price:

$$X_{t+1}^e = D_{t+1}^e + P_{t+1}^e.$$

Consequently,

$$P_t^e = \mathbb{E}_t [M_{t,t+1}(D_{t+1}^e + P_{t+1}^e)] , \quad (12.5)$$

$$1 = \mathbb{E}_t [M_{t,t+1}R_{t+1}^e] . \quad (12.6)$$

**Risky excess returns.** For any risky return  $R_{t+1}$ ,

$$1 = \mathbb{E}_t[M_{t,t+1}R_{t+1}] = \mathbb{E}_t[M_{t,t+1}]\mathbb{E}_t[R_{t+1}] + \text{Cov}_t(M_{t,t+1}, R_{t+1}), \quad (12.7)$$

$$1 = R_{t+1}^f \mathbb{E}_t[M_{t,t+1}]. \quad (12.8)$$

Therefore,

$$\frac{\mathbb{E}_t[R_{t+1}] - R_{t+1}^f}{R_{t+1}^f} = -\text{Cov}_t(R_{t+1}, M_{t,t+1}), \quad (12.9)$$

or, equivalently,

$$\mathbb{E}_t[R_{t+1}] - R_{t+1}^f = \frac{\sigma_t(M_{t,t+1})}{\mathbb{E}_t[M_{t,t+1}]} [-\text{Corr}_t(R_{t+1}, M_{t,t+1})\sigma_t(R_{t+1})]. \quad (12.10)$$

The first factor,

$$\frac{\sigma_t(M_{t,t+1})}{\mathbb{E}_t[M_{t,t+1}]},$$

is the *market price of risk*. The term in brackets is the quantity of risk: it depends on return volatility and on the return's correlation with the SDF. Because the SDF is high in bad times, an asset that pays low returns in bad times has a negative covariance with the SDF and must offer a positive expected excess return. For equity, this excess return is the equity premium.

## 12.2 The standard consumption-based model

Consider the two-period problem

$$\begin{aligned} \max_{\theta} \quad & u(C_t) + \beta \mathbb{E}_t[u(C_{t+1})] \\ \text{subject to} \quad & C_t + \theta P_t = Q_t, \quad C_{t+1} = \theta X_{t+1} + Q_{t+1}, \end{aligned} \quad (12.11)$$

where the household chooses asset holdings  $\theta$ . The first-order condition is

$$P_t u'(C_t) = \beta \mathbb{E}_t [u'(C_{t+1})X_{t+1}]. \quad (12.12)$$

Hence the consumption-based SDF is

$$M_{t,t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}. \quad (12.13)$$

For CRRA utility,

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0, \quad (12.14)$$

the SDF becomes

$$M_{t,t+1} = \beta \left( \frac{C_{t+1}}{C_t} \right)^{-\gamma}. \quad (12.15)$$

Because  $\gamma > 0$ , the SDF is high when consumption growth is low. Those are bad times, when one additional unit of consumption is especially valuable. Importantly, this moment condition requires only

optimal asset choice and CRRA preferences; it does not depend on many other details of the household's income process or the rest of the model.

### 12.2.1 Joint lognormality

Let

$$g_{t+1} \equiv \Delta \log C_{t+1} = \log C_{t+1} - \log C_t.$$

Assume

$$\begin{pmatrix} g_{t+1} \\ \log R_{t+1} \end{pmatrix} \stackrel{\text{iid}}{\sim} \mathcal{N} \left[ \begin{pmatrix} \mu_C \\ \mu_R \end{pmatrix}, \begin{pmatrix} \sigma_C^2 & \sigma_{CR} \\ \sigma_{CR} & \sigma_R^2 \end{pmatrix} \right]. \quad (12.16)$$

**Risk-free rate.** Using  $\mathbb{E}[e^Z] = e^{\mu_Z + \sigma_Z^2/2}$  for a normal variable  $Z$ ,

$$(R_{t+1}^f)^{-1} = \beta \mathbb{E}_t[e^{-\gamma g_{t+1}}] = \beta e^{-\gamma \mu_C + \frac{1}{2} \gamma^2 \sigma_C^2}, \quad (12.17)$$

$$\log R_{t+1}^f = -\log \beta + \gamma \mu_C - \frac{\gamma^2 \sigma_C^2}{2}. \quad (12.18)$$

**Risky return.** The Euler equation is

$$1 = \beta \mathbb{E}_t [e^{-\gamma g_{t+1} + \log R_{t+1}}].$$

Joint normality implies

$$-\gamma g_{t+1} + \log R_{t+1} \sim \mathcal{N}(-\gamma \mu_C + \mu_R, \gamma^2 \sigma_C^2 - 2\gamma \sigma_{CR} + \sigma_R^2).$$

After applying the lognormal expectation formula and using

$$\log \mathbb{E}_t[R_{t+1}] = \mu_R + \frac{\sigma_R^2}{2},$$

one obtains

$$\begin{aligned} \log \left( \frac{\mathbb{E}_t[R_{t+1}]}{R_{t+1}^f} \right) &= \gamma \text{Cov}_t(\Delta \log C_{t+1}, \log R_{t+1}) \\ &= \gamma \text{Corr}_t(\Delta \log C_{t+1}, \log R_{t+1}) \sigma_t(\Delta \log C_{t+1}) \sigma_t(\log R_{t+1}). \end{aligned} \quad (12.19)$$

This is the CRRA-lognormal specialization of (12.10).

## 12.3 US asset-return facts

Observed financial returns are nominal. If  $\Pi_{t+1} = P_{t+1}/P_t$  is gross inflation, then

$$R_{t+1}^{\text{nominal}} = R_{t+1} \Pi_{t+1}. \quad (12.20)$$

For net rates, the Fisher approximation gives

$$r_{t+1} \simeq r_{t+1}^{\text{nominal}} - \pi_{t+1}. \quad (12.21)$$

The notes use the core PCE deflator to convert nominal financial returns into real returns, and compare:

- the annualized real return on the Wilshire 5000 total stock market;

- the annualized real return on a one-year US Treasury bond;
- annualized real consumption growth.

For 1972Q2–2016Q1, the reported moments, in percentage units, are:

| Series $X$                               | $\mathbb{E}(X)$ | $\sigma(X)$ | $\text{Corr}(X, \Delta \log C)$ |
|------------------------------------------|-----------------|-------------|---------------------------------|
| Stock-market return $\log R_{t+1}$       | 6.4210          | 17.9316     | 0.2608                          |
| Risk-free return $\log R_{t+1}^f$        | 2.0287          | 2.4322      | 0.3548                          |
| Consumption growth $\Delta \log C_{t+1}$ | 2.9281          | 1.8663      | 1.0000                          |

Equity returns are much more volatile than risk-free returns, are procyclical, and exceed the risk-free return by about 4.4 percentage points on average.

## 12.4 The equity premium puzzle

The positive equity premium is qualitatively consistent with the model: equity pays poorly in bad times and is risky, so investors require compensation. Quantitatively, however, the observed premium is too large for the standard CRRA-lognormal model.

Using the moments above in (12.19) requires approximately

$$\gamma \simeq 50. \quad (12.22)$$

This value is implausibly high for two related reasons:

1. An agent with  $\gamma = 50$  facing a lottery that pays 100 with probability 1/2 and 500 with probability 1/2 has a certainty equivalent of only about 101.
2. With  $\beta \simeq 1$ , the risk-free-rate formula (12.18) implies a risk-free return of roughly 103% per year in the calculation reported in the notes, far above the observed value.

The same CRRA parameter controls two objects:

- relative risk aversion is  $\gamma$ ;
- the intertemporal elasticity of substitution is  $1/\gamma$ .

Raising  $\gamma$  helps generate a large equity premium, but simultaneously makes households extremely unwilling to substitute consumption across time, requiring a very high risk-free return to induce saving. The equity-premium puzzle can therefore also be viewed as a risk-free-rate puzzle: within the standard CRRA-lognormal model, one cannot match both moments simultaneously.

### 12.4.1 Possible resolutions

#### Wrong preferences.

- Epstein–Zin preferences separate risk aversion from willingness to substitute consumption intertemporally. The extra flexibility can match the equity premium without forcing the same parameter to explain both margins.
- Habit formation makes utility depend on consumption relative to a habit formed from past consumption. This produces a more volatile SDF and therefore a higher market price of risk.

**Wrong shock process.** Returns exhibit crashes and negative skewness, which joint lognormality omits. Rare-disaster models allow consumption growth and returns to be approximately lognormal in normal times but to collapse with small probability. With disaster risk, moderate CRRA risk aversion can generate a much larger equity premium.

**Wrong model.** The representative-agent framework implicitly abstracts from incomplete markets, heterogeneous exposure to risk, limited information, and borrowing constraints. Departures along these dimensions can change asset prices and the relevant SDF.

#### 12.4.2 Asset-pricing block: equations to retain

$$\begin{aligned}
P_t &= \mathbb{E}_t[M_{t,t+1}X_{t+1}], & 1 &= \mathbb{E}_t[M_{t,t+1}R_{t+1}], \\
R_{t+1}^f &= \frac{1}{\mathbb{E}_t[M_{t,t+1}]}, & M_{t,t+1} &= \beta \left( \frac{C_{t+1}}{C_t} \right)^{-\gamma}, \\
\mathbb{E}_t[R_{t+1}] - R_{t+1}^f &= \frac{\sigma_t(M_{t,t+1})}{\mathbb{E}_t[M_{t,t+1}]} [-\text{Corr}_t(R_{t+1}, M_{t,t+1})\sigma_t(R_{t+1})], \\
\log R_{t+1}^f &= -\log \beta + \gamma\mu_C - \frac{\gamma^2\sigma_C^2}{2}, \\
\log \left( \frac{\mathbb{E}_t[R_{t+1}]}{R_{t+1}^f} \right) &= \gamma \text{Cov}_t(\Delta \log C_{t+1}, \log R_{t+1}).
\end{aligned}$$